



SHIP CONSTRUCTION PROCESS

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Overview & Insights From The BBI Dev Team,
September 26, 2022

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Important Disclaimer & Context

As the community of Hardspace: Shipbreaker enthusiasts grew throughout Early Access and launch, we were fortunate to see modders try to engage with our game. While making ships for the game is no small endeavour - making it unlikely that we will ever be in a position to release proper modding tools, as we explained in [further details in our FAQ](#) - we have been blown away by some of the attempts we have seen from the community.

Although we are still skeptical about the possibility of modding new ships into the game, history has proven time and time again that there is no telling what the collective intelligence of passionate and talented modders can do! So we figured the least we could do was pull the curtain back and provide some extra info to the brave souls willing to take up the challenge.

This document you are reading is taken straight from our internal technical documentation for making the game. It is the blue-print to our ship making (and ship testing) process, outlining step-by-step what is needed to create a basic ship and ensure it functions as intended in Unity.

We are not really in a capacity to provide any further support or documentation to modders beyond what has been gathered here - but we hope it might be enough to rekindle the fire, or at least satiate the curiosity of inquisitive and technically-minded fans.

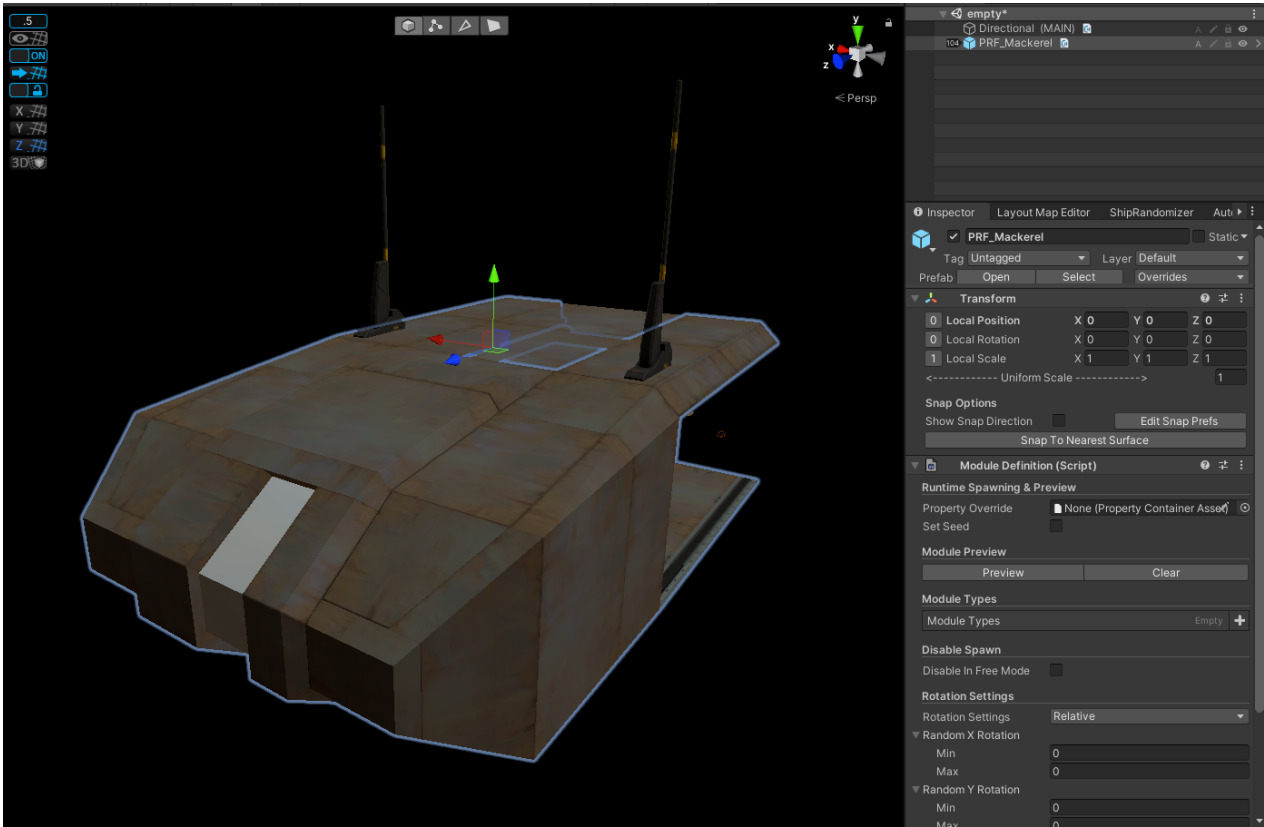
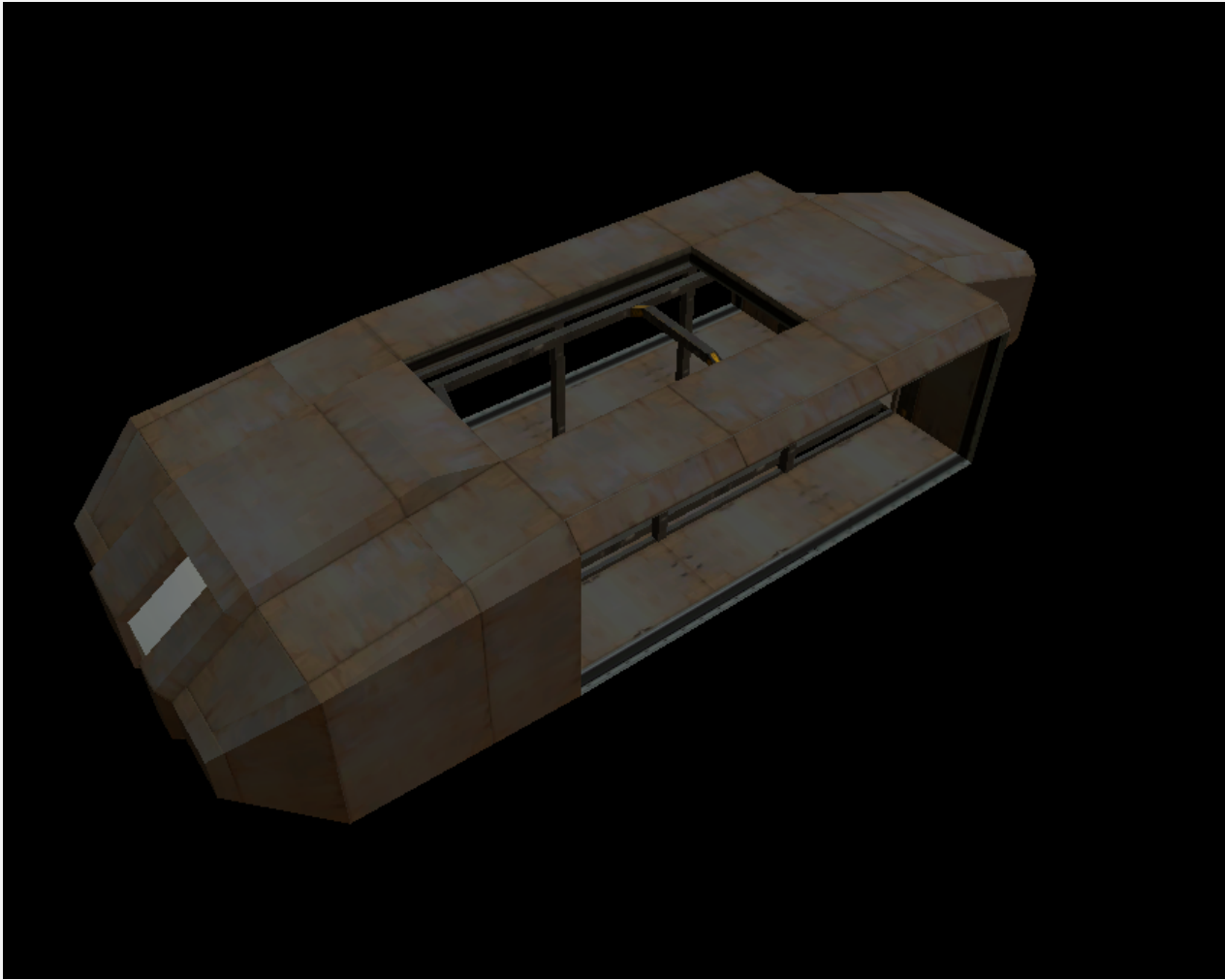
Thank you so much again for supporting us on our journey and making the game a success,

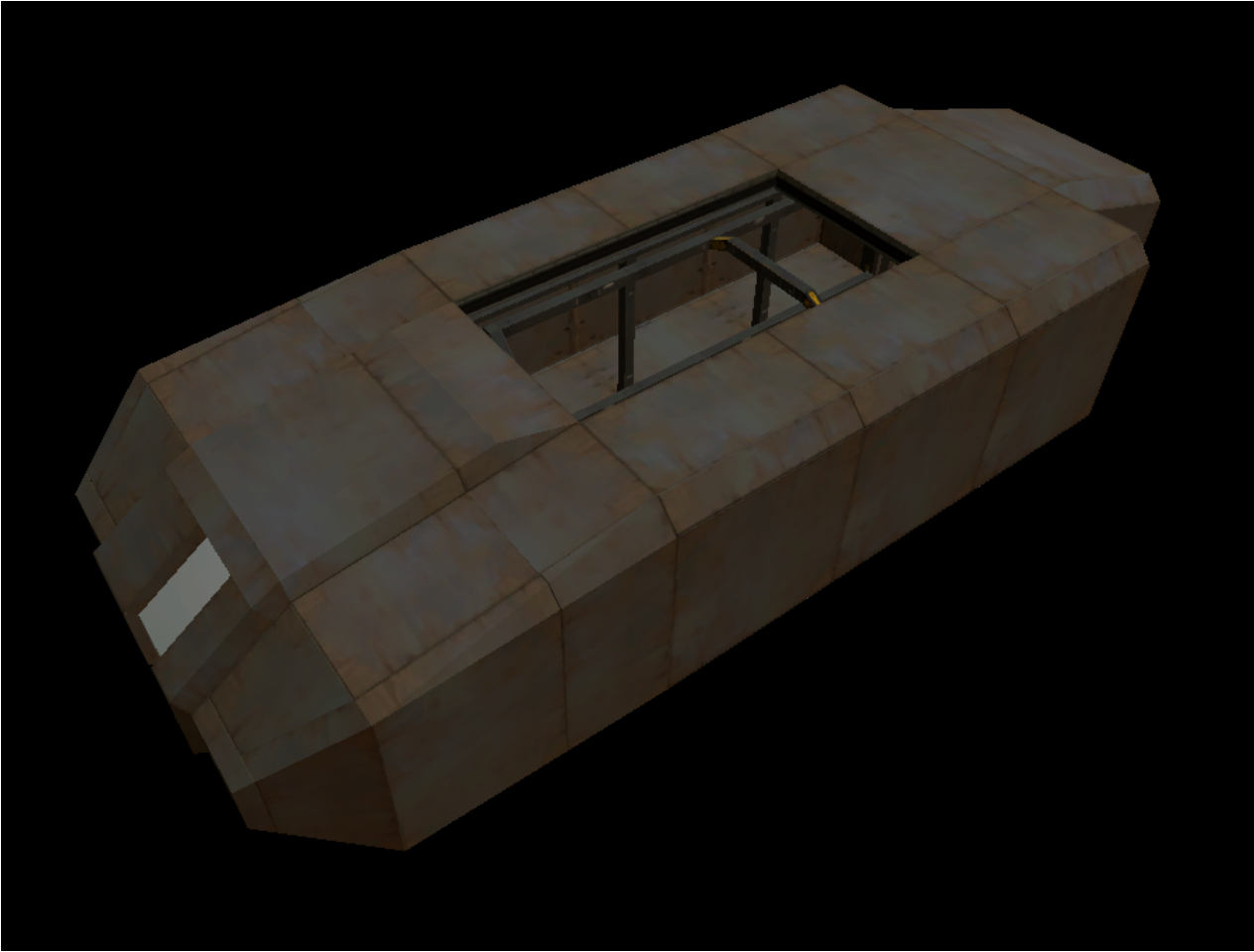
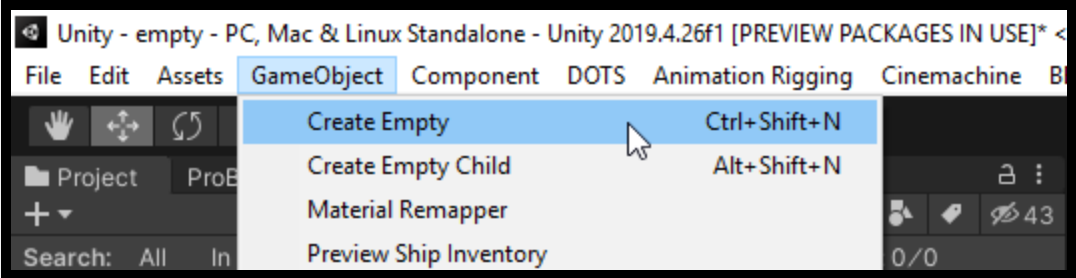
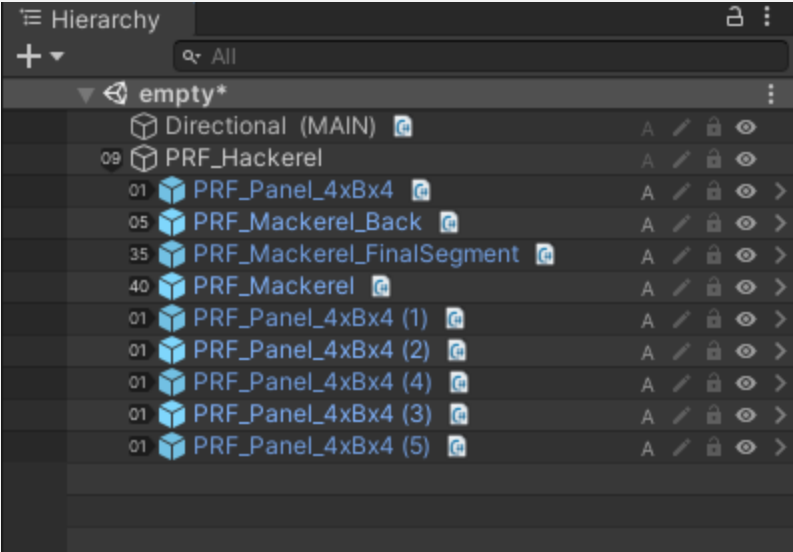
– The Shipbreaker Team at Blackbird Interactive

A Note About Testing

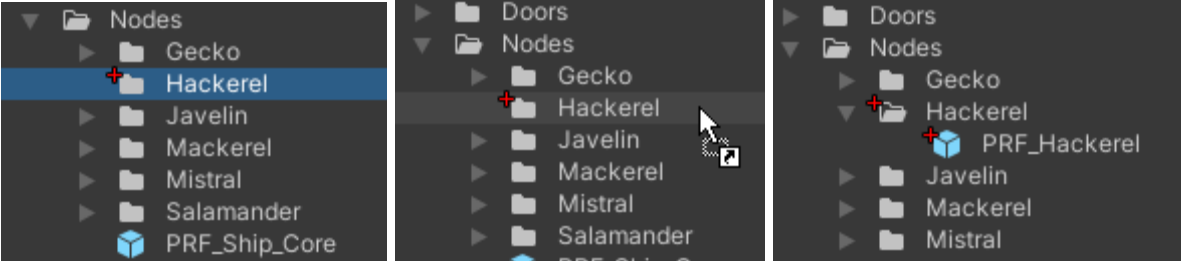
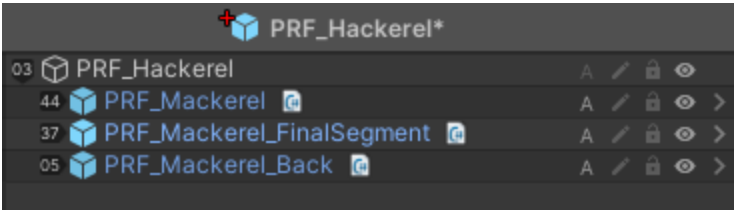
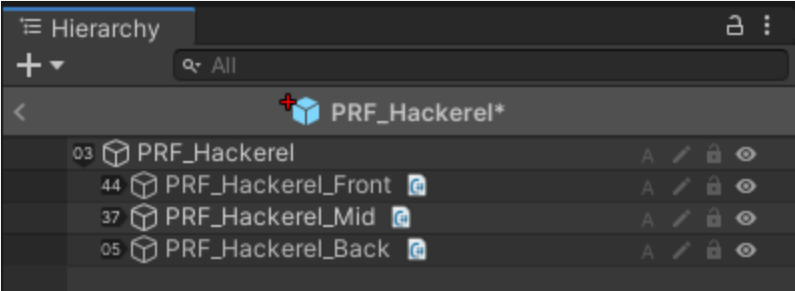
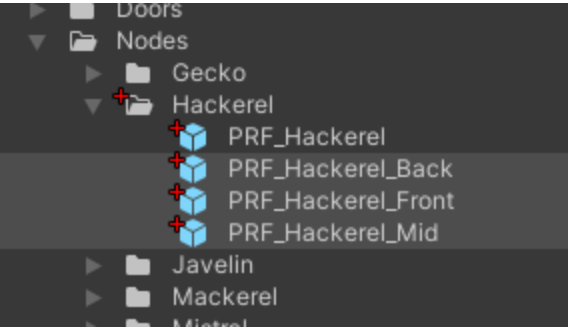
This document should also serve as an instructional process for testing all the systems involved in making a ship for the Adressables Stream - which is at moderate risk of having bugs or failures in one or more related tools. Ideally, every system should be tested by these steps, so as to eliminate any questions around the construction of future ships, or the elimination of bugs for Release and Console Builds.

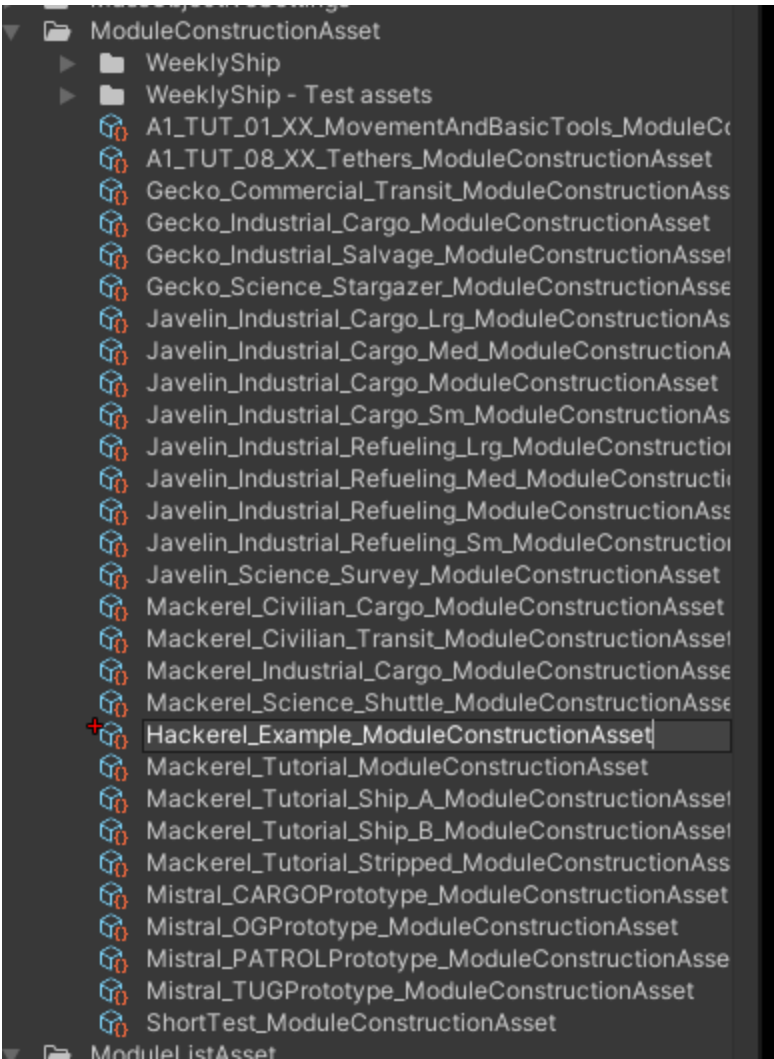
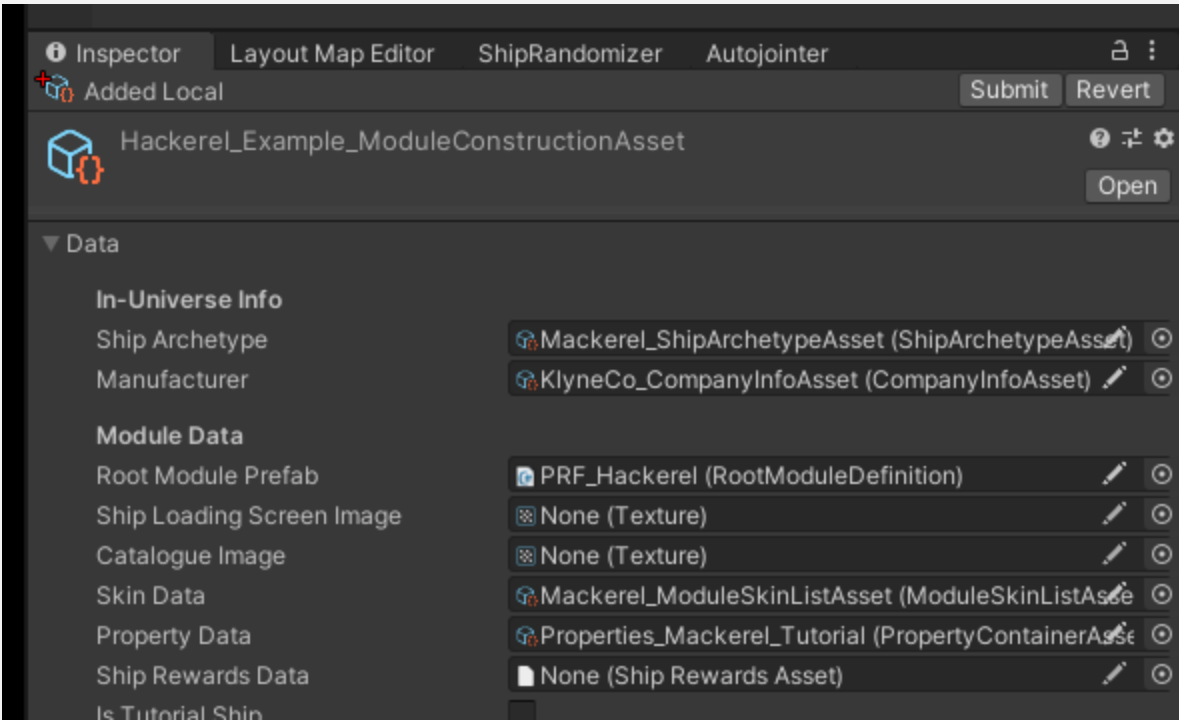
Process

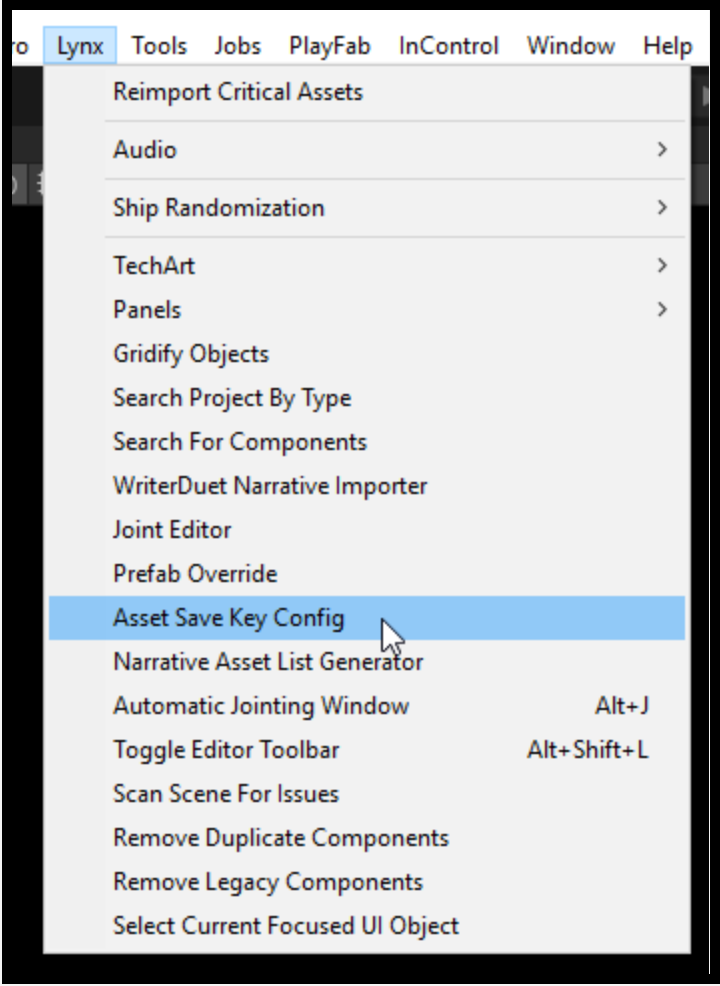
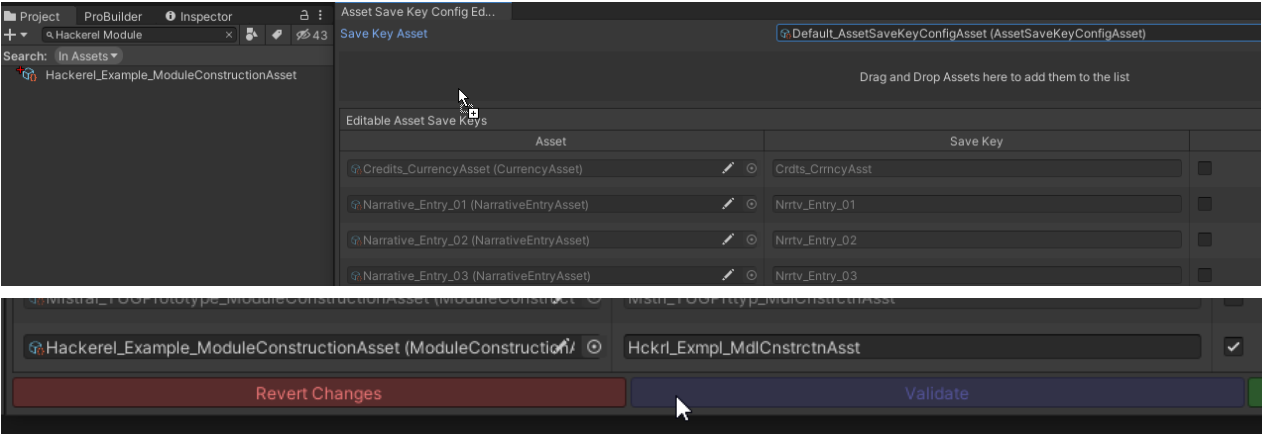
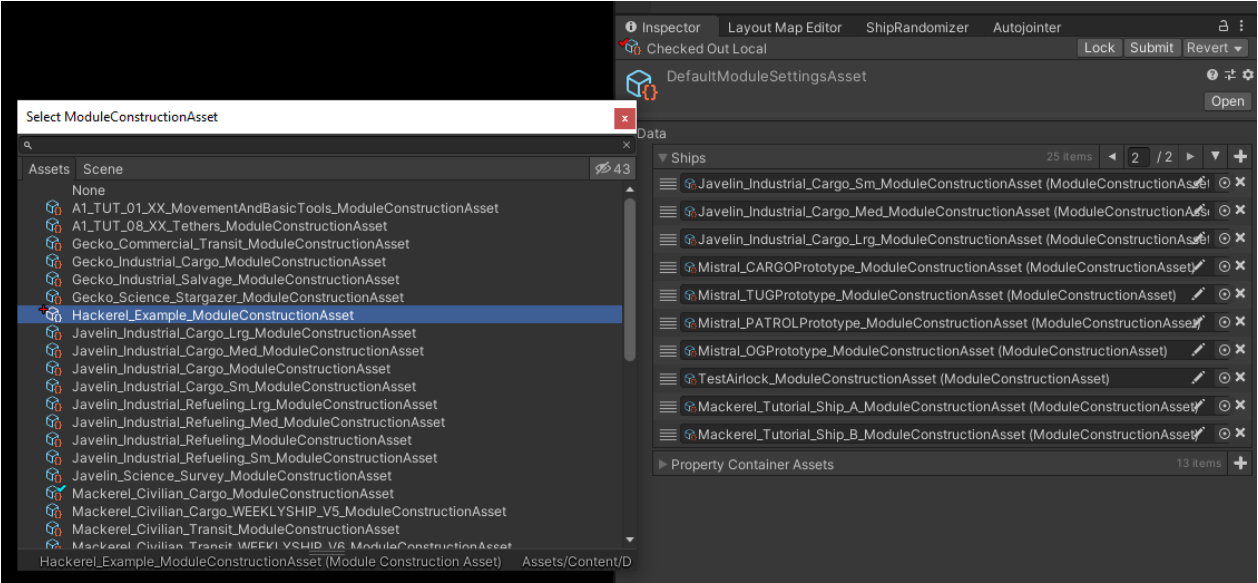
Step #	Involved Systems	Description
1	Basic Competency	Navigate the editor to a work scene or Map_Empty.
2	Basic Assembly	Assemble the most basic exterior shape and structure of a ship. For testing, this can be expedited by following steps 2a - 2c
2a	Basic Assembly	Place PRF_Mackerel in the scene and 0-out its transform. 
2b	Basic Assembly	Place PRF_Mackerel_Final and PRF_Mackerel_Back into the scene. 0-out their transforms and then vertex-snap them into position to the back of PRF_Mackerel . You may need to adjust snap settings via ProGrids to achieve this. 

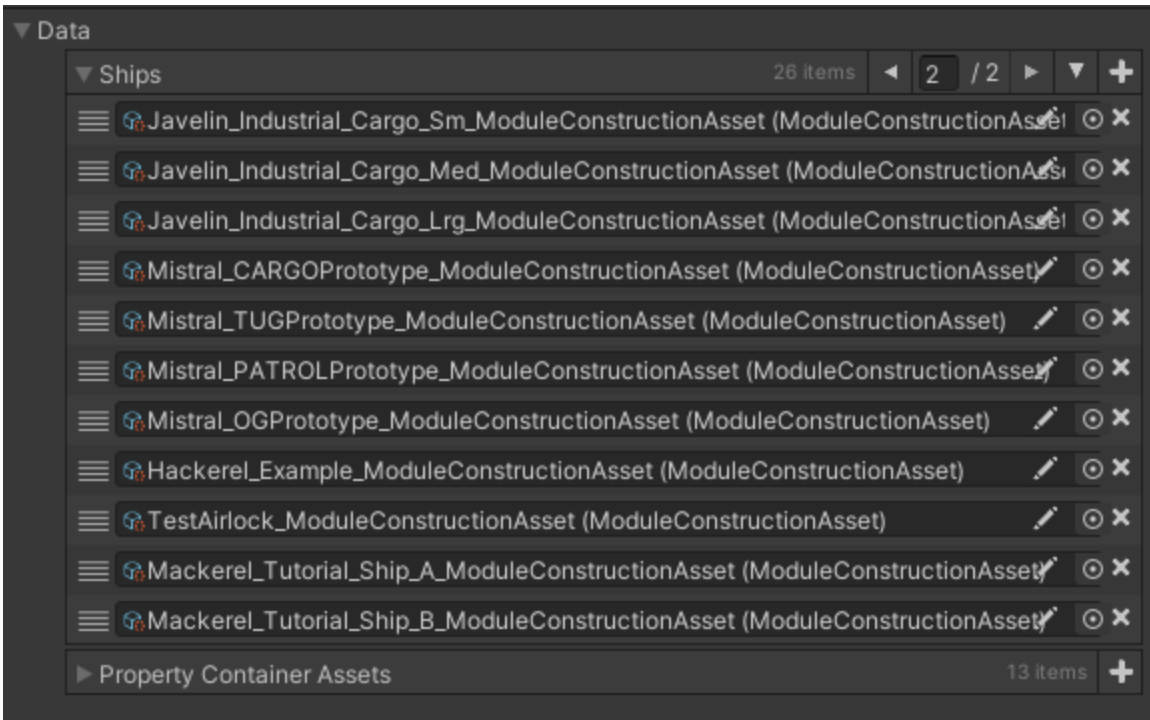
2c	Basic Assembly	Locate PRF_Panel_4xBx4 and place, and snap 6 instances alongside the exterior walls of the pseudo-Mackerel being created. Including panels spawned in via hardpoints, this is the completed construction of our new ship, the "<i>Hackere!</i>". 
3	Basic Assembly	Create a new empty gameobject in the scene. 0-out it's location. 
3a	Basic Assembly	Rename the new gameobject to follow the convention for high-level ship structures. For prototyping and testing, this name can be very general and changed to be more accurate as time goes on.
3b	Basic Assembly	Move all relevant objects under the new empty game object you named. 
3c	Basic Assembly	Add a module definition component to this root object. Note: for rapid testing replace this Module Definition with a Root Module component.

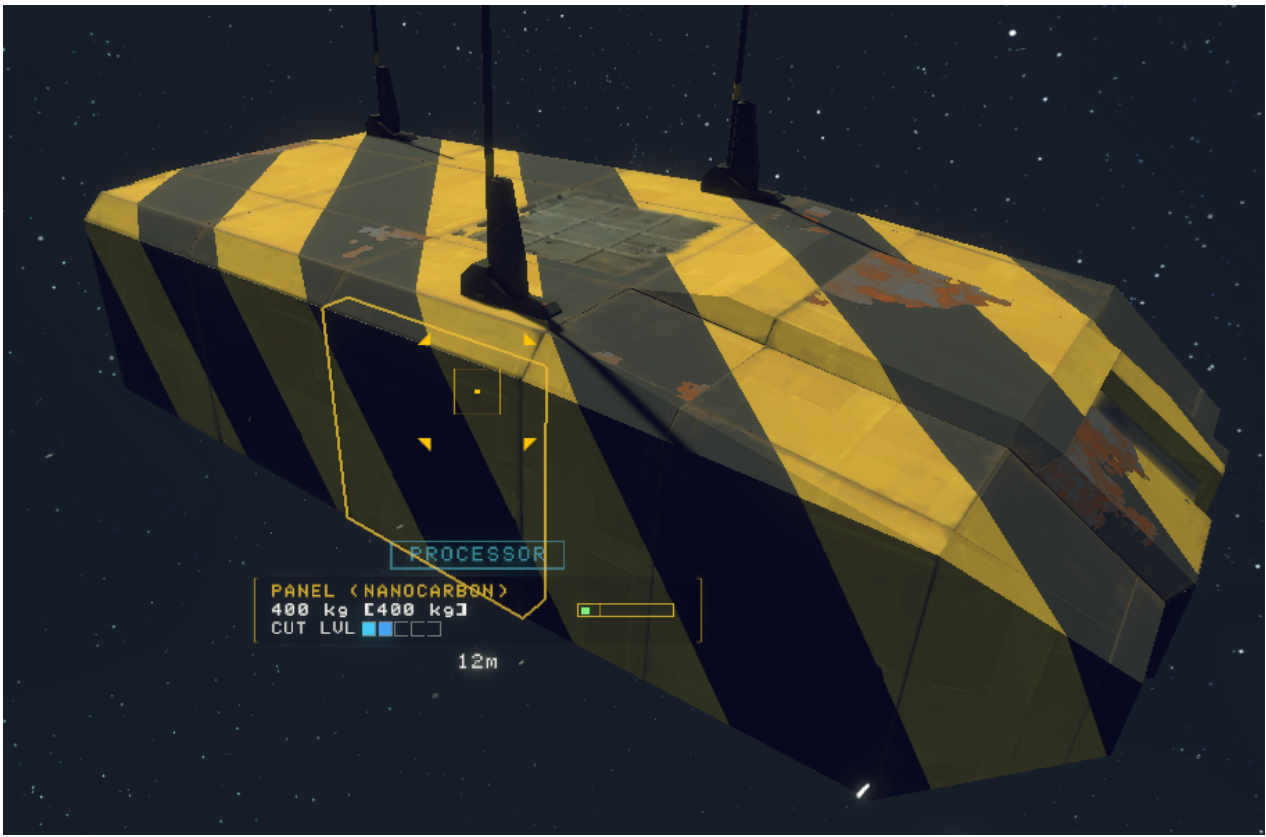
		Inspector Layout Map Editor ShipRandomizer Aut▶⋮ ✓ PRF_Hackerel Static▼ Tag Untagged Layer Default ▼ Transform ? ⌵ ⋮ 0 Local Position X 0 Y 0 Z 0 0 Local Rotation X 0 Y 0 Z 0 1 Local Scale X 1 Y 1 Z 1 <----- Uniform Scale -----> 1 Snap Options Show Snap Direction ☐ Edit Snap Prefs Snap To Nearest Surface Add Component 🔍 Module × Search 📄 Module Definition 🔌 Module Preview Component 🔌 Module Spawn Point 🔌 InControl Input Module 🔌 Orca Standalone Input Module 🔌 Root Module Definition 🔌 Sell Module Action 🔌 Spawn Module 👤 Standalone Input Module 👤 Touch Input Module 🔌 Triggerable Module Spawner New script >
4	Basic Assembly	Locate and create a new folder within the Nodes design folder for your ship to go in. This can be found at: Lynx\main\Lynx\Unity\Assets\Content\Prefabs\Ship Kit ▼ Assets ▶ Code ▼ Content ▶ Animation ▶ Art ▶ Audio ▶ Data ▶ Fonts ▶ Light Cookies ▶ Materials ▶ PhysicsMaterials ▼ Prefabs ▶ _Unsorted ▶ Audio ▶ Campaign ▶ Core Systems ▶ Narrative ▶ Objects ▶ Player ▶ QA Testing ▶ Salvage Bay ▼ Ship Kit ▶ Cut Points ▶ Decals ▶ Doors ▼ Nodes ▶ Gecko ▶ Javelin ▶ Mackerel ▶ Mistral ▶ Salamander ▶ PRF_Ship_Core
4a	Basic Assembly	Create and name a new folder for your test ship nodes. Then drag the WIP object into that folder to create a new prefab. The prefab in the scene should now be identified with a blue icon in the Hierarchy.

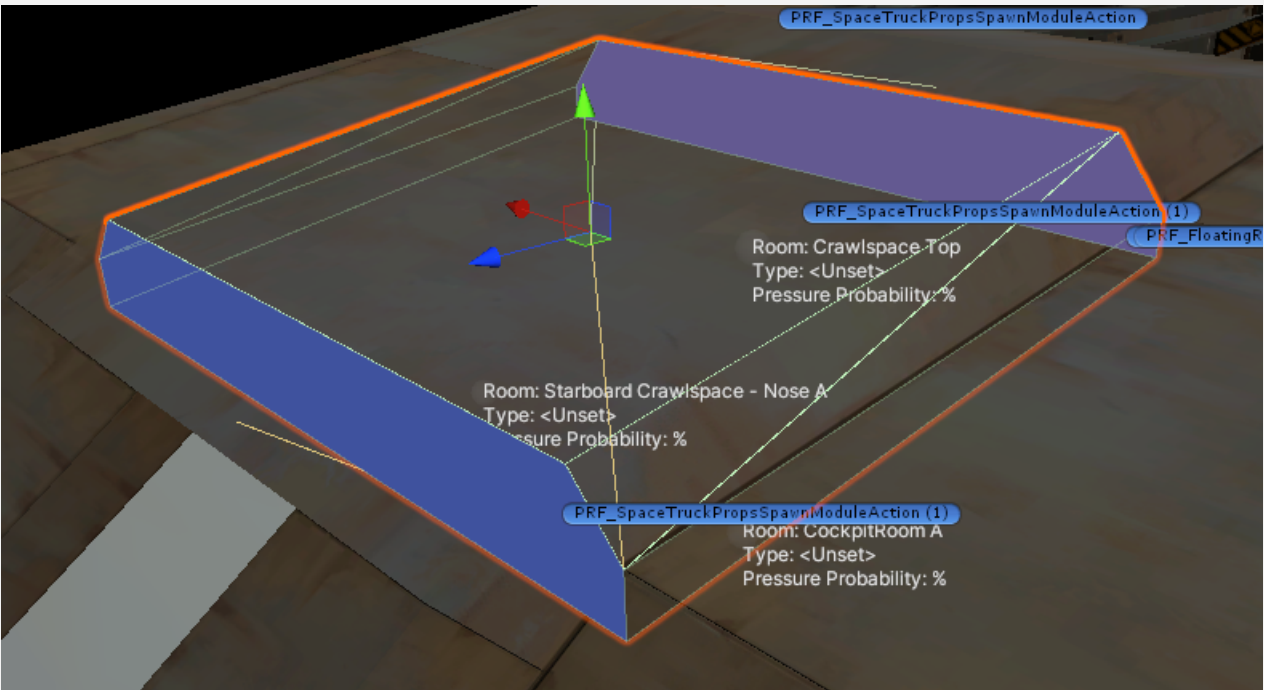
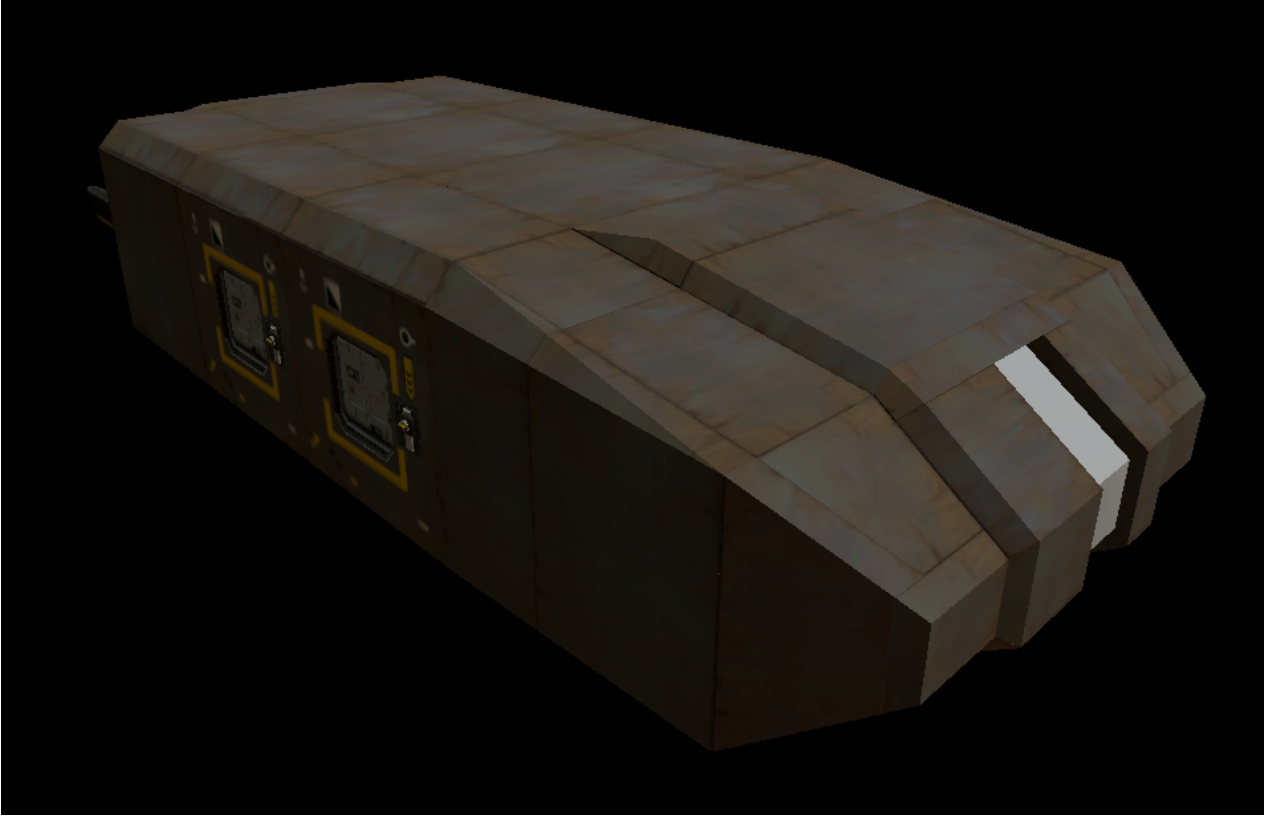
		
5	Basic Assembly	Open the new prefab and proceed to organize its pieces into logical nodes with names that denote their areas. These could include node / front / middle / back / tail / wings / etc... For testing, this should be done by following steps 5a - 5b
5a	Basic Assembly	The back 2 panels of the 6 placed earlier should be sorted under PRF_Mackerel_FinalSegment and the front 4 should be placed under PRF_Mackerel. 
5b	Basic Assembly	Unpack all Mackerel prefabs and rename accordingly. 
6	Basic Assembly	Turn all new segments into separate prefabs by dragging their names into the Project window. 
6a	Basic Assembly	Add module definition components to the root level of all new prefabs (unless they already have them).
7	Module Construction Assets	Locate the Module Construction Assets folder. This should be located in: Lynx\main\Lynx\Unity\Assets\Content\Data\ModuleConstructionAsset. Copy one of the existing module construction assets and rename appropriately.

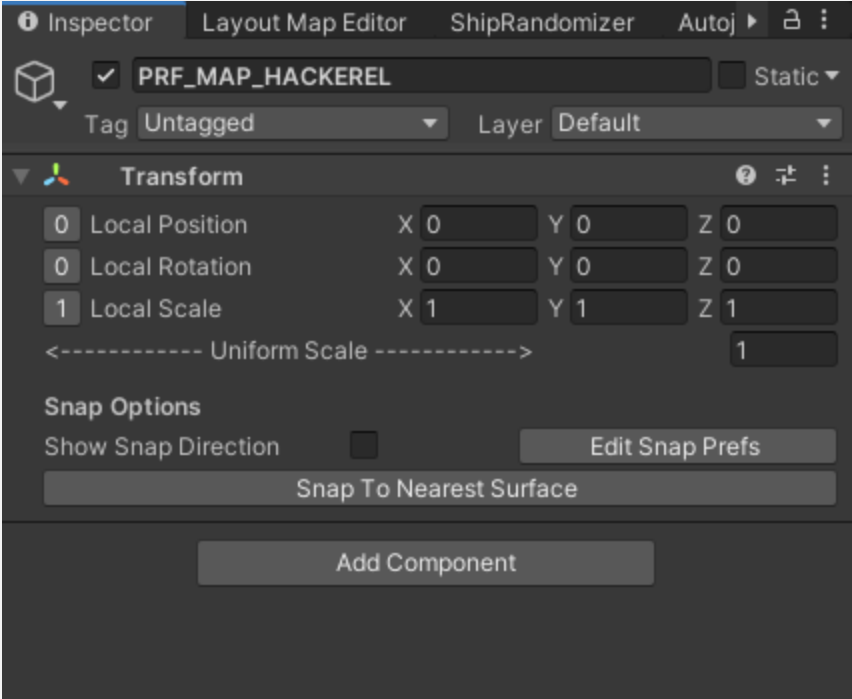
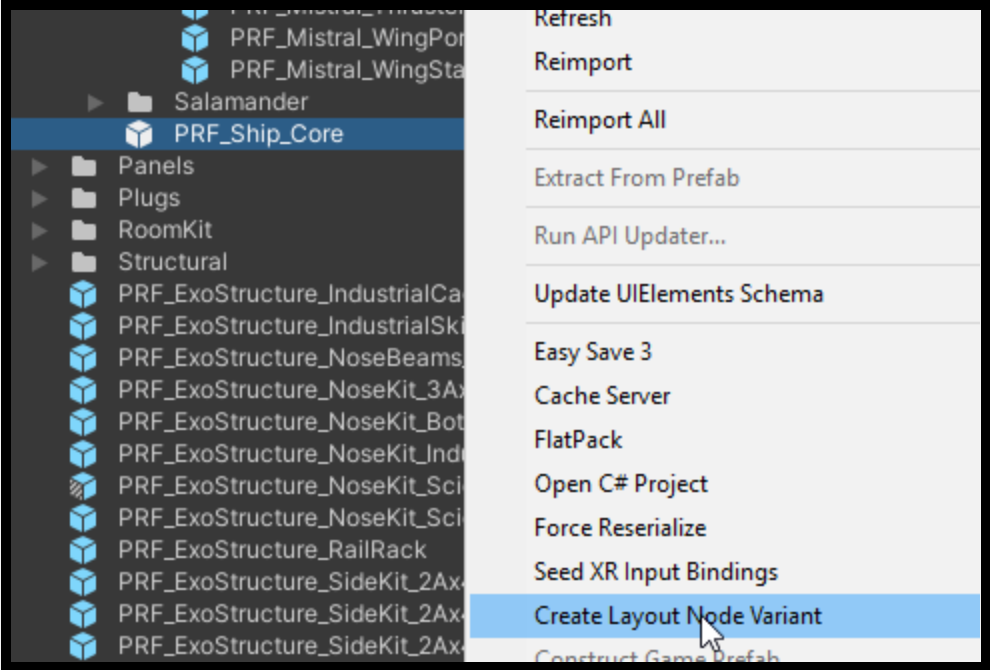
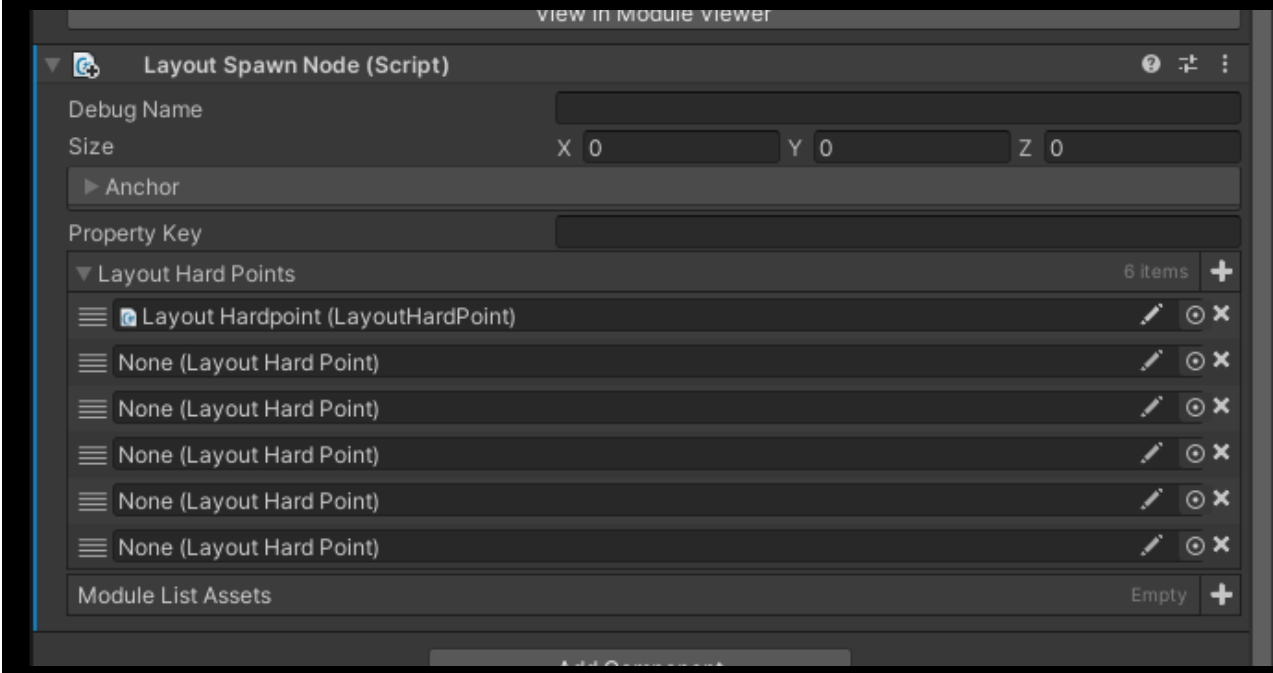
		
8	Module Construction Assets	Within the module construction asset, replace the contents in the Root Module field with your new high level prefab. If this cannot be done, and the field will not accept the new asset, make sure it has a Root Module on it and not a Module Construction Asset. See: Step 3c 
9	Module Construction Assets	This is an optimal time to create new property data for the Module Construction Asset, but it can occur at any point afterwards. This is also an optimal time to remove all restrictions from the duplicated asset.
9a	Module Construction Assets Save Keys	To avoid spawn errors, locate the Asset Save Key Config .

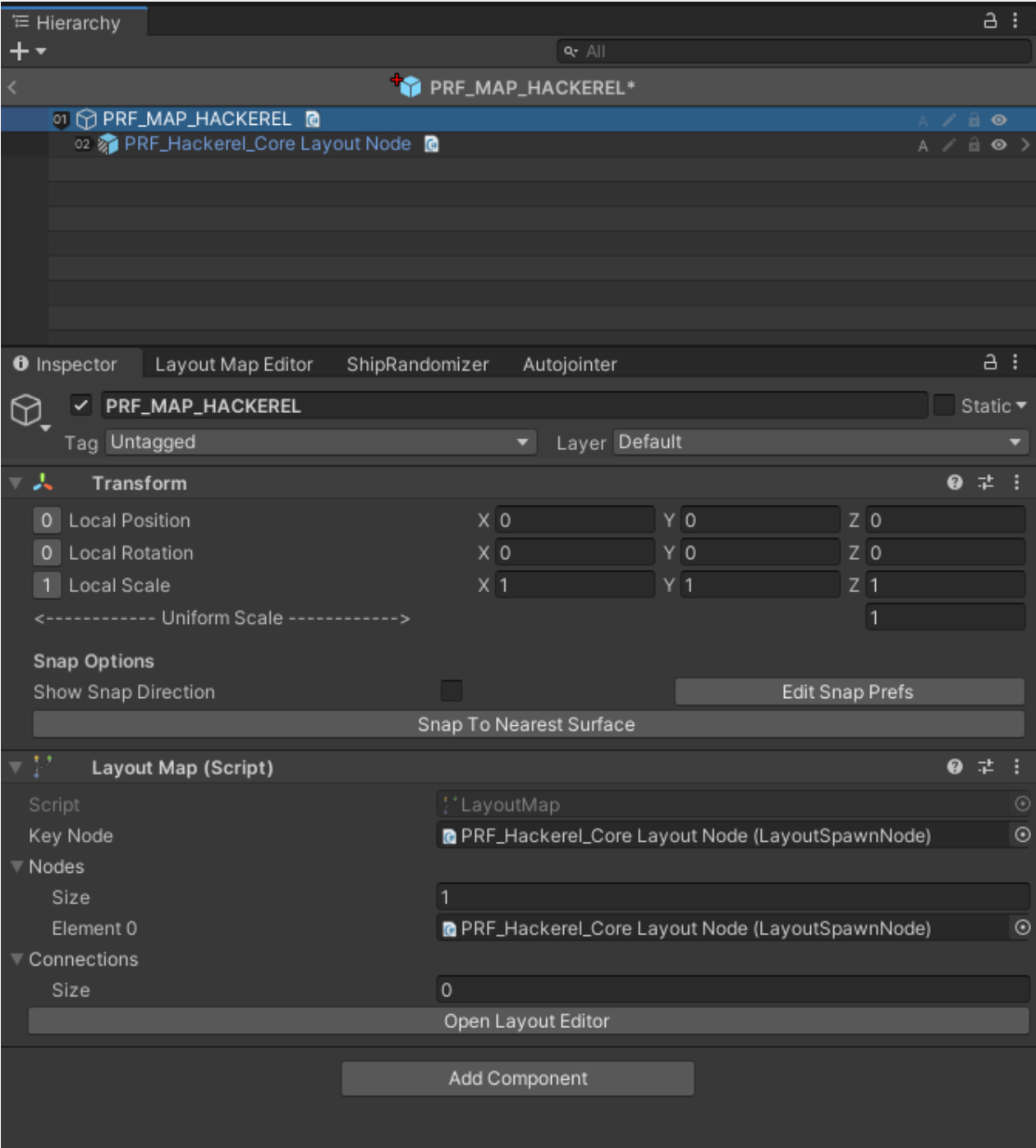
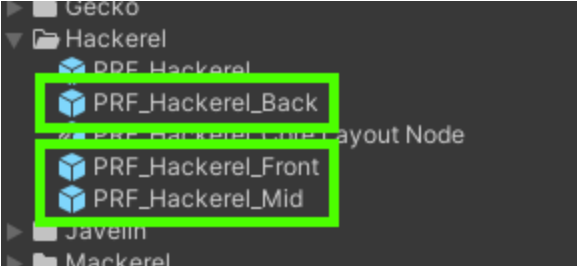
		
9b	Module Construction Assets Save Keys	With the Asset Save Key Config open, add the new construction asset to the save key. Click validate and then save. This will correct several different potential bugs. 
10	Ship Spawning	Locate DefaultModuleSettingsAsset. This can be found at: Lynx\main\Lynx\Unity\Assets\Content\Data\ModuleSettingsAsset .
10a	Ship Spawning	Add the new Construction Asset to the DefaultModuleSettingsAsset. 

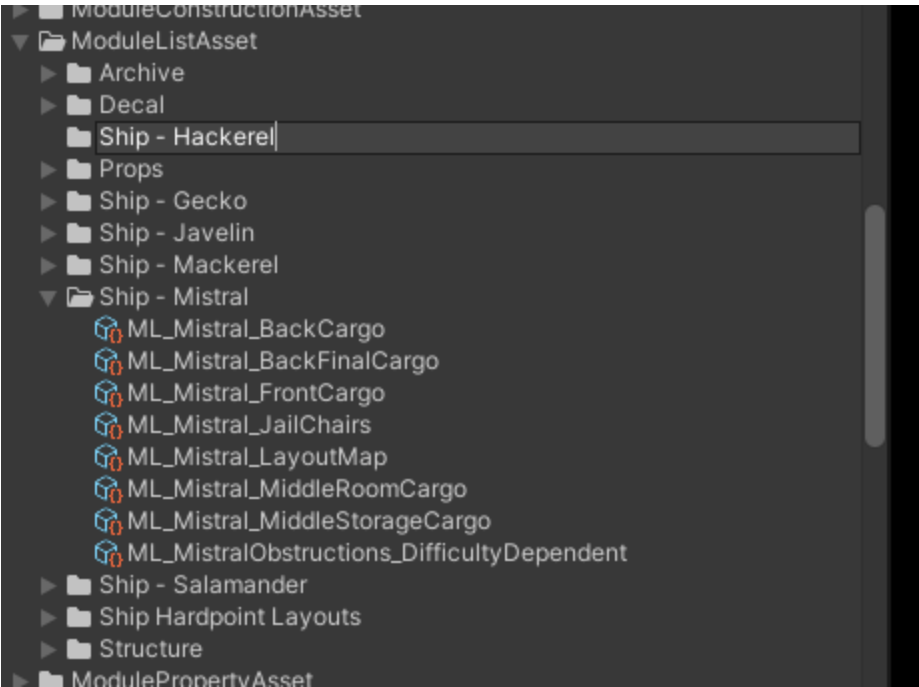
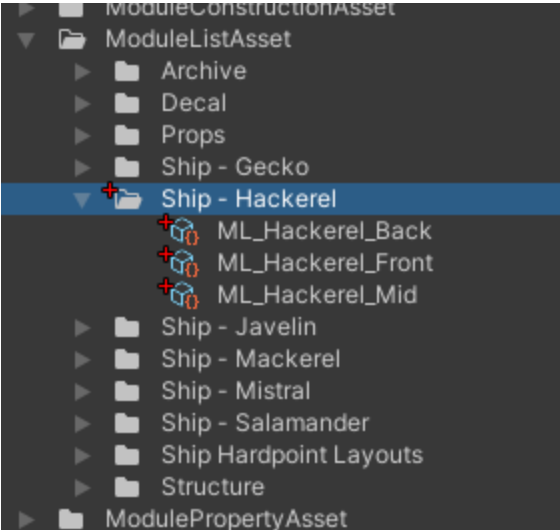
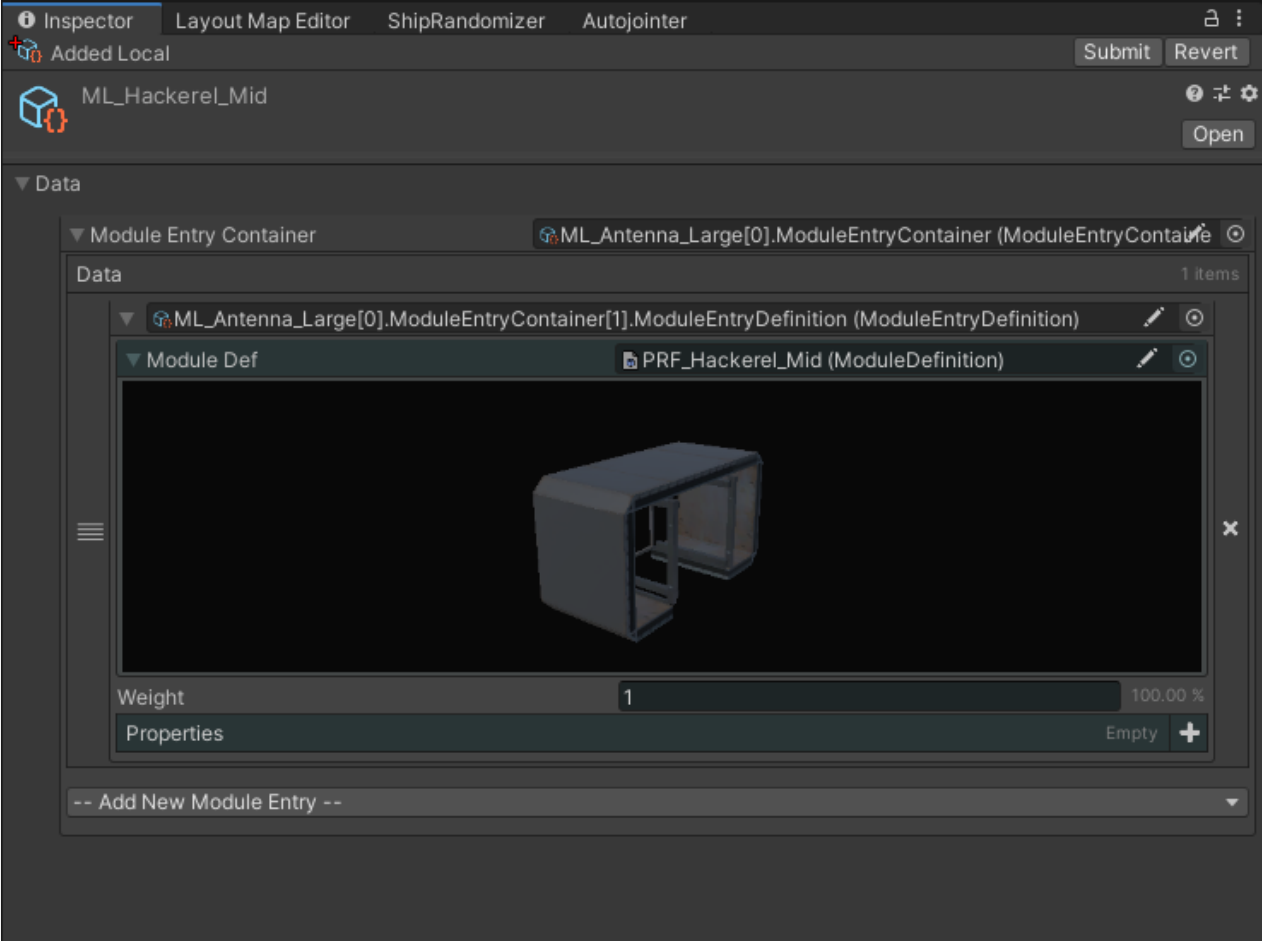
		
11	Ship Spawning	From this point on, the ship should be able to be spawned from the debug menu at any time. 
11a	Ship Spawning	Take this time to iterate on the ship's design, and engineer a basic jointing and salvaging solution.

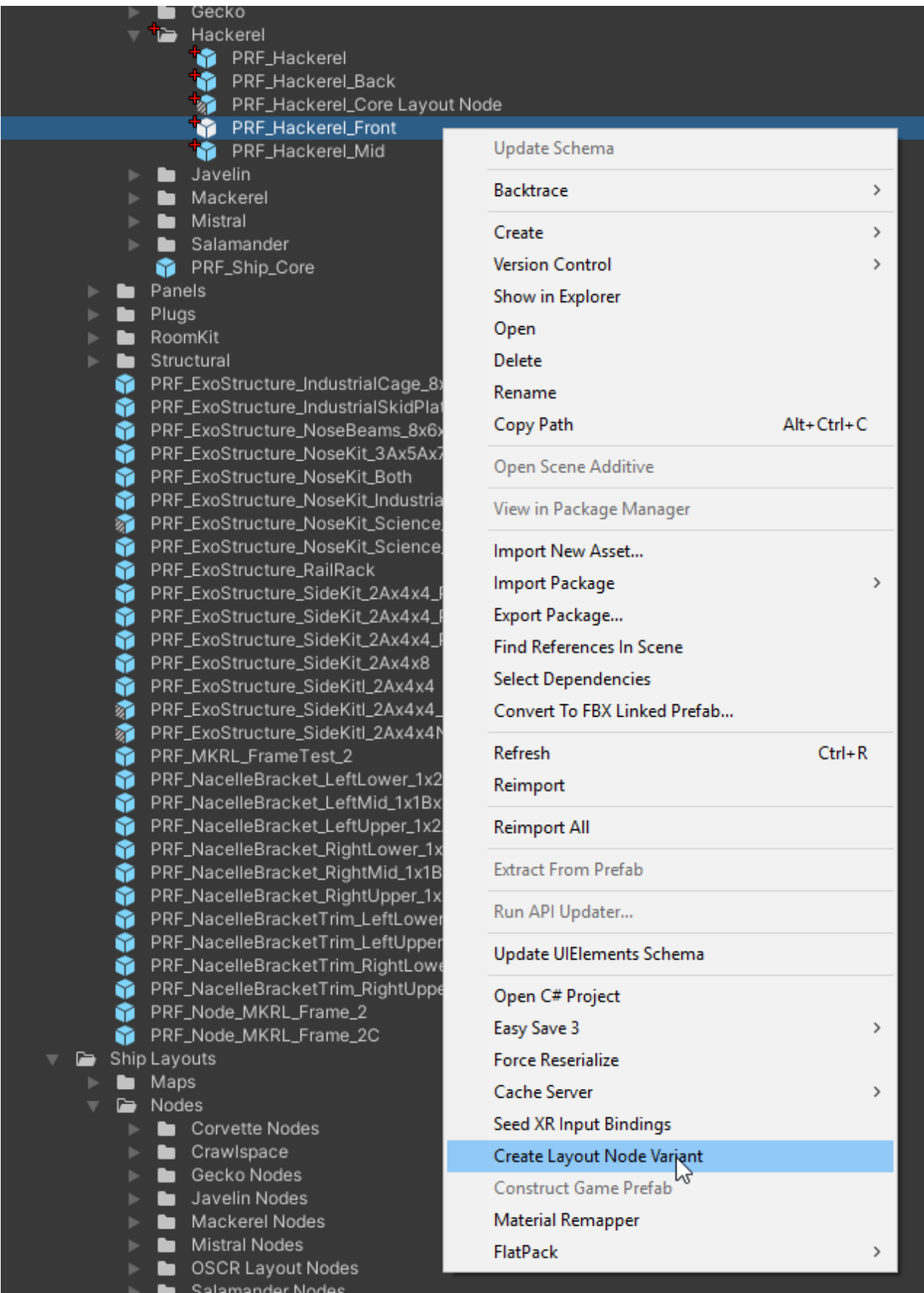
		
11b	Automatic Jointing Window	To test jointing in-editor, place the highest level prefab into the scene and 0-out it's position. Open the Automatic Jointing window and generate joints for the object. With Gizmos turned on, a mesh can be clicked on (not just a prefab) and it's joints can be visualized. Turning off Hardpoints in Gizmos may make this easier. LynxToolsJobsPlayFabInControlWindowHelp Reimport Critical Assets Audio> Ship Randomization> TechArt> Panels> Gridify Objects Search Project By Type Search For Components WriterDuet Narrative Importer Joint Editor Prefab Override Asset Save Key Config Narrative Asset List Generator Automatic Jointing WindowAlt+J Toggle Editor ToolbarAlt+Shift+I spectorLayout Map EditorShipRandomizerAutojoiner Joint Detection Co-planar Dot Threshold 0.05 Co-directional Dot Threshold 0.9 Bounds Search Bias 1.1 Generate Joints Candidate Joints: 258 Clear All Joint Data Show Candidate Joints ☐ Debug Selected Parts ☐ ! Operation Complete: 0.09399414s

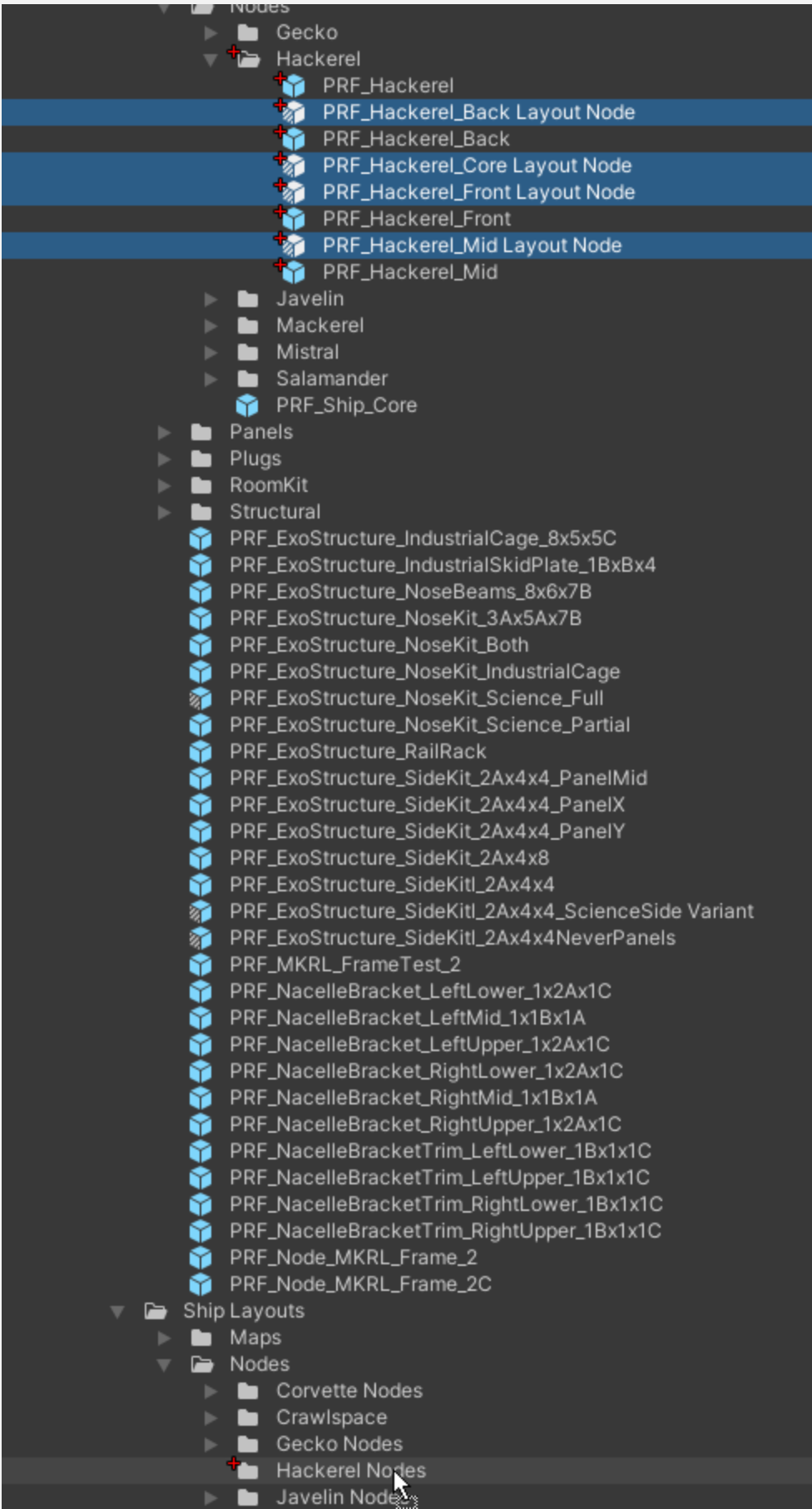
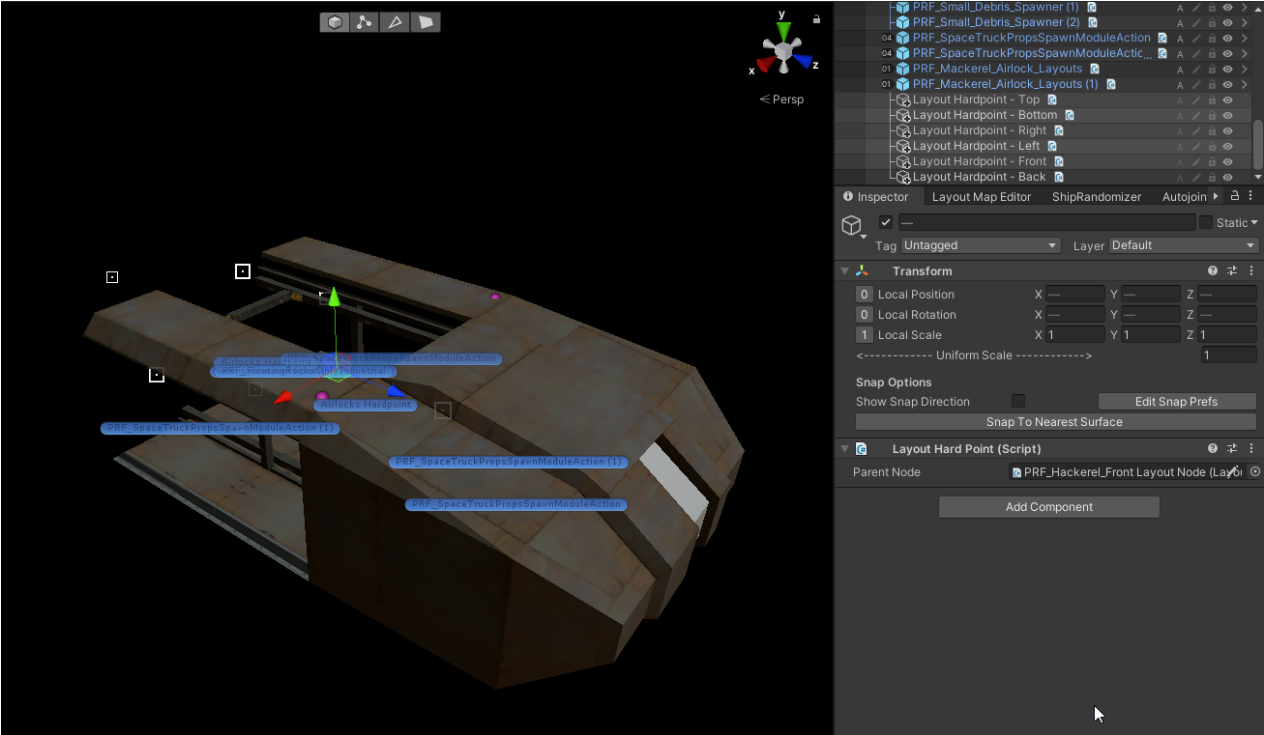
		 PRF_SpaceTruckPropsSpawnModuleAction PRF_SpaceTruckPropsSpawnModuleAction (1) PRF_FloatingR Room: Crawlspace Top Type: <Unset> Pressure Probability: % Room: Starboard Crawlspace - Nose A Type: <Unset> Pressure Probability: % PRF_SpaceTruckPropsSpawnModuleAction (1) Room: CockpitRoom A Type: <Unset> Pressure Probability: %
11c	Basic Assembly	For the <i>Hackerel</i>, it will need 3 instances of PRF_Mackerel_Interior_Layout placed to correspond to the 6 panels added earlier. One or more of these can be replaced by an instance of PRF_Mackerel_Airlock_Layouts though in this case the ship will need to have Mackerel-compatible Roles set in its Construction Asset properties (eg: Science theme, Light Cargo Role, Station Hopper Role).   Perfect.
12	Layout Map	Clear out all related prefabs in the scene, and spawn a new empty gameobject. 0-out it's location. This new object does not need a module definition to its root. This newest object will be the main

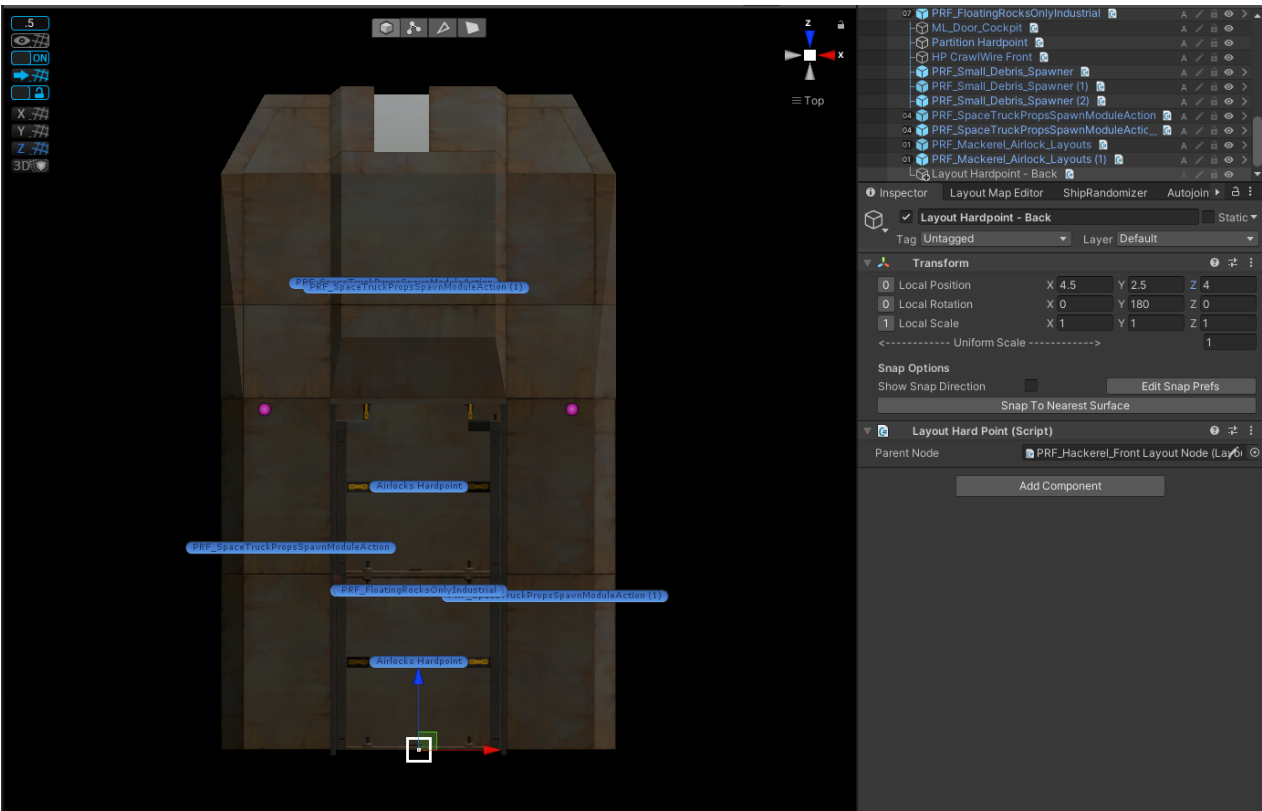
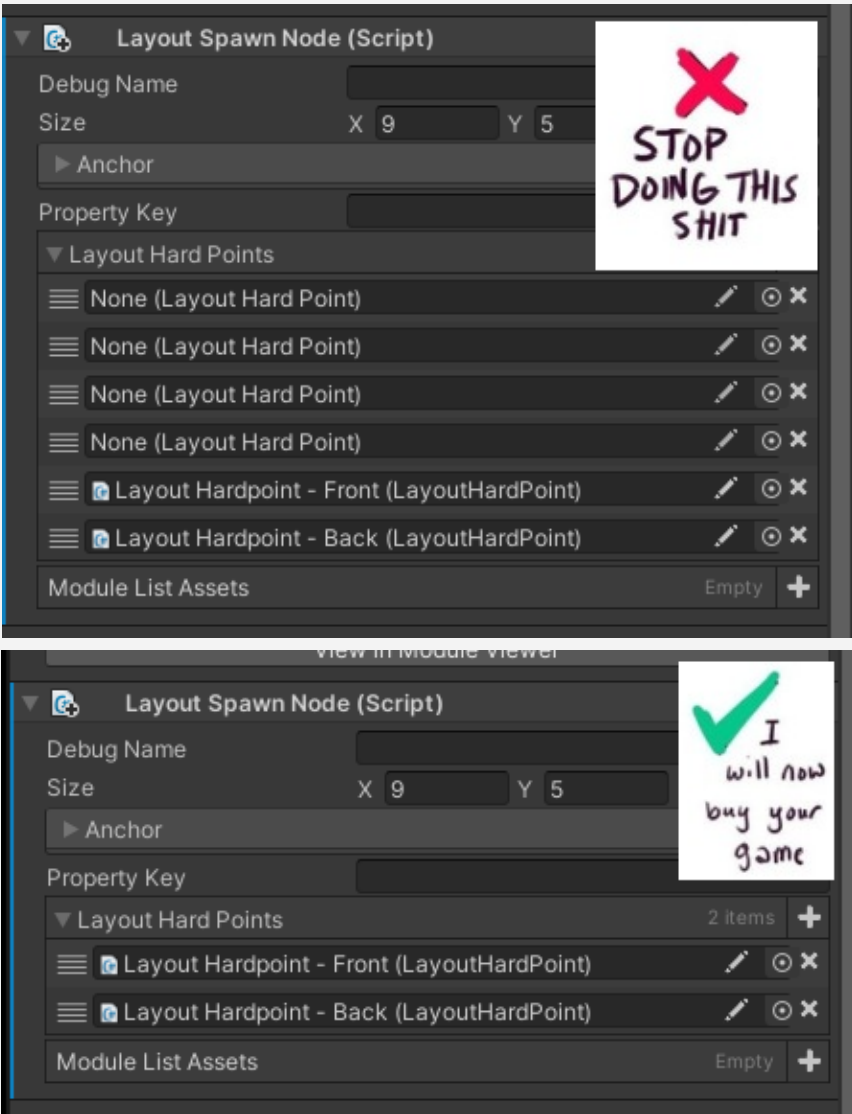
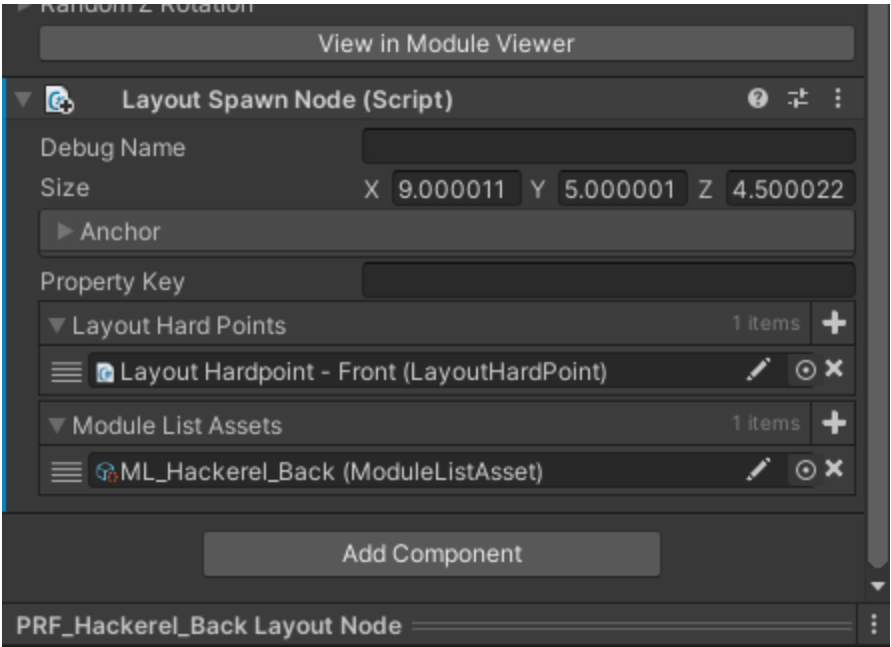
		Layout Map. Rename it accordingly, maps all have an all-caps naming convention. Don't ask why. 
12a	Layout Map	Add the Layout Map component to this root object and make it into a prefab. Layout maps can be located at: Lynx\main\Lynx\Unity\Assets\Content\Prefabs\Ship Layouts\Maps .
12b	Layout Map	Open the new Layout Map prefab. Locate PRF_Ship_Core within the project folder. Right click it and create a new Layout Node Variant. Rename this (Hackerel will be called PRF_Hackerel_Core Layout Node), and move this to the appropriate folder and open it up. 
13	Layout Map	Within the new Core layout map. Find and delete all but one layout map hardpoint. Rename the sole survivor, and 0-out it's position. You may wish to give it a rotation of 0,180,0 depending on your construction. Navigate to the root of the layout node and find the Layout Spawn Node script. It is very important that every time Layout Hard points are deleted, the Layout Hard Points drop-down be opened up and all empty fields be deleted. Without this step, the layout editor will commit ritual sudoku every time it's enabled and your editor will become unresponsive every time it thinks about the Layout Map. You will have to unhook the layout map editor, or restart the editor. This layout node does <i>not</i> need a module list asset. 

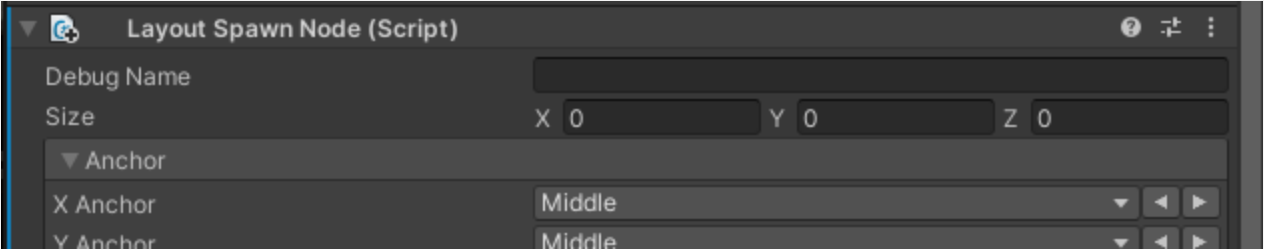
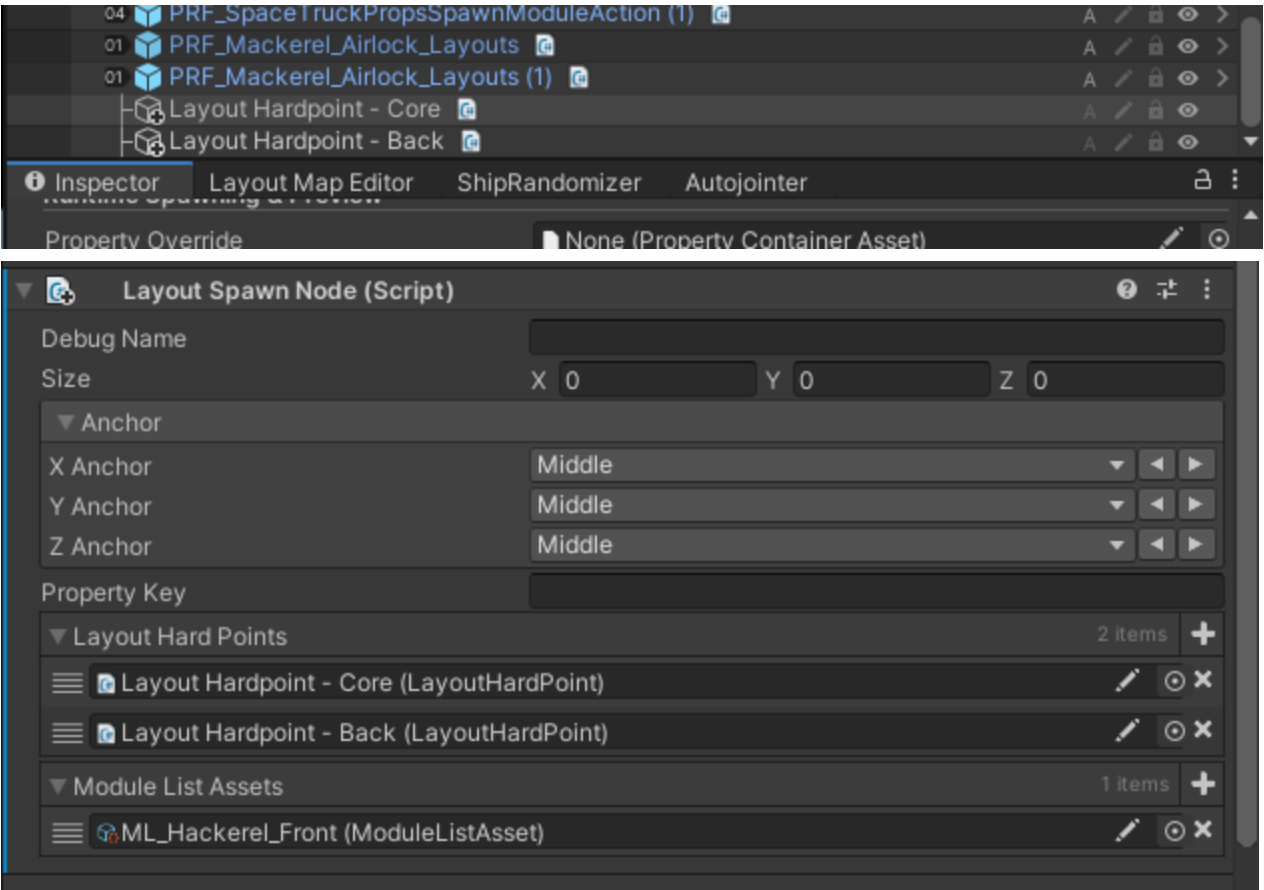
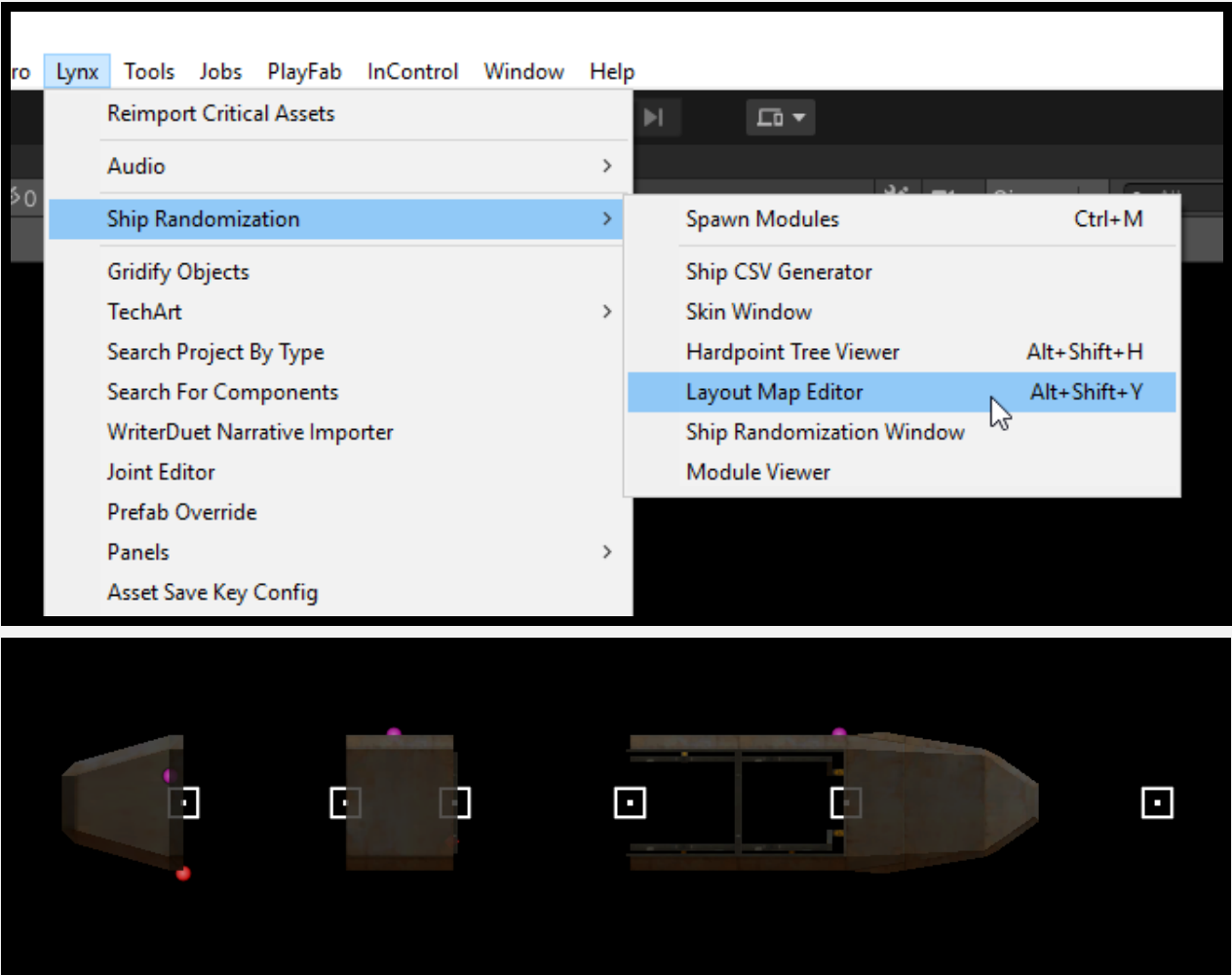
14	Layout Map	Back within the Map, place the new core layout node and 0-out its position. With the root selected, add the new Core Layout Node to the Layout Map script as both an element in its nodes, and as the Key Node. 
15	Layout Map, Module Lists	The next few steps will seem to be out of order, but it's only because there's a number of requirements involved in assembling a proper layout map. Make note of the number of different nodes present to assemble the ship in question. Do not include any currently existing layout nodes, or "assembly" nodes that just correlate many things together (like PRF_Hackerel). For each of them (3 for the Hackerel), construct a dedicated module list asset that corresponds to a node. Make sure this is located in its own module list folder; module lists can be found at: Lynx\main\Lynx\Unity\Assets\Content\Data\ModuleListAssets . 

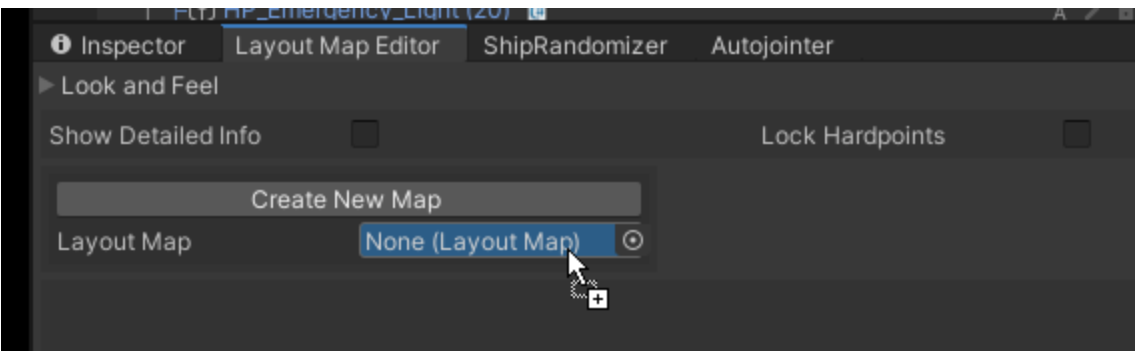
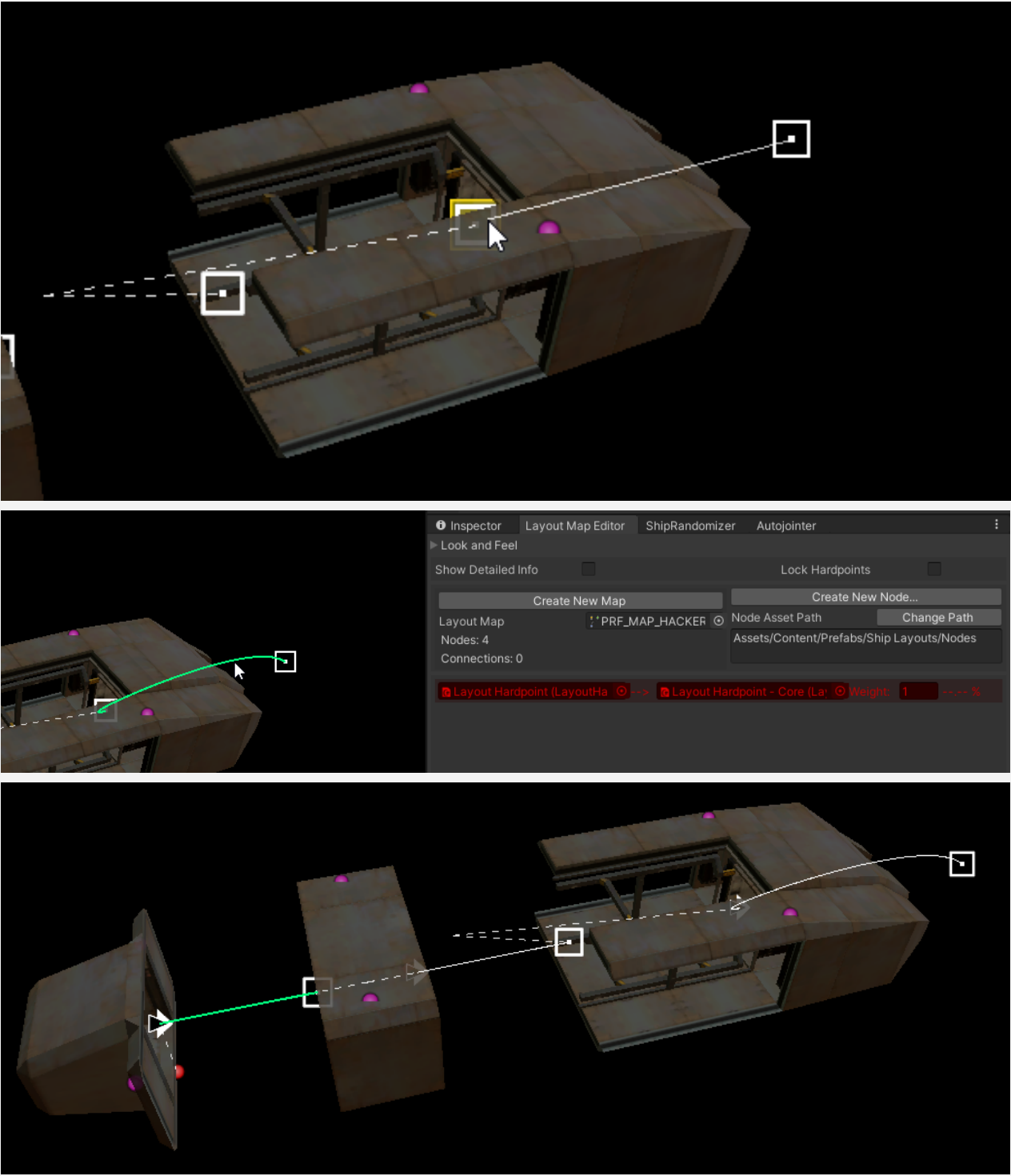
		 
15a	Layout Map, Module Lists	Populate each module list with their corresponding prefab. These module lists can be used to introduce variation in ships <i>inside</i> the layout map without having to introduce more nodes. However this technique is recommended only for special cases, as it is difficult to visualize and predict what a node may be doing, if the actual variation it causes is performed within a hardpoint. 
16	Layout Map, Layout Nodes	In a similar process earlier, create layout node variants of all the main nodes of your ship.

		
16a	Layout Map, Layout Nodes	This is an optimal time to move all relevant layout nodes to a new, dedicated folder. Layout nodes are stored in a nearby folder: Lynx\main\Lynx\Unity\Assets\Content\Prefabs\Ship Layouts\Nodes . Construct a new folder to contain all of your ship's nodes. (Image below.)

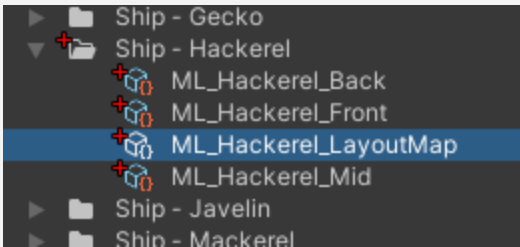
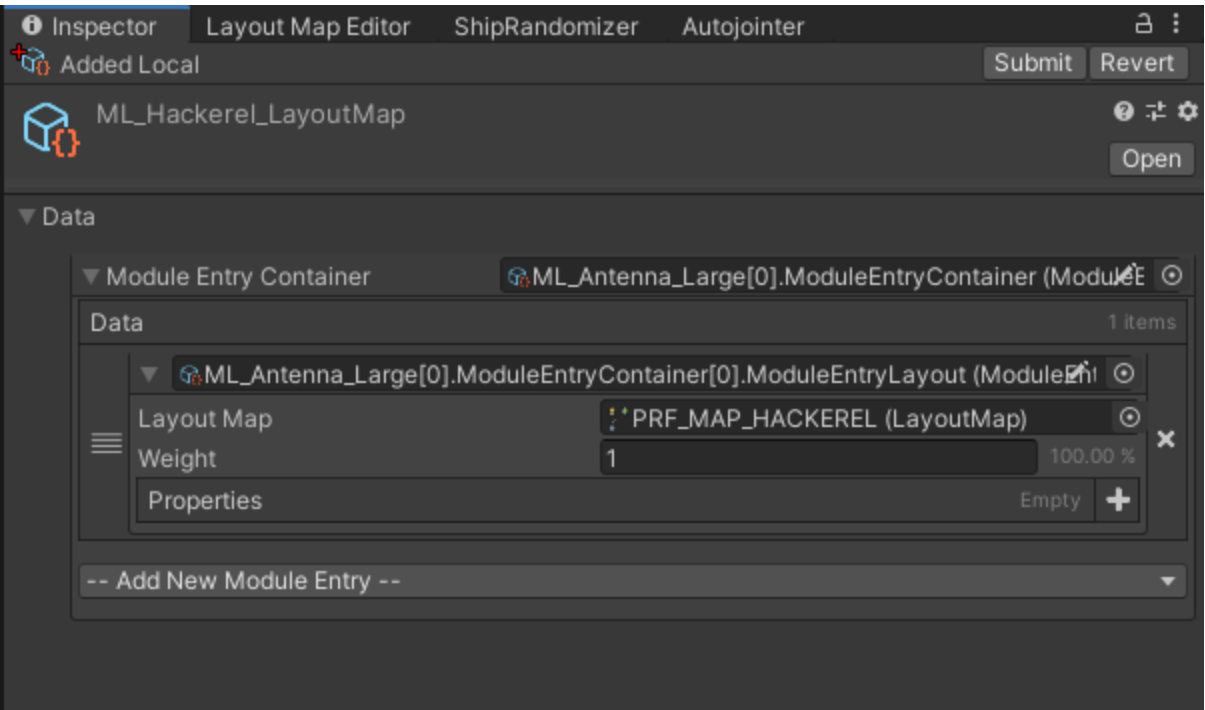
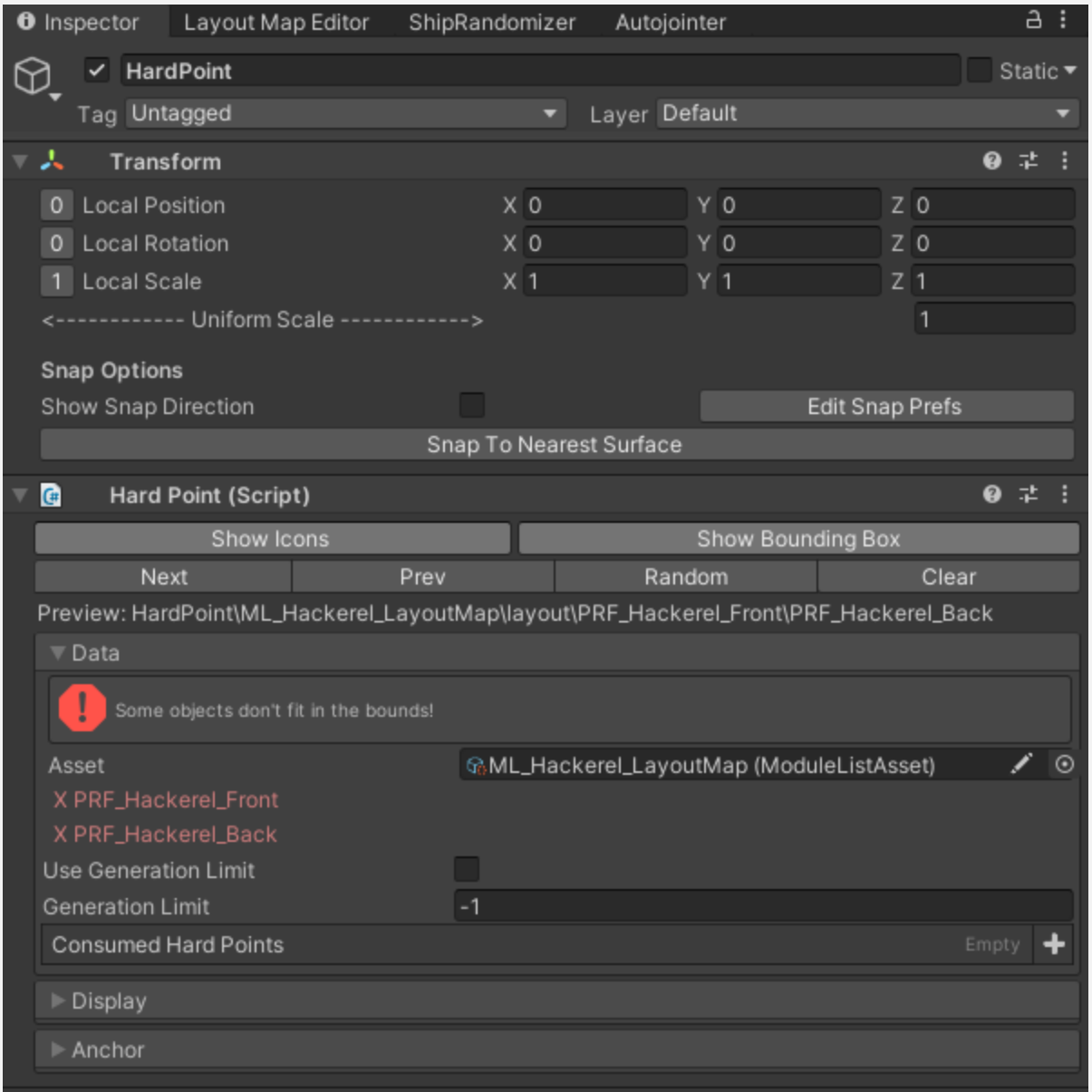
		
17	Layout Map, Layout Nodes	For each layout node, the layout hardpoints will need to be carefully curated. Most do not need more than 1 or 2, and may need their node positions adjusted. If layout hardpoints do not appear, try opening the Layout Map Editor with the prefab you're working with opened. This can be found from the topbar menu in Lynx Ship Randomization Layout Map Editor. 

		
17b	Layout Map, Layout Nodes	Just as before, clear all empty Layout Hard Points from the root's Layout Spawn Node script of all affected prefabs. 
17c	Layout Map, Layout Nodes, Hardpoints	(This is an important step that probably should be relegated to something called 17c.) For all layout prefabs, populate the Module List Asset array with its corresponding module list. 

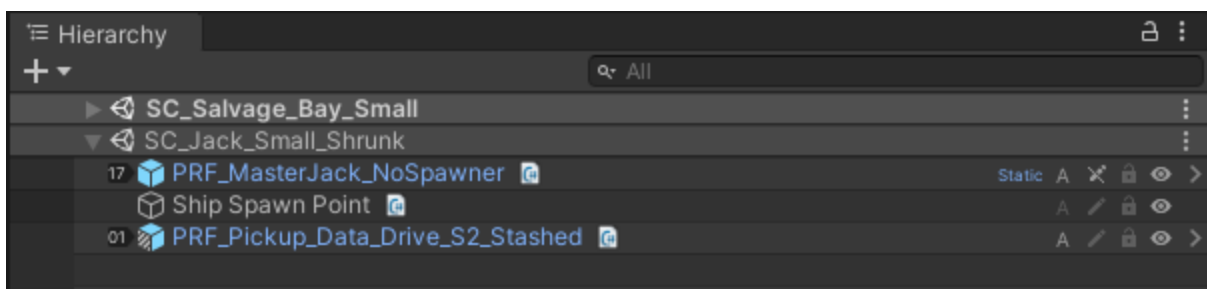
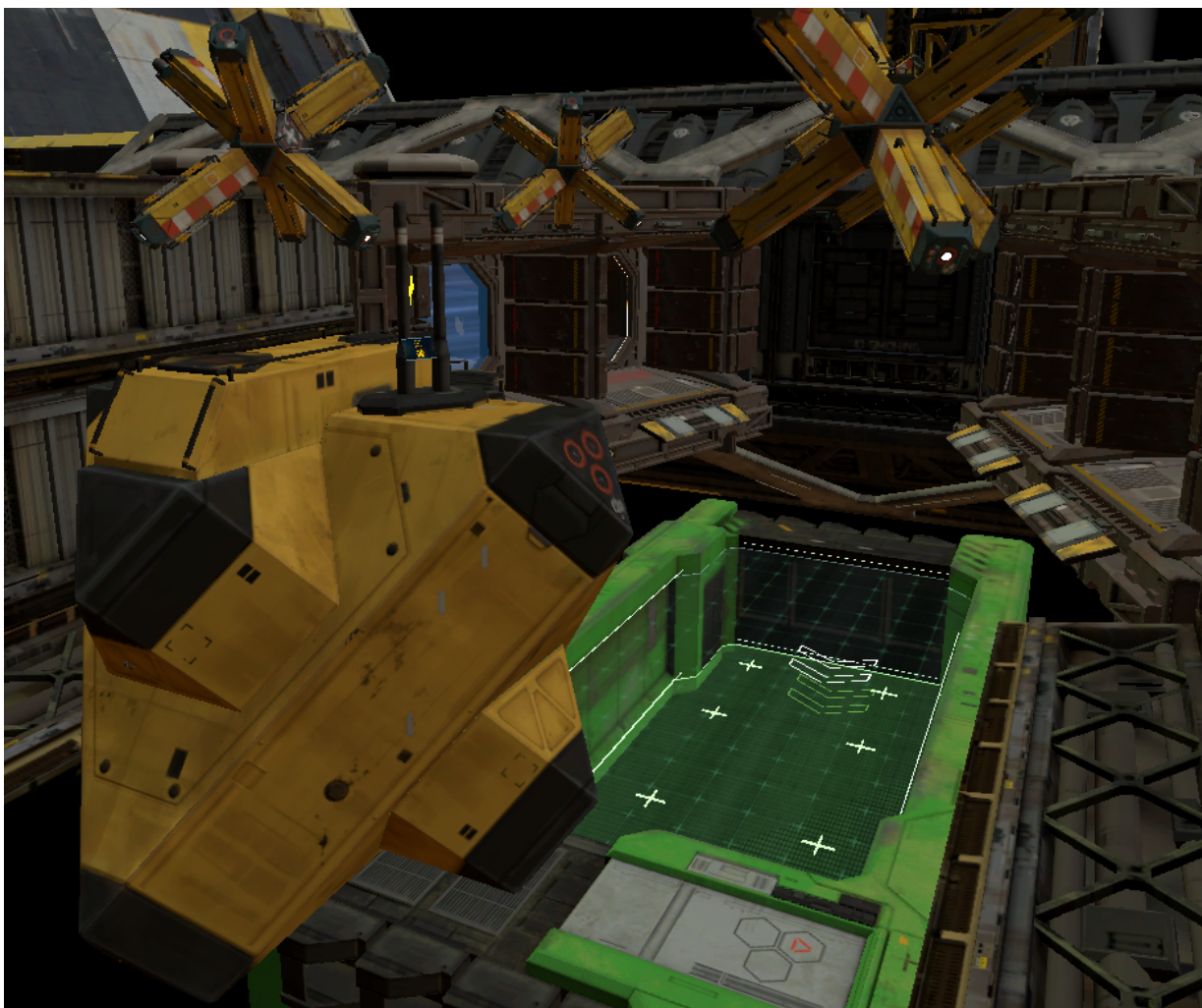
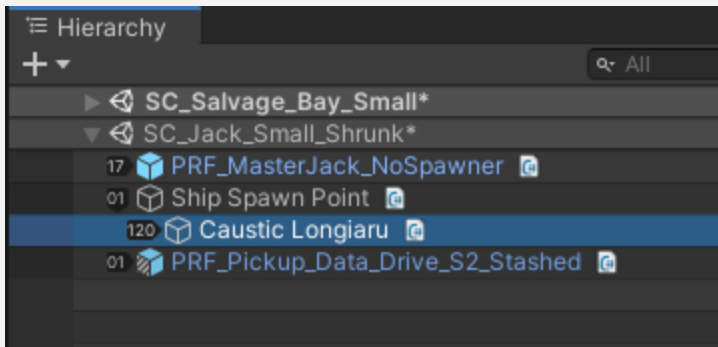
17c	Layout Map, Layout Nodes	Optional: 0-out the size volumes for all related Layout Nodes. These have no function and are often only visually obstructive. 
17d	Layout Map, Layout Nodes, Hardpoints	For the node which you will consider the starting node for the ship, create an extra Layout Hardpoint named something like the "Core" node. 0-out its location (or center it), and add it to the Layout Spawn script. 
18	Layout Map, Layout Nodes	Return to the Layout Map. Place all related layout nodes inside. For the full visualization, open up the Layout Map editor from the Lynx Ship Randomization Layout Map Editor topbar menu. 
18a		Assign the current Map to edit in the Layout Map Editor. This is done by dragging the object into the main Layout Map field.

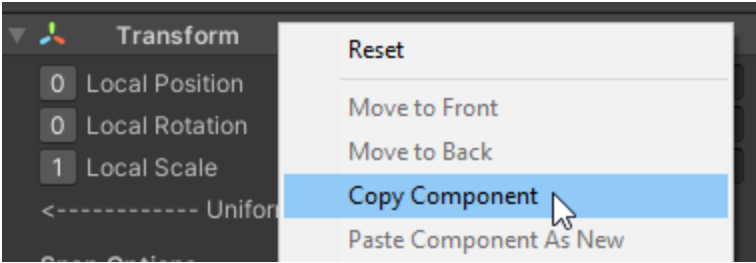
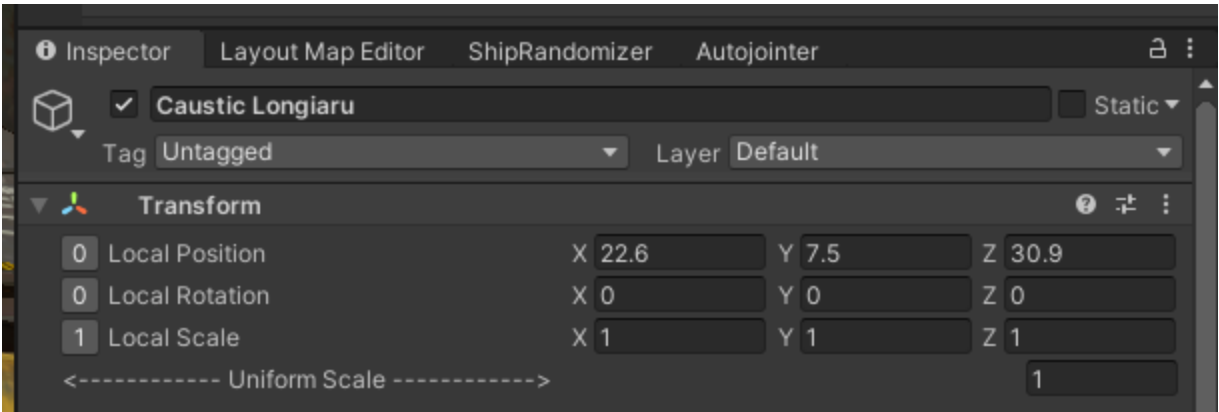
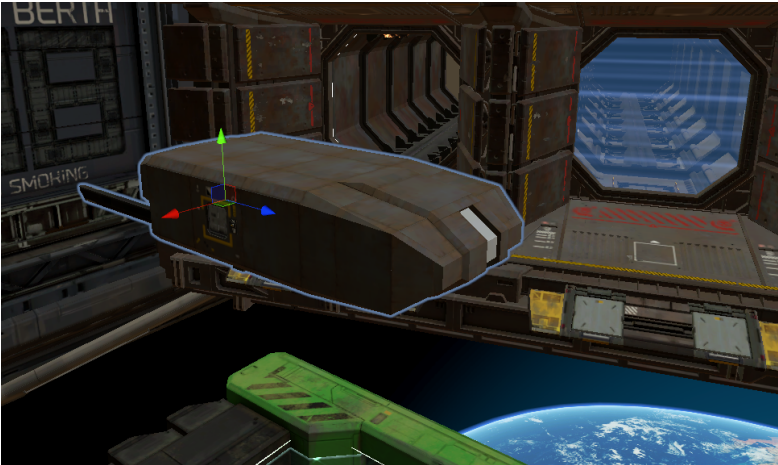
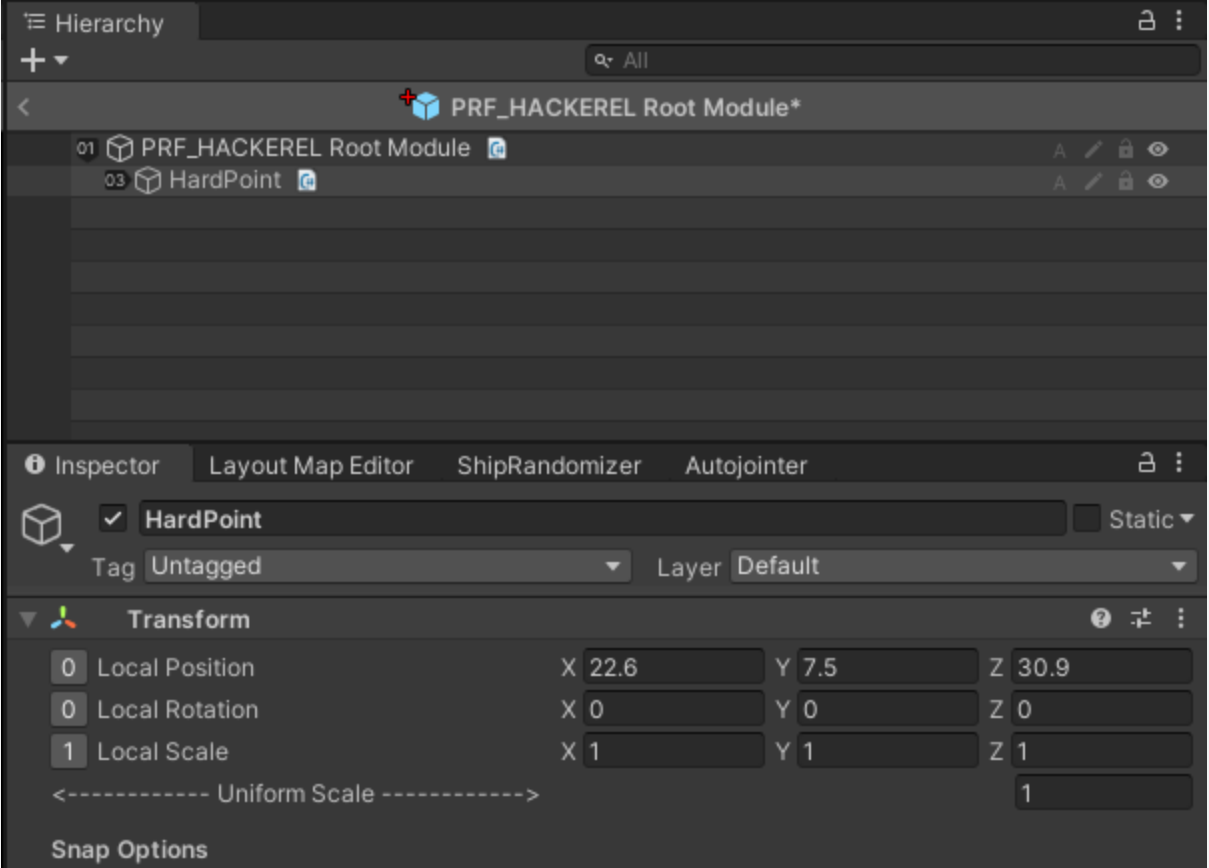
		
18b	Layout Map, Layout Nodes	Proceed to connect all layout hardpoints. This is done by holding CTRL, and click-dragging from one Layout Hardpoint to another, a wire should be drawn. If a connection needs to be broken, hover your mouse over a wire, hold CTRL and (I'm not kidding here) mash left and right click. This should break the connection. 

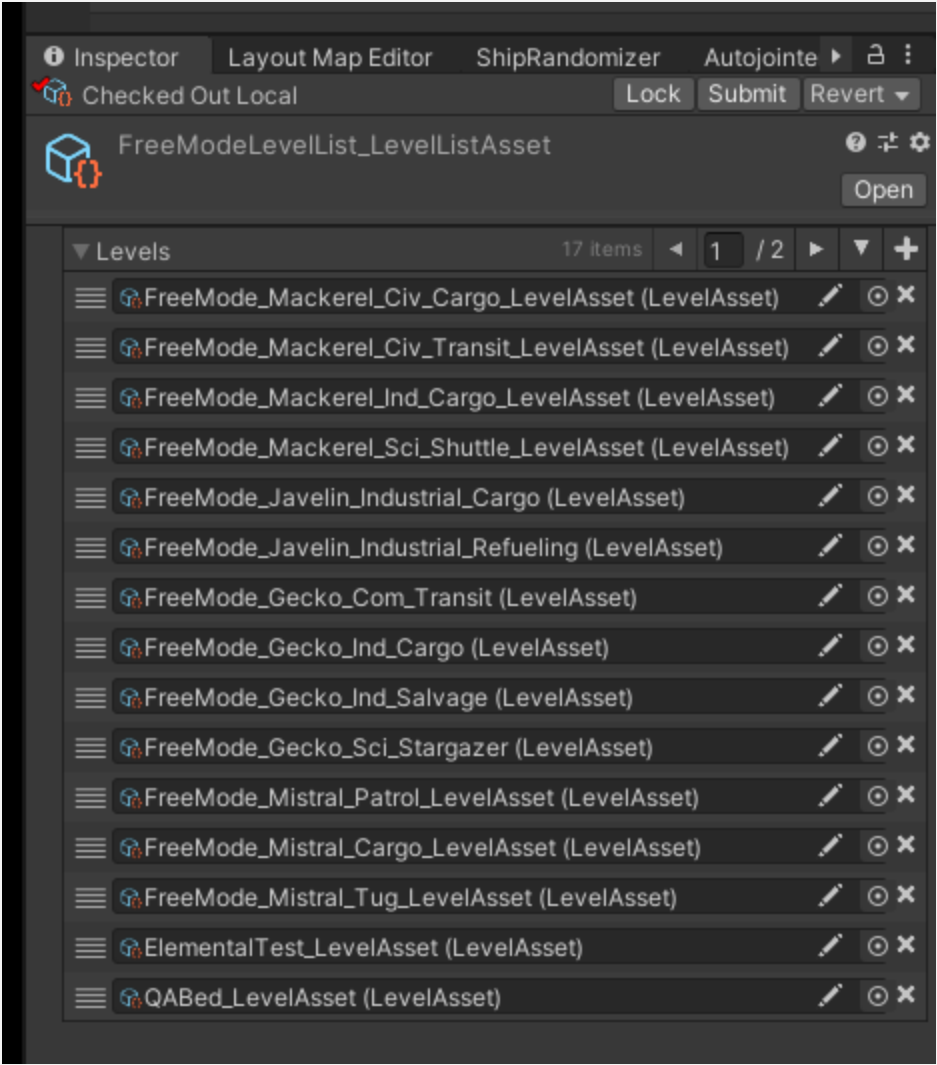
		Layout Map (Script) Script Key Node ▼ Nodes Size Element 0 Element 1 Element 2 Element 3 ▼ Connections Size ▼ Element 0 Weight Hard Point 1 Hard Point 2 ▼ Element 1 Weight Hard Point 1 Hard Point 2 ▼ Element 2 Weight Hard Point 1 Hard Point 2 Open Layout Editor LayoutMap PRF_Hackerel_Core Layout Node (LayoutSpawnNode) 4 PRF_Hackerel_Core Layout Node (LayoutSpawnNode) PRF_Hackerel_Front Layout Node (LayoutSpawnNode) PRF_Hackerel_Mid Layout Node (LayoutSpawnNode) PRF_Hackerel_Back Layout Node (LayoutSpawnNode) 3 Layout Hardpoint - Back (LayoutHardPoint) Layout Hardpoint - Front (LayoutHardPoint) 1 Layout Hardpoint - Back (LayoutHardPoint) Layout Hardpoint - Front (LayoutHardPoint) 1 Layout Hardpoint (LayoutHardPoint) Layout Hardpoint - Core (LayoutHardPoint)
19	Root Module	With the layout map complete, all that remains is implementing it. Locate the Root Module assets, found in the same folder as the Maps. Duplicate to create a new one, and name it accordingly. The "Hackerel" benefits from duplicating off the Mackerel. Ship Layouts ▼ Maps PRF_GECKO Root Module PRF_HACKEREL Root Module PRF_JAVELIN Root Module PRF_MACKEREL Root Module PRF_MAP_GECKO Mk 2 PRF_MAP_HACKEREL PRF_MAP_JAVELIN Mk2 PRF_MAP_MACKEREL PRF_MAP_MISTRAL PRF_MAP_SALAMANDER IndustrialCargo PRF_MAP_SALAMANDER SciResearch PRF_MISTRAL Root Module PRF_SALAMANDER IndustrialCargo Root Module PRF_SALAMANDER SciResearch Root Module Nodes
20	Root Module	The Root Module has within it, a single hardpoint innocuous placed at an offset. This offset is actually one of the most important numbers in the game, as it determines where in the bay a ship spawns. In the meantime, find the Hardpoint, locate, and duplicate its Module List asset. Move this new module list into the appropriate folder. Scene Game Asset Store Shaded 2D 3D PRF_HACKEREL Root Module Save Auto Save 4 100 X Y Z 3D Hierarchy PRF_HACKEREL Root Module HardPoint Inspector Layout Map Editor ShipRandomizer Autojoiner PRF_HACKEREL Root Module Tag Untagged Layer Default Transform Local Position X 0 Y 0 Z 0 Local Rotation X 0 Y 0 Z 0 Local Scale X 1 Y 1 Z 1 Uniform Scale 1 Snap Options Show Snap Direction Snap To Nearest Surface Edit Snap Prefs Root Module Definition (Script) Runtime Spawning & Preview Property Override None (Property Container Asset) Set Seed Module Preview Preview Clear Module Types Module Types Empty Disable Spawn Disable In Free Mode

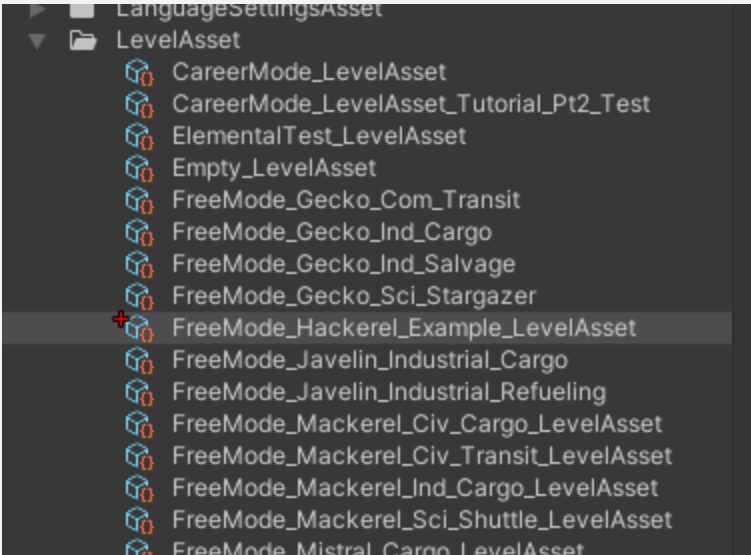
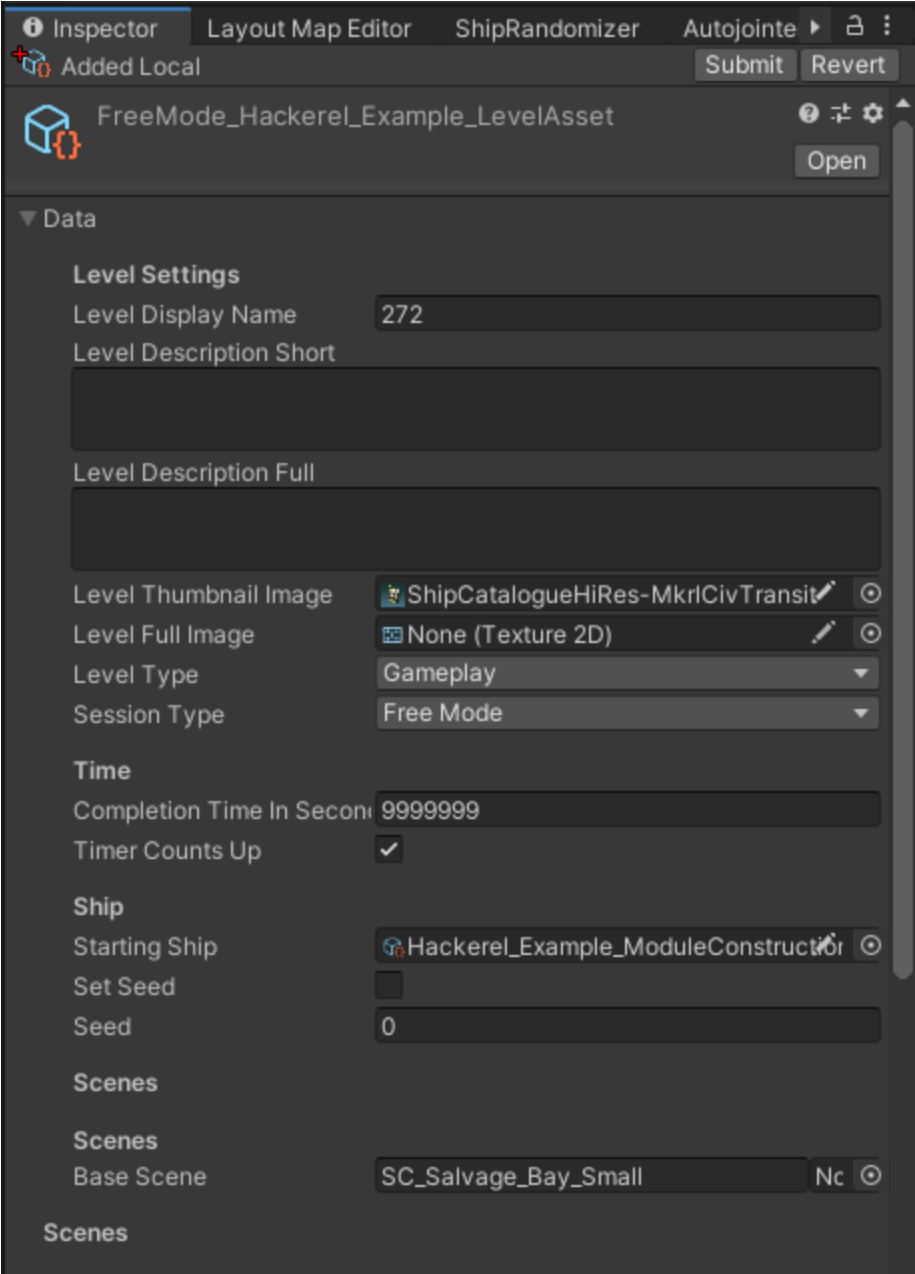
		
21	Root Module, Module Lists	Replace the value of the Module entry with the new ship. 
22	Root Module	Replace the Module List in the new Root Module. 
23	Construction Asset	Return to the ship's construction asset, and replace the Root Module Prefab field with the new Root Module ship. From this point forward, the assembly prefab can be deleted, or kept around to aid in testing. Spawning this ship now from the Ship Randomizer, or from the in-game menu, should now spawn a ship fully constructed from the Layout Map.

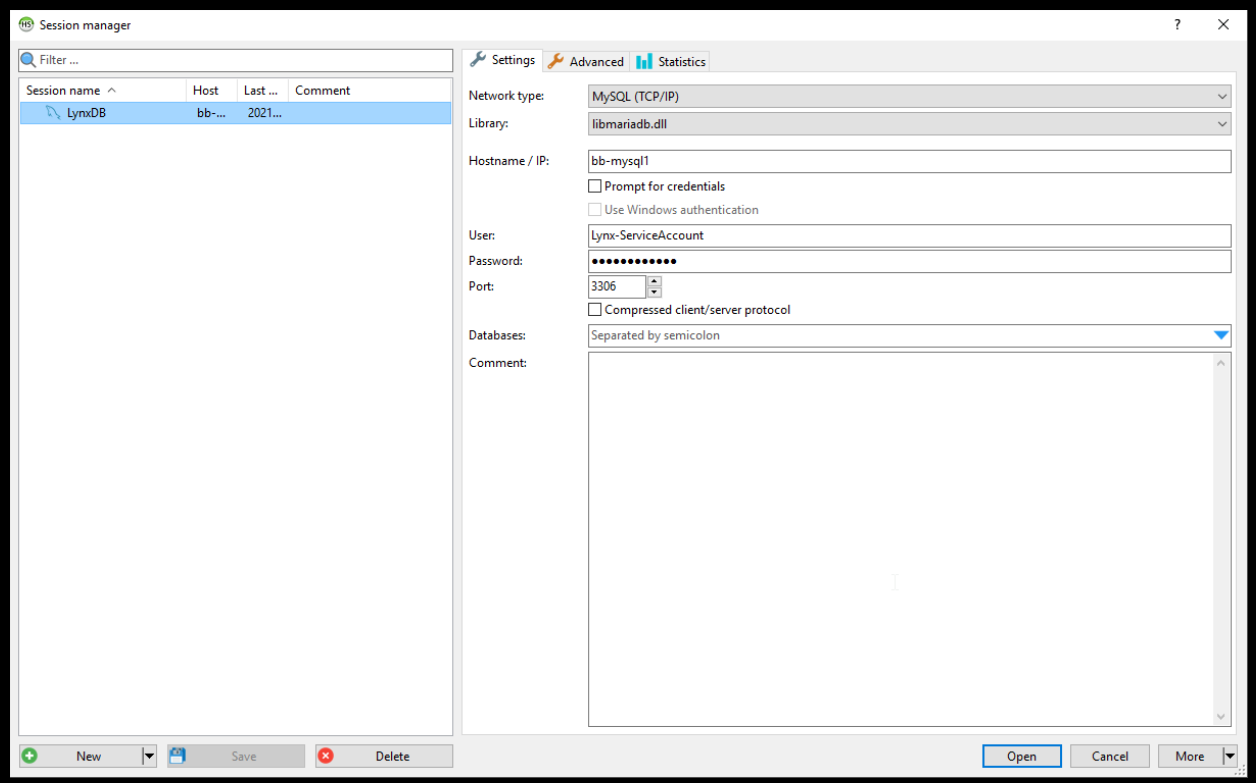
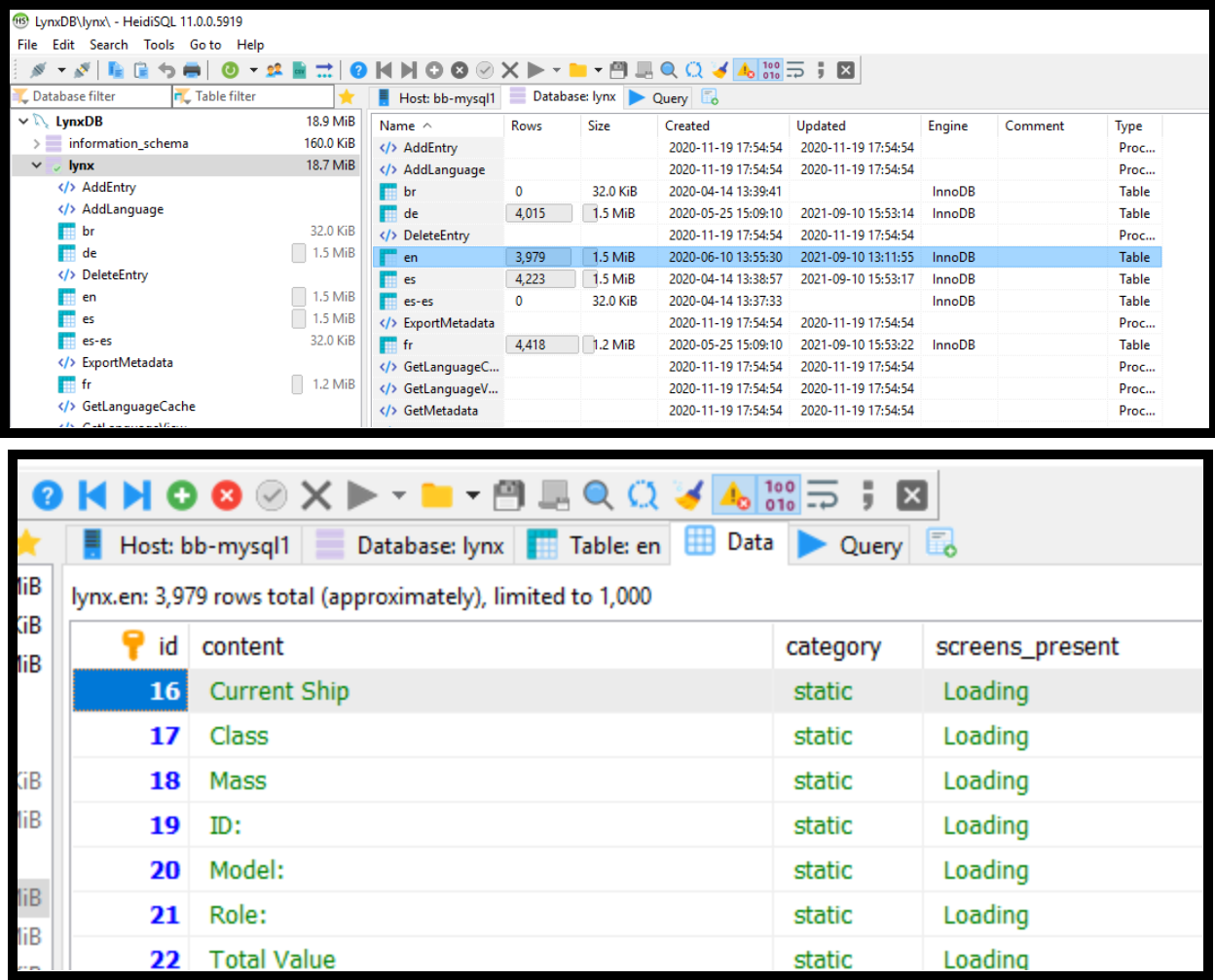
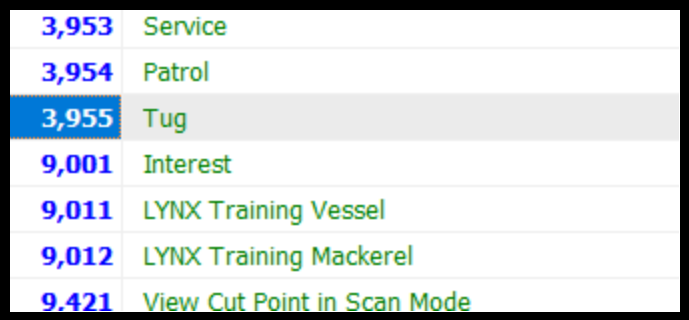
		InspectorLayout Map EditorShipRandomizerAutojoiner Added Local SubmitRevert HackereI_Example_ModuleConstructionAsset Open Data In-Universe Info Ship ArchetypeMackereI_ShipArchetypeAsset (ShipArchetypeAsset) ManufacturerKlyneCo_CompanyInfoAsset (CompanyInfoAsset) Module Data Root Module PrefabPRF_HACKEREL Root Module (RootModuleDefinition) Ship Loading Screen ImageNone (Texture) Catalogue ImageNone (Texture) Skin DataMackereI_ModuleSkinListAsset (ModuleSkinListAsset) Property DataProperties_MackereI_Science_Shuttle (PropertyContainer) Ship Rewards DataNone (Ship Rewards Asset) Is Tutorial Ship
24	Ship Randomizer	The ship randomizer can be used to visualize the ship without entering gameplay, and is invaluable for saving time while developing. Ships spawned via the ship randomizer are not valid for testing gameplay on, as they do not initialize data properly and do not spawn objects from Spawn Module Actions. To open the ship Randomization Window, locate it from the topbar menu: Lynx Ship Randomization Ship Randomization Window. With the window open, select the type of ship you want to spawn from the Module Construction Asset. Then, preview, and then spawn a ship to have it appear in a scene. Ships spawned with the same seed should always give the same results, unless active development has caused a ship "seed shift". If nothing spawns, there is likely an error in the layout map. Check for missing module lists in layout nodes, (17d), or broken connections (18b). If the editor is having conniptions, there's a Layout Node with empty Layout Hard Point entries (13, 17b). You will also have to slap yourself in penance for this crime. LynxToolsJobsPlayFabInControlWindowHelp Reimport Critical Assets Audio Ship Randomization Gridify Objects TechArt Search Project By Type Search For Components WriterDuet Narrative Importer Joint Editor Prefab Override Spawn ModulesCtrl+M Ship CSV Generator Skin Window Hardpoint Tree ViewerAlt+Shift+H Layout Map EditorAlt+Shift+Y Ship Randomization Window Module Viewer  The 'Caustic Longiaru'. Terrific.
25	Ship Positioning in the Bay	In order to test the positioning of the ship... go to the scene SC_Salvage_Bay_Small and drag into its hierarchy SC_Jack_Small_Shrunk.

		 
26	Ship Positioning in the Bay	Spawn a ship via the ship spawner (step 24) place it, in the hierarchy, under the Ship Spawn Point in the SC_Jack_Small_Shrunk scene. 0-out its location. The ship will probably spawn in one corner of the back, near the right-hand furnace. If the ship appears at the wrong rotation, the rotational values in the core node likely need to be changed.  
26a	Ship Positioning in	Reposition the ship in the bay so that it appears in a nice location. Make note of the final

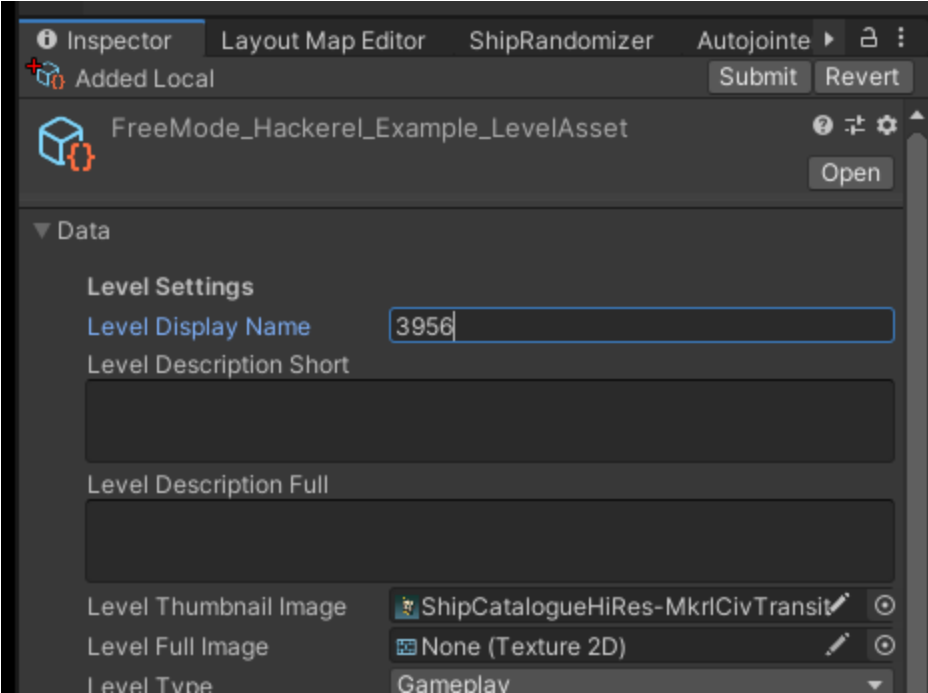
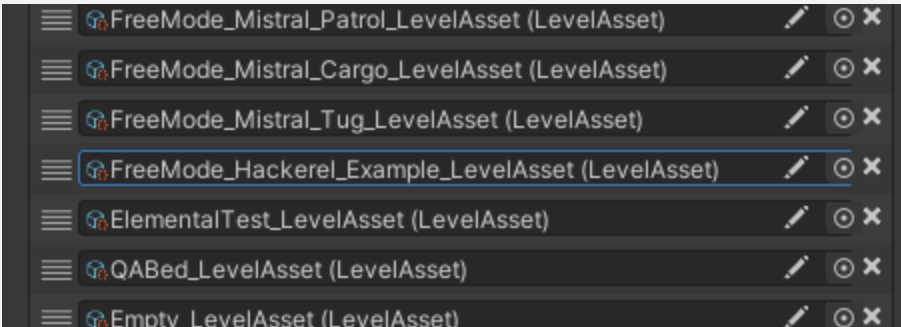
	the Bay	transform, this value is what should be used for the ship's offset in the root module. Copy the values of the Transform component. Do not save any changes to the scenes. 
26b	Ship Positioning in the Bay	Paste the copied values into the hardpoint. This process can be re-tested by spawning a ship into the bay, moving the ship under the ship spawn point in the bay, and 0'ing-out the location. 
26c	Ship Positioning in the Bay	The ship should now spawn properly in an empty map, and in-bay.

		
27	Free Play Menu	In order to introduce this new ship to the Free Play menu, locate FreeModeLevelList_LevelListAsset. Expand your windows to see the whole list, as flipping between pages is not ideal. Alternatively, hit the down arrow on the array to list all pages in a single continuous list. 
27a	Free Play Menu	Add a new item to the list, and position it above the testing levels. 

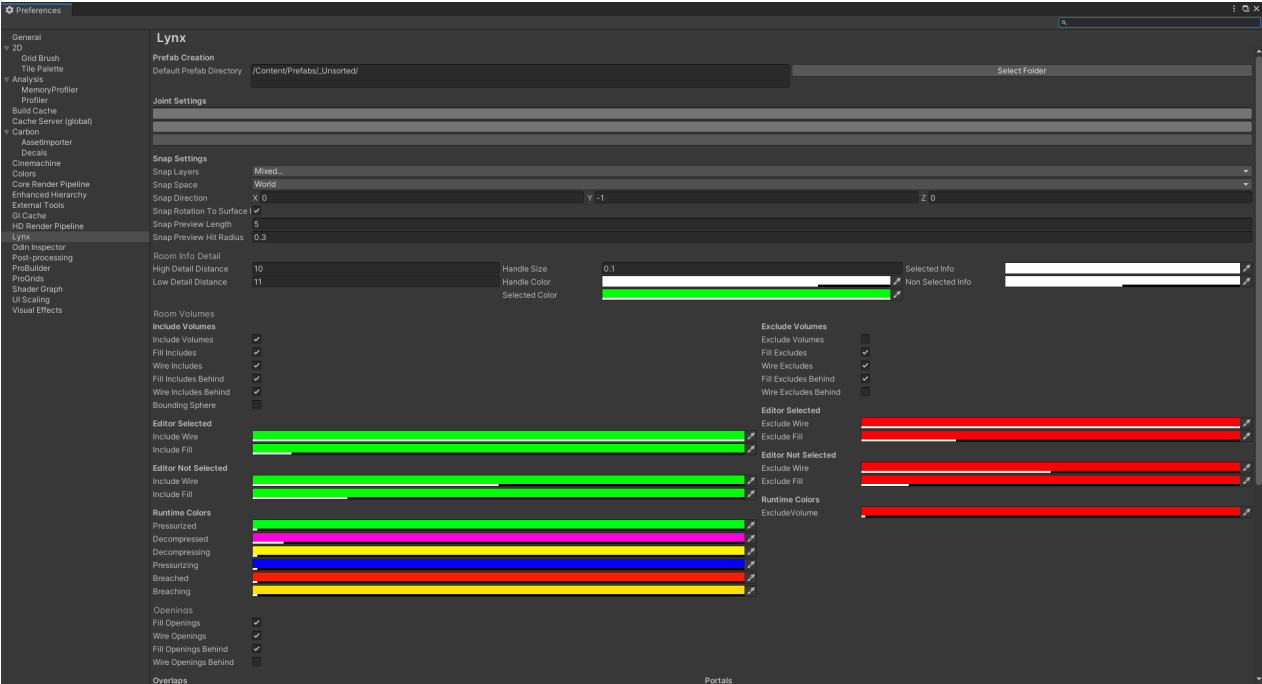
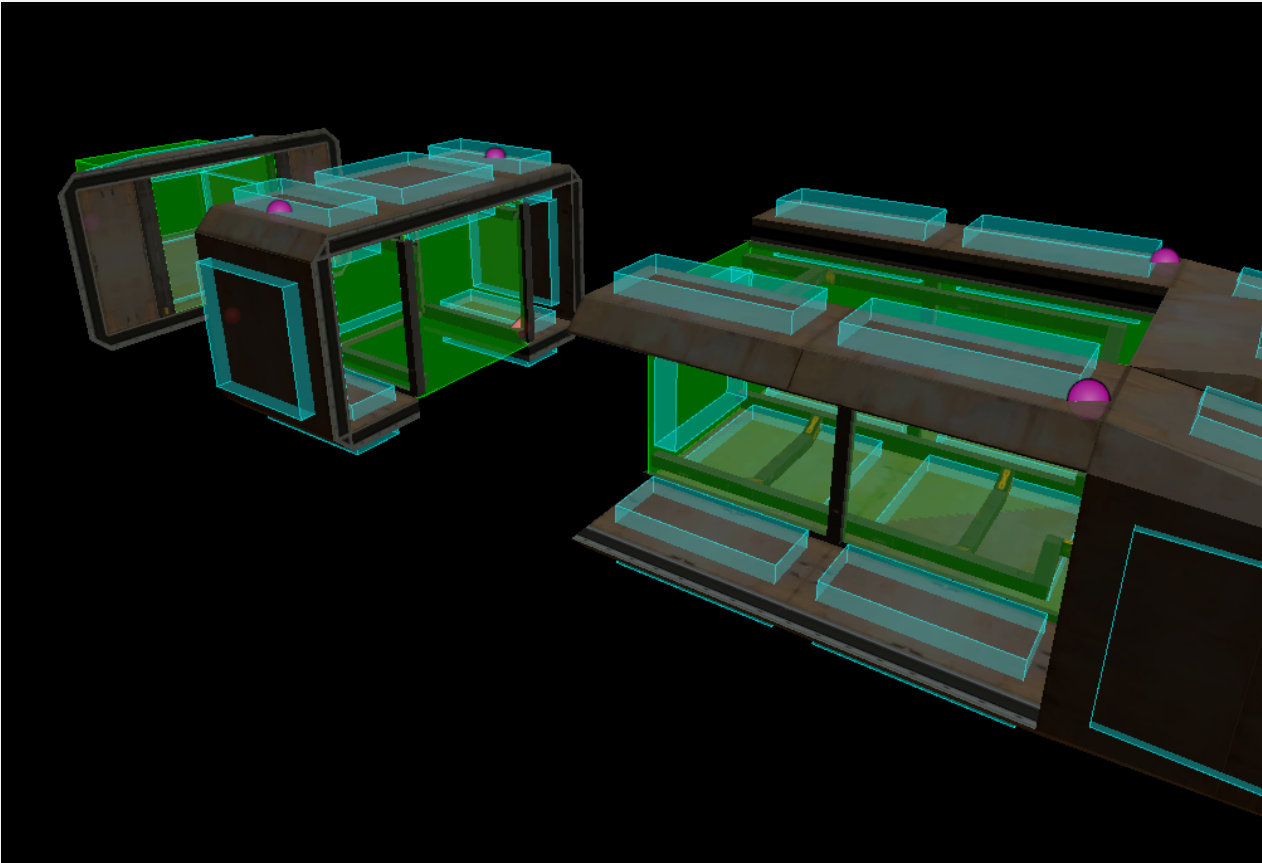
28	Free Play Menu	Locate the Free Mode Level assets, duplicate one, and rename it accordingly. 
28a	Free Play Menu	Replace the Starting Ship field with the ship you're using. 
29	Localization	These next steps will involve localizing text. Open HeidiSQL and open the Lynx project.

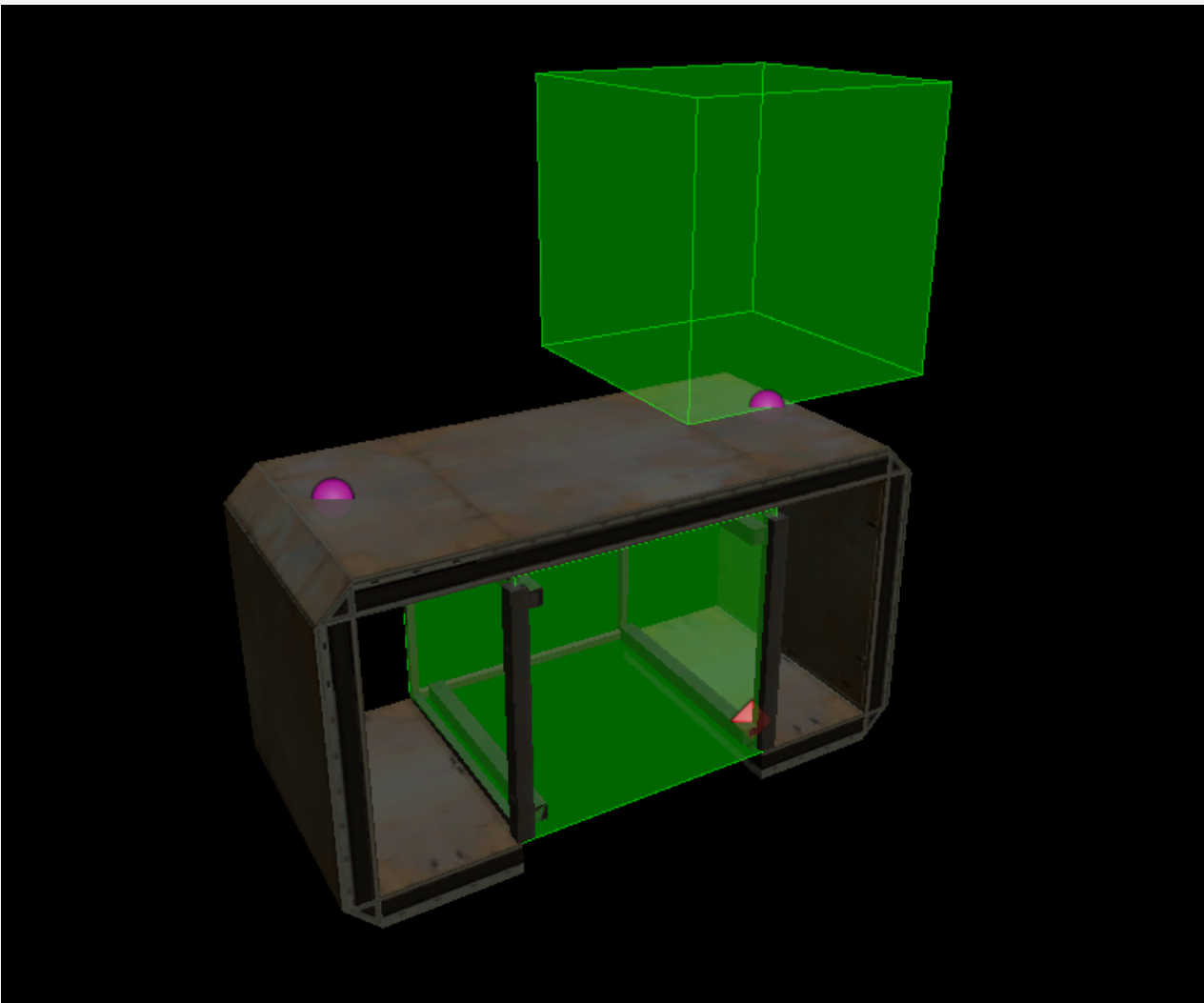
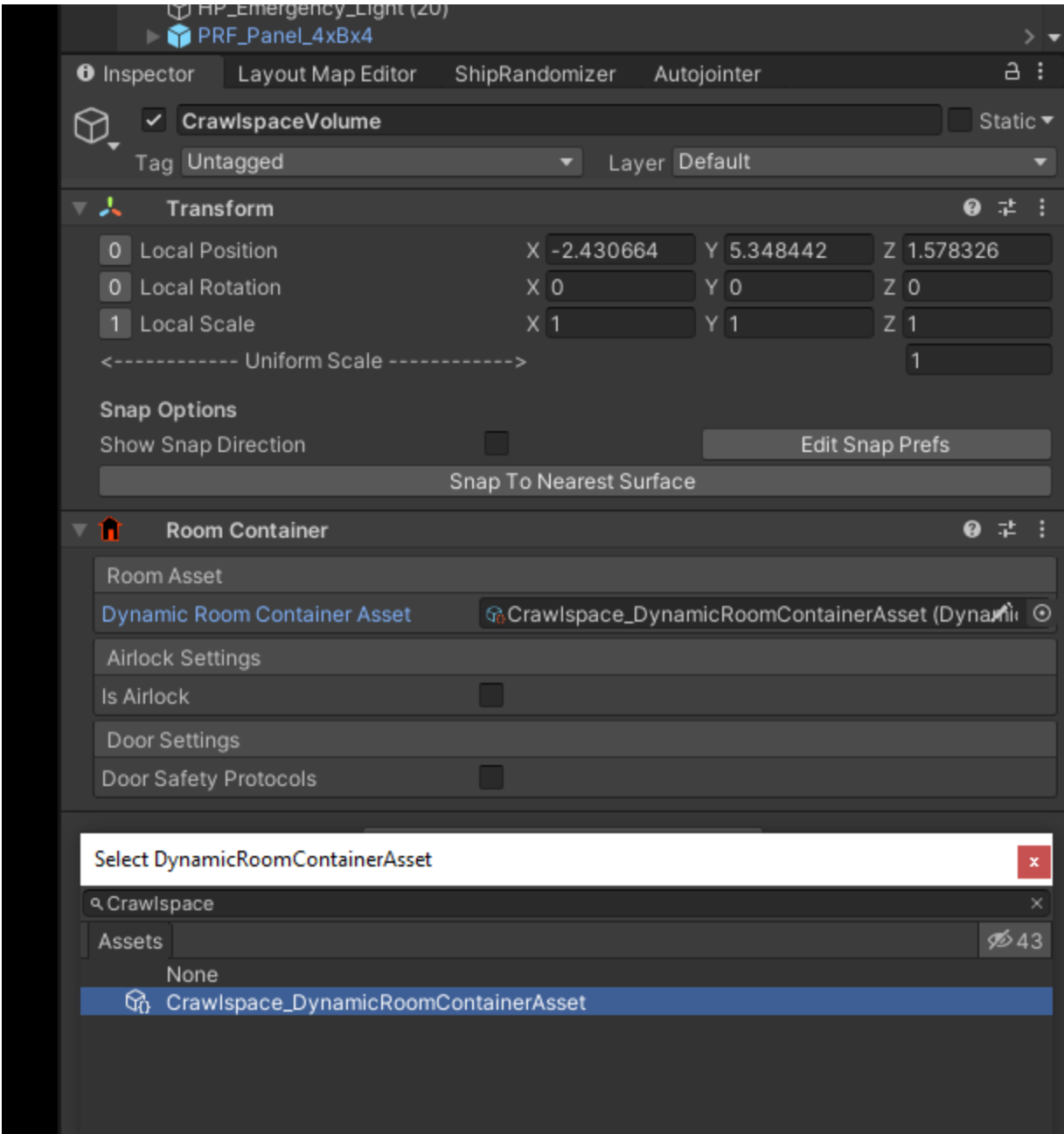
		
29a	Localization	Double click to enter the EN (English) database and then navigate to the data tab. 
29b	Localization	Press CTRL-End to go to the final cell of the database. If this is not done, the database will not be completely populated. Press CTRL-Home to return to the first cell in the database.
29c	Localization	Scroll down through the database until you reach the first "tail" in the database. Due to several implementation errors in the implementation process of localizable text over the years, the numbering system has several especially large gaps within it, that ideally, would be filled over time. The first major gap occurs between ~4000 and 9001 (don't ask). Make note of the last number before the "tail" ends- in this instance, it's 3955. 
30	Localization	Locate a similar entry to what we're looking to write, here it's a ship name, at entry 272. Locate an entry similar in function (maybe even like 272). Note that CTRL-F needs proper settings to locate an entry above the current cell and does not usually find anything when not searching "from cursor". Once located, right-click it and duplicate it <i>without</i> keys. Before clicking off, change the datatable row to <i>one higher</i> than the tail found afterwards.

		Search and replace text Search in:Data grid Text to find:272 Replace: Replacement can have \n for new lines and \t for tabs Options Case sensitive Regular expression Whole words Prompt on replace Direction Forward Backward Origin From cursor Entire scope Scope Global Selection Find Replace all Cancel 271Mackerel - Civilian CargodynamicGameplay - HUD 272Mackerel - Civilian TransitdynamicGameplay - HUD 273Mackerel - Industrial CargodynamicGameplay - HUD 274Salamander - SciencedynamicGameplay - HUD 275Elemental TestdynamicGameplay - HUD 276Salamander - IndustrialdynamicGameplay - HUD 277The Processor is named after Eugene L. ProcdynamicGameplay - HUD 278Cutting a Reactor is considered hazardous for dynamicGameplay - HUD 279Suit Patch KitdynamicGameplay - HUD 280Fuel CanisterdynamicGameplay - HUD 281Repair KitdynamicGameplay - HUD 282OxygendynamicGameplay - HUD CopyCtrl+C Copy selected rowsShift+Ctrl+C PasteCtrl+V Insert value Insert files into TEXT/BLOB fields... Save BLOB to file ... Grid view options Open URL Insert rowIns Duplicate row without keysCtrl+Ins Duplicate row with keys 271Mackerel - Civilian CargodynamicGameplay - HUD 272Mackerel - Civilian TransitdynamicGameplay - HUD 3,956Mackerel - Civilian TransitdynamicGameplay - HUD 273Mackerel - Industrial CargodynamicGameplay - HUD 274Salamander - SciencedynamicGameplay - HUD
30a	Localization	Realign the array by clicking on the key at the top of the database, until the key appears again. Return back down to the end of the tail, and locate the entry you'd edited. Host: bb-mysql1 lynx.en: 4,237 rows total idcontent 260Mackerel 261Salamander
31	Localization	Modify the field and clear the two date fields (CTRL-Shift-N) at the end of the row. Then click upon the "last edited" field. A lot of work to just write a single line of text, isn't it? Try writing about the process of doing it, you'll start questioning your mortality. HeidiSQL's localized database is "hot" and does not require saving. Close the program when you are done with the editing. 3,954PatroldynamicShip Role 3,955TugdynamicShip Role 3,956HackereldynamicGameplay - H 9,001IntereststaticGeneral 9,011LYNX Training VesseldynamicGameplay - T ty of ShipSuch as emergency services, or military service.2021-09-09 15:25:06(NULL) 2021-09-09 15:27:54(NULL) 2021-09-09 15:27:59(NULL) DO NOT LOCALIZE, EXAMPLE FOR DEVELOPERS2020-04-30 12:15:232020-05-26 17:28:50 nancial inter...2020-05-01 17:06:072020-05-26 17:28:50 Ship(NULL)2020-05-01 17:06:07(NULL) Shin(NULL)2020-05-01 17:06:07(NULL)
32	Localization, Free	Back within the editor, replace the level display name with the new, localized name field. This

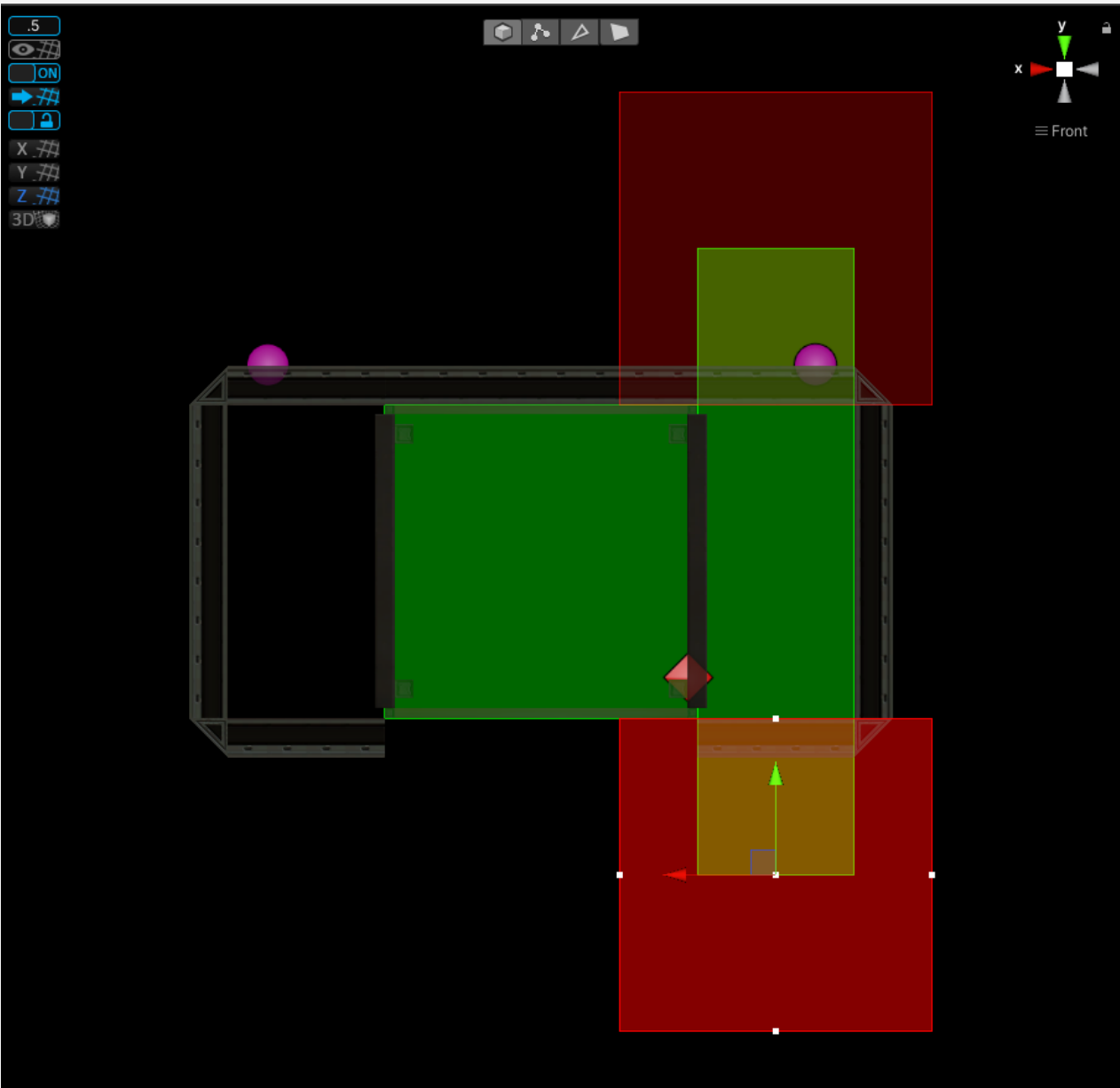
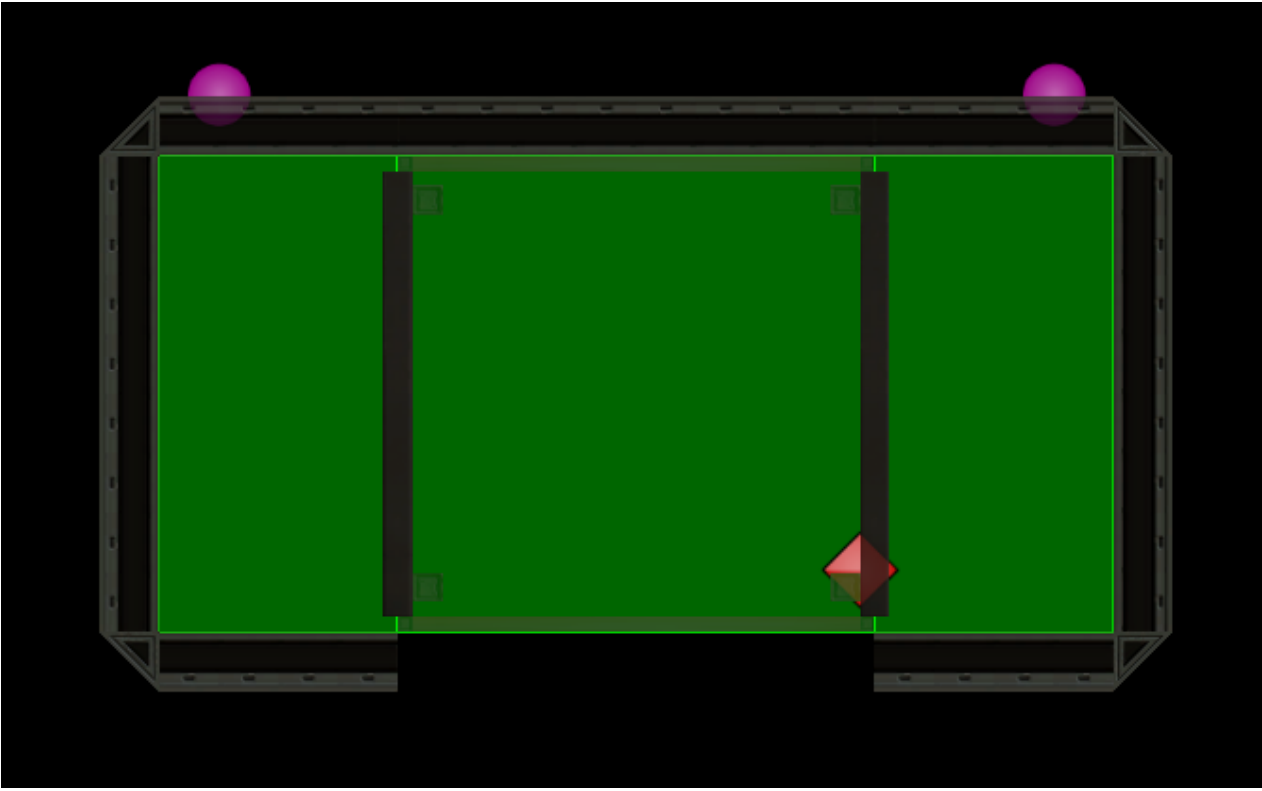
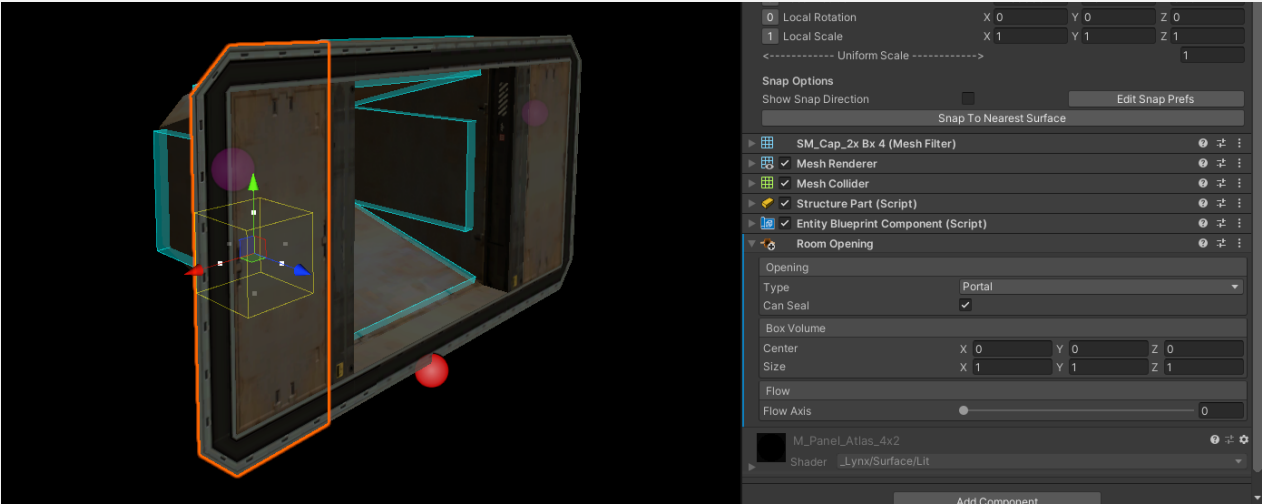
	Play Menu	localized entry will need to be included in a database update by code, but the work on Design's side for this is done. 
33	Free Play Menu	Finally, return to the FreeModeLevelList_LevelListAsset, and in the empty array entry, place your newest free mode level asset. 
34	Pressurization	Now the ship is spawning properly in the bay, most of what remains is related to gameplay. However at this point, the Hackerel, and most ships, will suffer from pressurization issues. First and foremost, the ship decompresses on spawn. Investigating pressurization is a difficult and complicated job, likely more complicated than anything that could be fit into this document- but an easy way to narrow down where pressurization errors manifest is by flying around a ship in-game and making note any spaces missing room data, or that cannot be pressurized with the debugger. 

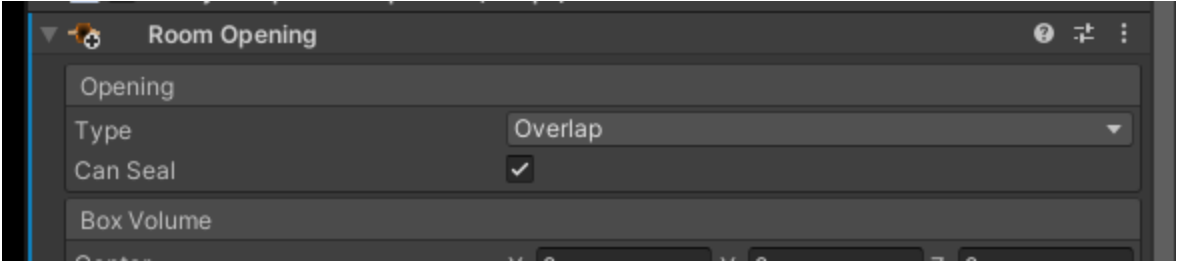
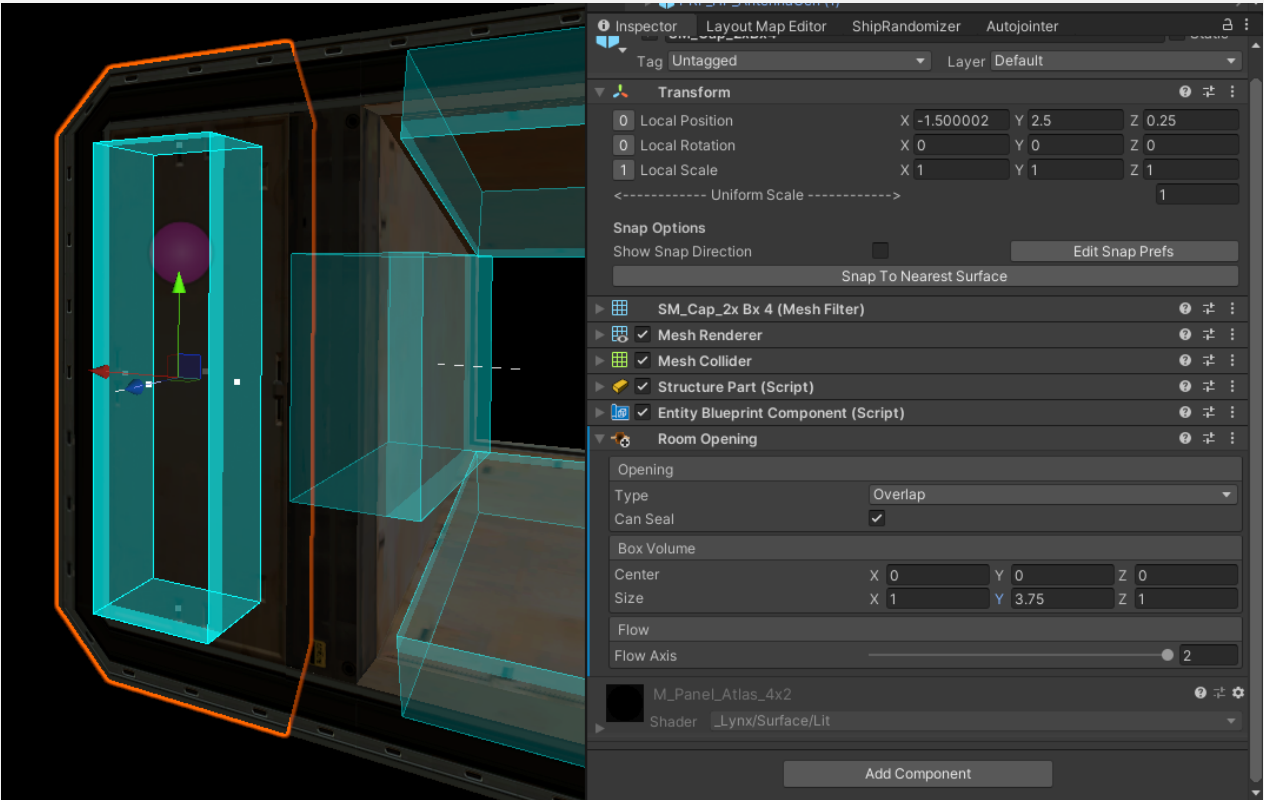
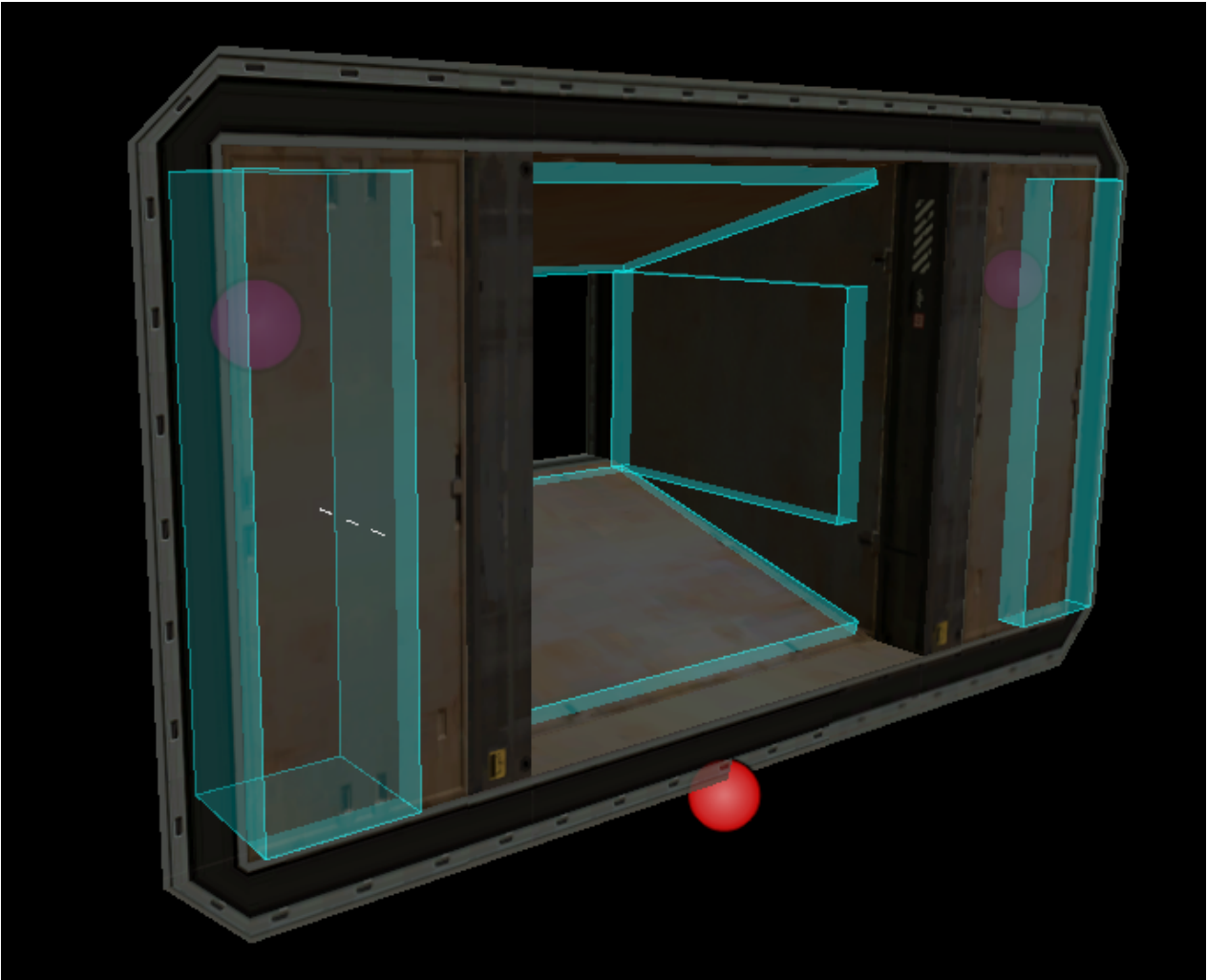
		Audio Debugger Audio WAAPI Background sequence viewer Crash Test Debugger Durability Debugger Effects Viewer Glitch Debugger Haunting Debugger Joint Debugger Localization Debugger Lynx Audio Debugger MonoBehaviour Debugger Narrative Debugger Player Debugger Position (18.7, 8.1, 25.7) No Clip Mode God Mode Trout Mode Air Pressure XP Info: 0 / 50 XP Amount 100000 Add XP Bonus XP Reserved: --Currencies-- > None SHIP NAME SALVAGED: 100% \$1000000 RE UNPRESSURIZED
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35a	Pressurization	Rather than using dedicated gizmos, the Lynx tab is home to the switches for all Room and Overlap visualizers. Note that turning these on is often very hard on performance, especially overlaps, so it's recommended to also turn off Hardpoints (gizmos) while working with rooms. Turn on rooms and overlaps and locate the ship you wish to modify. 
35b	Pressurization	To even the untrained eye, there should be something rather obvious wrong with the <i>Hackerel</i>. It's completely missing rooms (green) in the crawlspaces, and has no overlaps (blue) between any of its sections- harming air flow. (For those curious, this problem is solved in the Mackerel via the "x3" and "x4" layout prefabs.) 
35c	Pressurization	Part of the difficulty in room volumes is that the work has to be performed predictively, in a similar manner that hardpoints are. Game knowledge is very important in determining what may spawn where, and what it may interact with. Room volumes are extremely sensitive, and slight misalignment or extraneous volumes can (and will) absolutely annihilate a ship with 0 indication as to where the problem is, and why it's happening. It's horrendously common for a room overlap to be placed between two parts of a contiguous room, and silently instigate 6 hours of room volume debugging. This is a long way of saying that for the purposes of the document, I'll be fixing this problem in the <i>middle</i> segment of the Hackerel, as I've placed airlocks that already spawn with overlaps in the front section. There are problems with that setup, but they're far less than trying to correct all the problems that could be noted in 35b.
36	Pressurization	Ideally, one would construct room volumes by creating empty prefabs and assigning components to them, but traditionally room volumes are all just copies of other volumes. Select a room such as ReactorVolume in the middle of PRF_Hackerel_Mid and duplicate it. Mode the new volume aside.

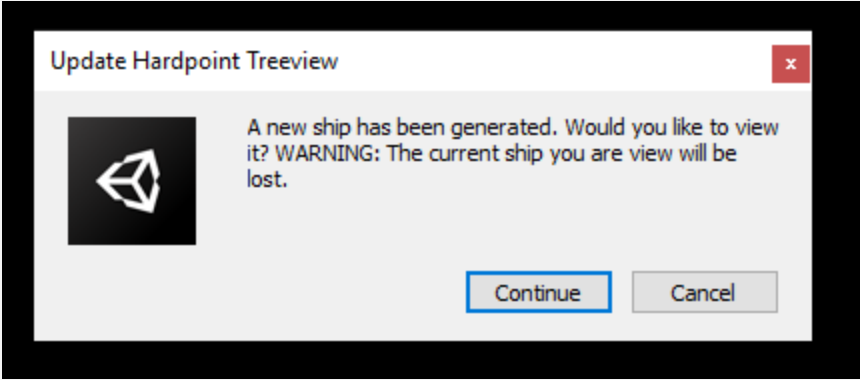
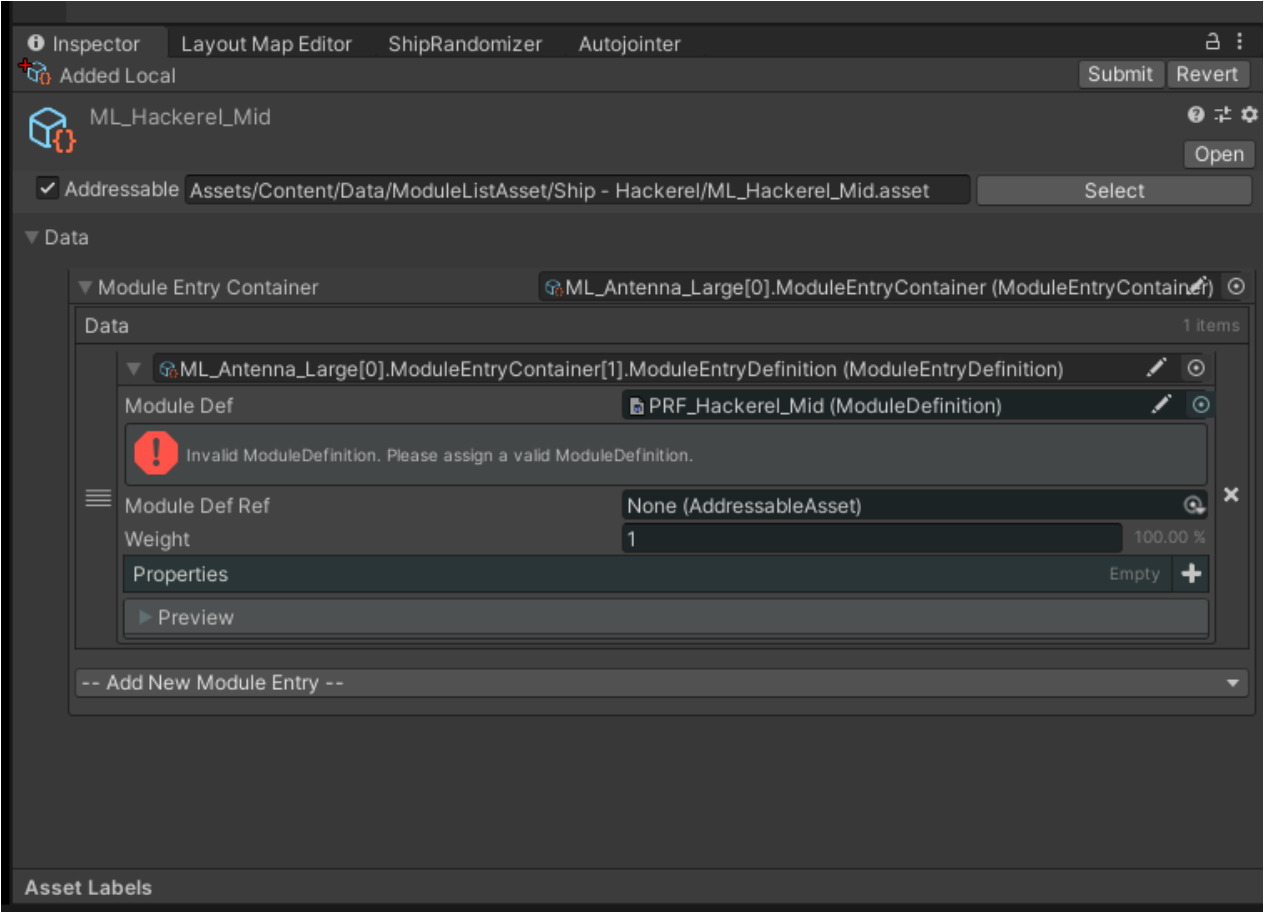
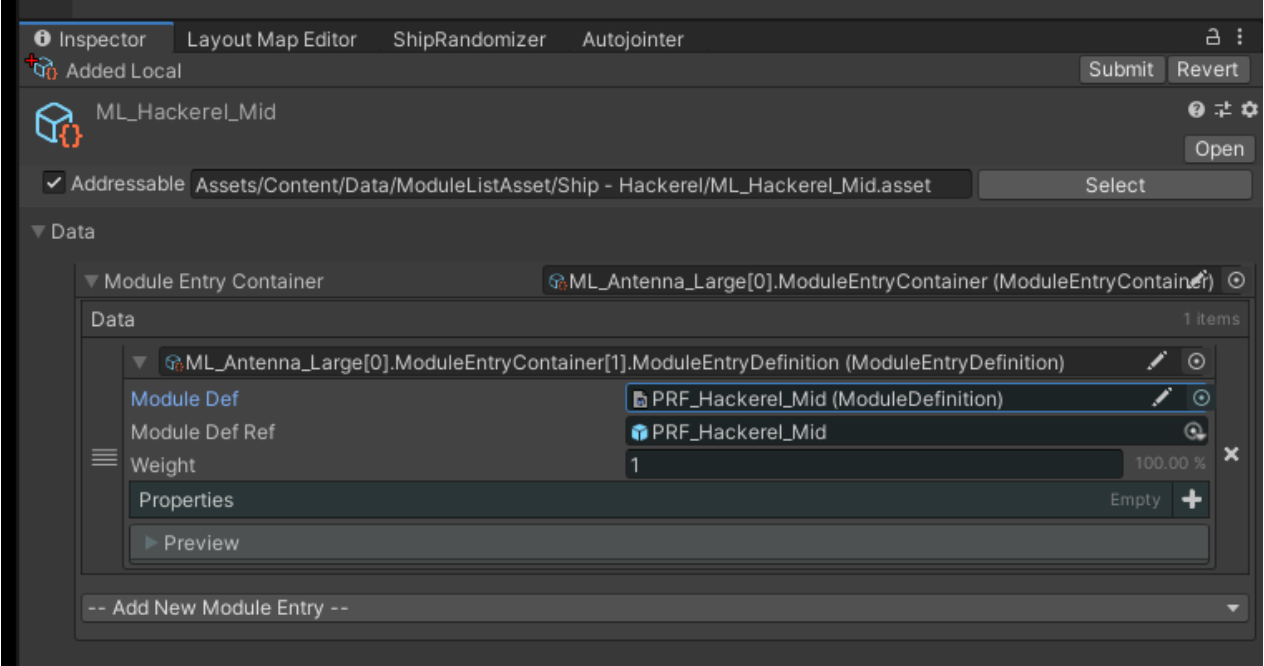
		
36a	Pressurization	Rename the new volume. Replace the Dynamic Room Container Asset with a new asset appropriate for what room the new volume will be representing, such as a Crawlspace or Bathroom. 
36b	Pressurization	Now is time to move the new room volume. Switch the camera to Orthographic. It's quite important that when the room volume is <i>moved</i> it is the root object of the room volume that is moved. However, when the object is resized, it is the subj-volume's Room Sub Volume component values that are modified. Ensure that the volume is lined up in all Orthographic views, Progrids can be very helpful.

36c	Pressurization	In the event you need to make a complicated shape for a room, more than one sub-volume can be used. Be wary that this is very prone to complication-related errors and should be used sparingly.
36d	Pressurization	In addition to adding more sub volumes, Room Sub Volume components can be set to exclude. This will exclude the contents in this volume that may have been added by a standard include volume. This does not modify the include volumes of any Room volume other than the room volume it is an exclude volume of.

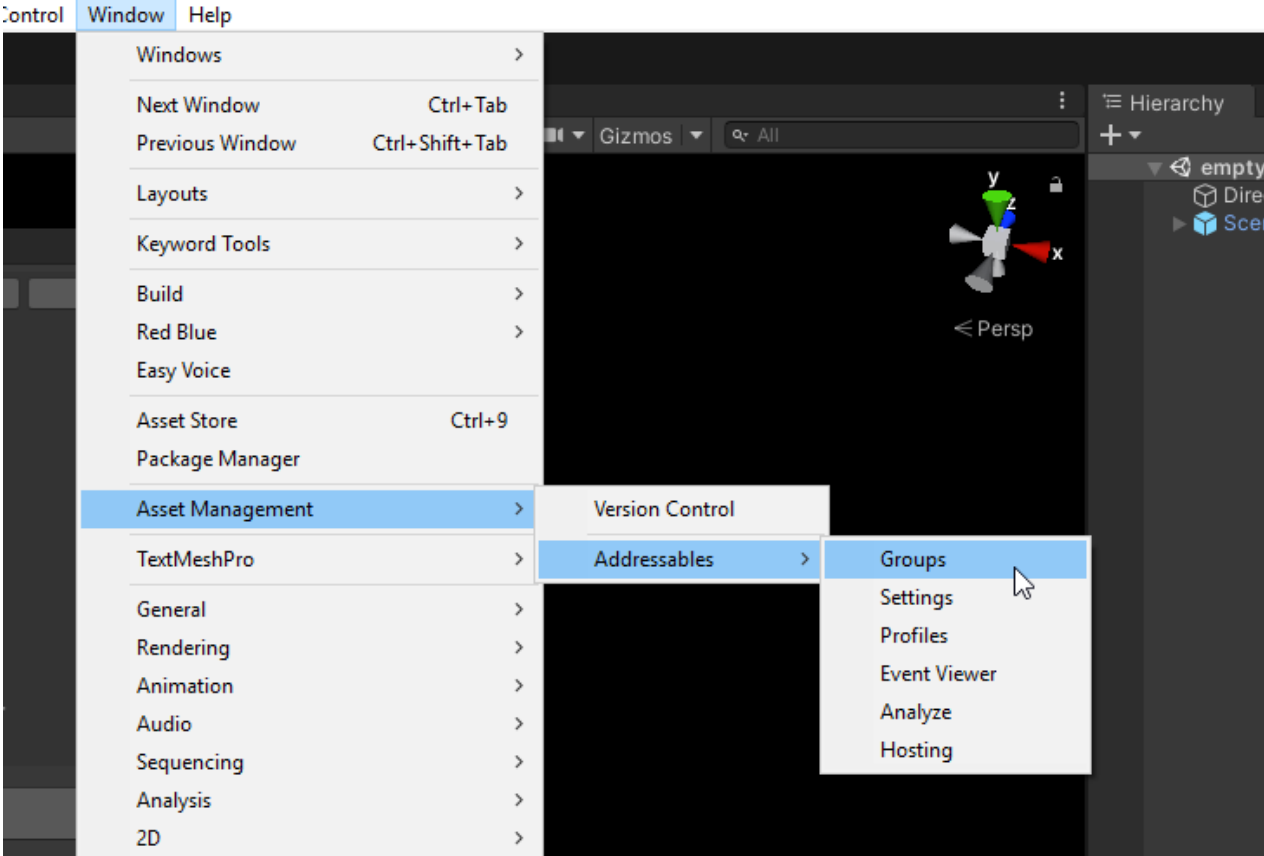
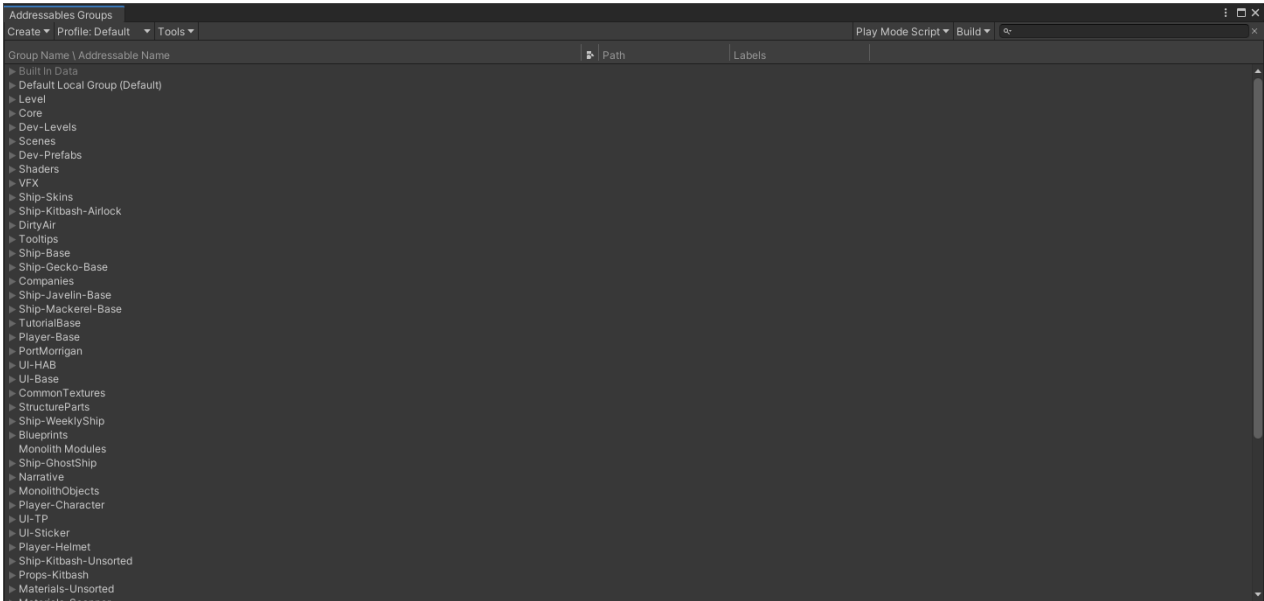
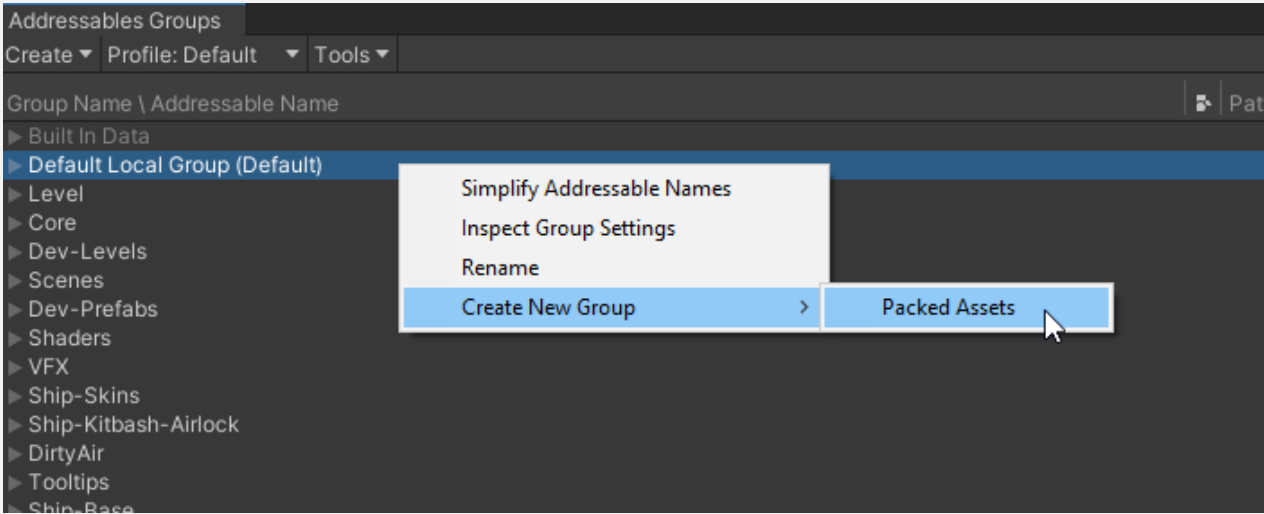
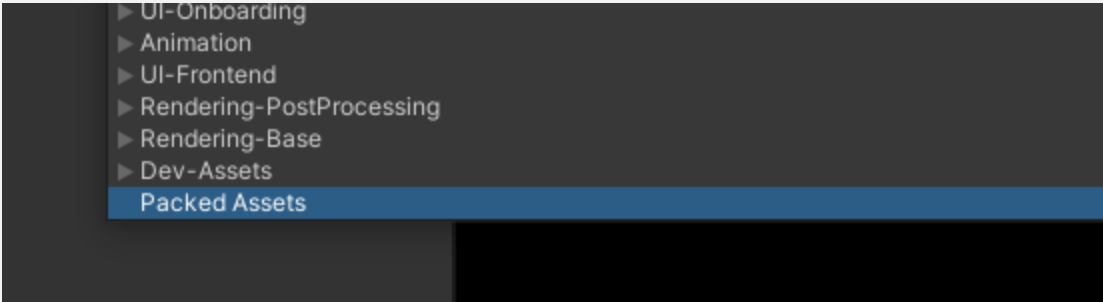
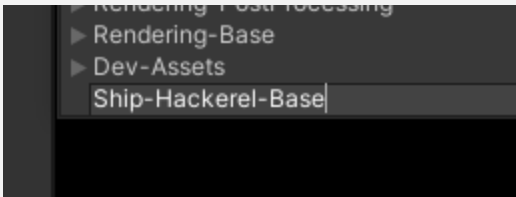
		
36e	Pressurization	The Hackerel's crawspace is completed by duplicating the already adjusted volume from one side to the other. 
37	Pressurization	Room overlaps manage the "flow" of air between rooms and how air is sealed by walls and doors. PRF_Hackerel_Back needs overlaps to properly connect to the back of the ship. Unlike Room Volumes, Room Overlaps can also be applied to pre-existing prefabs and meshes. Apply the Room Opening component to an empty gameobject or existing prefab. 

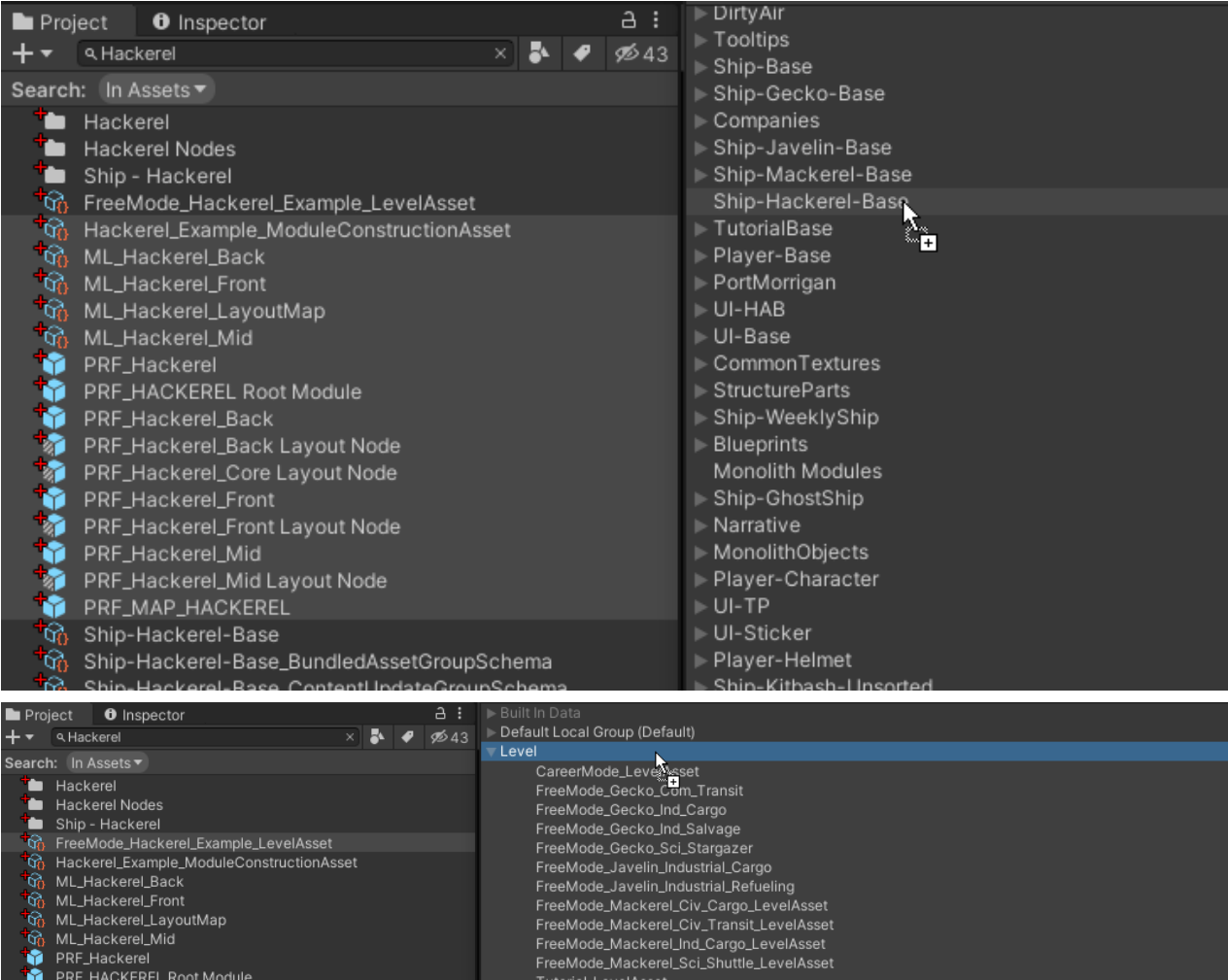
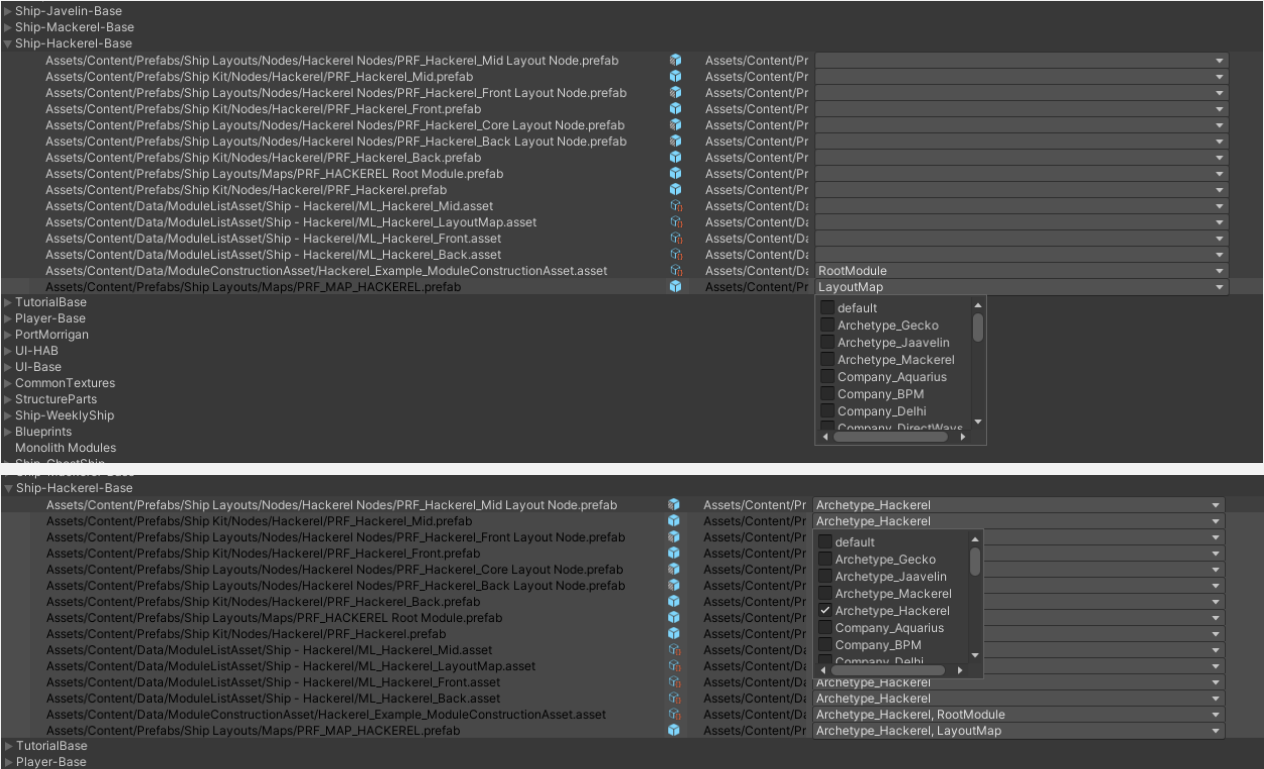
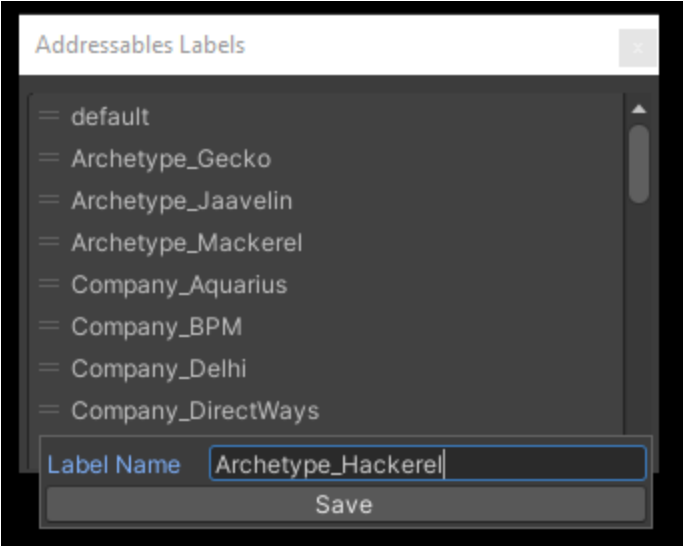
37a	Pressurization	By default, Room Openings are set to be "portals" which are room volumes used for doors. In most cases this should be set to "Overlap". 
37b	Pressurization	Adjust the size of the overlap. This should always be within the bounds of at least one Room volume. Be sure to also adjust the Flow Axis, this determines the direction in which air will flow if the attacked panel or area is breached (such as with a splitsaw, or demo charge). 
37c	Pressurization	For the Hackerel Back, copy this component to the other panel missing an overlap. While simple and possibly still error-prone, most of the major problems plaguing this ship should now be solved. 
38	Hardpoint Tree Viewer	In the event you need to study the hierarchy of a spawned ship, there is no better tool than the Hardpoint Tree Viewer. Locate it from the topbar menu: Lynx Ship Randomization Hardpoint Tree viewer.

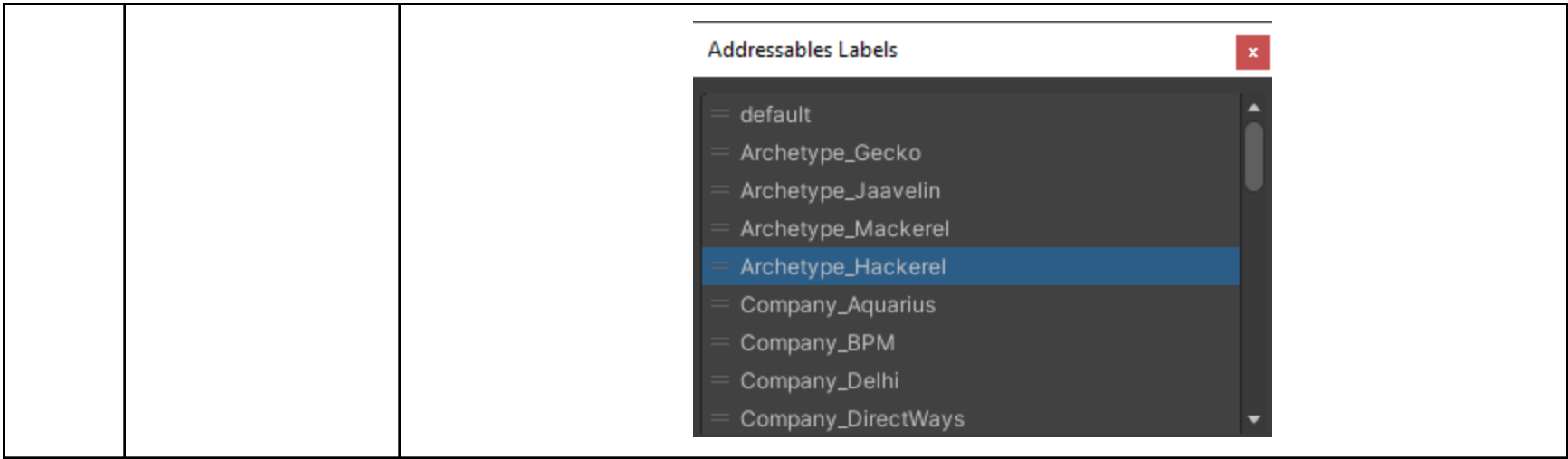
		ook Pro Lynx Tools Jobs PlayFab InControl Window Help Reimport Critical Assets Audio Ship Randomization Gridify Objects TechArt Search Project By Type Search For Components WriterDuet Narrative Importer Joint Editor Prefab Override Panels Spawn Modules Ship CSV Generator Skin Window Hardpoint Tree Viewer Layout Map Editor Ship Randomization Window Module Viewer Hardpoint Tree Viewer Refresh 0.6 Nothing selected <table><tr><th>Name</th><th>ModuleList</th><th>Result</th><th>Children</th><th>Generation</th></tr><tr><td>PRF_HACKEREL Root Module (RootModuleDefinition)</td><td></td><td></td><td>351</td><td></td></tr><tr><td>HardPoint</td><td>ML_HackereLayoutMap (ModuleListAsset)</td><td>Layout</td><td>350</td><td>-1</td></tr><tr><td>PRF_HackereFront (ModuleDefinition)</td><td></td><td></td><td>240</td><td></td></tr><tr><td>PRF_HackereMid (ModuleDefinition)</td><td></td><td></td><td>98</td><td></td></tr><tr><td>PRF_HackereBack (ModuleDefinition)</td><td></td><td></td><td>9</td><td></td></tr></table> <tr><td>38a</td><td>Hardpoint Tree Viewer</td><td> You can open the items on the menu to the left to locate every single hardpoint in a spawned ship. By digging into each hardpoint, you can uncover the module list's logic and related restrictions while determining what objects should spawn. HP Walkway (1) ML_Mackerel_Walkway3Cx2 (ModuleListA Module Definition 1 -1 HP_Emergency_Light (1) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP_Emergency_Light (2) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP_Emergency_Light (4) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP_Emergency_Light (7) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP_Emergency_Light (8) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP_Emergency_Light (10) ML_Hazard_Lights (ModuleListAsset) Module Definition 1 -1 HP Loose Junk ML_Mackerel_LooseJunk (ModuleListAsset) try is null, or properties/restrictions culled all entries 0 -1 Restrictions Ship Restrictions Current: 0 / Max: 0 Prop_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 0 FoundNarrativeDevice_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 1 PowerCell_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 3 Antenna_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 1 Reactor_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 0 InteriorDoor_ModuleTypeAsset (ModuleTypeAsset) Current: 0 / Max: 0 CrawlspaceElectricWires_ModuleTypeAsset (ModuleTypeAsset) No Group Restrictions Detailed Info Roll: 0.6251422 Total Weight: 1 PRF_Light_Running_C_Ghostship Result: Ineligible (Properties) Probability: 0% Weight: 0 Lower Bound: 0 Upper Bound: 0 Properties: GhostShip Types: PRF_Light_Running_C Result: Picked Probability: 100% Weight: 1 Lower Bound: 0 Upper Bound: 1 Properties: Difficulty_1HazardLevel, Difficulty_2HazardLevel, Difficulty_3HazardLevel, Difficulty_4HazardLevel, Difficulty_5HazardLevel, Difficulty_6HazardLevel, Difficulty_7HazardLevel, Difficulty_8HazardLevel, Difficulty_9HazardLevel Types: </td></tr> <tr><td>38b</td><td>Hardpoint Tree Viewer</td><td> It should be noted that if a new ship is spawned when the Hardpoint Tree Viewer is open, it will cause a warning which asks if the Tree Viewer page should be updated. Indicating "Continue" will </td></tr>	Name	ModuleList	Result	Children	Generation	PRF_HACKEREL Root Module (RootModuleDefinition)			351		HardPoint	ML_HackereLayoutMap (ModuleListAsset)	Layout	350	-1	PRF_HackereFront (ModuleDefinition)			240		PRF_HackereMid (ModuleDefinition)			98		PRF_HackereBack (ModuleDefinition)			9		38a	Hardpoint Tree Viewer	You can open the items on the menu to the left to locate every single hardpoint in a spawned ship. By digging into each hardpoint, you can uncover the module list's logic and related restrictions while determining what objects should spawn. 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		update the Tree Viewer with the new data. Hitting Cancel will cause the Tree Viewer to stop updating until it is manually refreshed. 
39	Adressables	When working in the adressables-compatible editor versions, there's a few things to make note of. The first and foremost is that module lists are set up differently. Rather than using Module Definitions, they utilize a different field called Module Def References. These can be easily solved by placing the same prefab within the field.  
39a	Adressables	Replacing many fields can be a time consuming process, so instead, it can be easily automated via the Adressables Converter tool. This tool can be found from the topbar menu under: Lynx Addressable Converter. Running this can take several minutes, but will be a comprehensive conversion.

		Look ProLynxToolsJobsPlayFabInControlWindowHelp Asset Stor Reimport Critical Assets Audio TechArt Ship Randomization Panels Gridify Objects Search Project By Type Search For Components Addressable Converter WriterDuet Narrative Importer Update TypeAsset GUIDs Joint Editor Prefab Override Addressable Converter Converters to RunAllNone Company Info Asset Addressable Converter Hard Point Addressable Converter Hierarchy Setting Addressable Converter Layout Spawn Node Addressable Converter Level Asset Addressable Converter Level Select Button Addressable Converter Module Construction Asset Addressable Converter Module List Asset Addressable Converter Module Skin List Asset Addressable Converter Narrative Entry Asset Addressable Converter Ship Class Asset Addressable Converter Triggerable Module Spawner Addressable Converter Triggerable Spawn Ship Addressable Converter Convert Objects Conversion Results Total Objects 0 Total Passed 0 Total Ignored 0
40	Adressables	All that remains is properly placing addressables in packages. Open up the Groups window, which can be found from the topbar menu at: Window Asset Managment Addressables Groups

		 
41	Adressables	Once open, create a new group by right clicking somewhere, and selecting the Create New Group Packed Assets buttons. The new category will spawn at the bottom of the menu, called "Packed-Assets". Rename that, following the same naming convention as the other ships. This should also position it properly in the list.   
42	Adressables	Now, begin the process of placing ship-relevant assets into these categories. Typically, the ship prefabs, map, root module, and construction asset can all go in the same category. FreeMode

		assets go in a separate folder called "Level". 
43	Adressables	Finally, assign categories to the assets you're working with by using the right-hand drop-down menus several can be. Updated at once by ctrl-clicking multiple entries and opening the menu at once. 
43a	Adressables	In the event a new label is needed, you can add a new category from the bottom of the scroll-list. The new label can then be moved using the small equals signs on the left-hand side of the list. 



Pressurization Setup

Additional info on how to set up Rooms in the ships and ensure proper function of the Pressurization System.

ROOM CONTAINER

The top-level object required to make a Room. Holds the definition to the type of room and its odds of being pressurized or not.

Parameter	Type	Description
Room Asset	Dynamic Room Container Asset	Defines what kind of room type this is and it's odds of pressurization.
Is Airlock	bool	Defines if this room is an Airlock or not, this is required for proper door and pressure cycling behavior.
Door Safety Protocols	bool	If <i>true</i> , doors in the room close when the pressure cycles.

ROOM SUB VOLUME

Room Sub Volumes must be placed beneath a Room Container, and collectively build the size and shape of that Room. Room Sub Volumes are typically applied as components to child actors to a Room Container parent actor.

Parameter	Type	Description
Center	Vector 3	The position of the volume (relative to the Transform position). It's advised to leave the center value of the volume at (0,0,0) and instead move the Transform position to move the position of the volume.
Size	Vector 3	Defines the size and shape of this volume, described as the <i>extents</i> of the Volume in <i>meters</i> .
Mode	Volume Type	Include - Defines the area that is to be included as part of the room. Exclude - Defines areas that are to be <i>excluded</i> from the room, effectively subtracting from a room's Include volumes. This is used primarily when the room shape is non-rectangular, for example rooms with sloped ceilings use an exclusion volume to <i>exclude</i> the portion of an Include volume that pokes out of the room. Due to the complicated math involved in handling include and exclude volumes, it is advised to try and keep rooms as square (or rectangular) as possible. Too many additional hooks or loops in room structure can create small invisible 'pockets' of depressurization inside normally fully pressurized spaced.

ROOM OPENING DEFINITION

Room Openings are used to define how Room Sub Volumes should connect to one another and to the vacuum. When a ship spawns, a Room Sub Volume with an Overlap Volume that intersects a Room Sub Volume in a separate room will be considered connected. No Openings are required between Room Sub Volumes that are children of the same Room Container.

Room Openings may only overlap **two** Room Volumes. If an Opening touches more than two volumes, it will cause unexpected pressurization behavior.

Parameter	Type	Description
Type	Opening Type	Portal - An Opening that is either continuously “open” to the connecting room (such as an open hatch), or can “open” and “close” during gameplay (such as a doorway <i>with</i> a door). If a Portal is <i>closed</i> and then breached (ie. the door is cut or dissolved), there is a unique breach distinct from breaching a normal overlap. Overlap - An Opening that is continuously “closed” to the connecting room (such as a wall). Objects inside the Opening that have “Can Seal” defined as <i>true</i> in their Joint Data will cause the Overlap to enter it's breach state when cut or destroyed. It is required for all exterior hull panels to have an Overlap connecting to only one interior Room Volume in order for a ship to understand how its internal pressurization relates to the vacuum outside the ship. If a room cannot be pressurized, even through debug, it is usually because an overlap (that appears to be functional) is actually bridging two Sub Volumes of one Room Container, rather than two rooms. This effectively results in the Opening becoming a transition to the void of space from <i>inside</i> the room.
Can Seal	bool	If <i>true</i> , this Opening will “seal” the pressure between the two connecting Volumes. If the seal is breached, a breach event will occur.
Center	Vector 3	The position of the Opening, relative to its Transform position. It's advised to leave the center value of the Opening at (0,0,0) and instead move the Transform position to move the position of the volume
Size	Vector 3	Defines the size and shape of the Opening, described as the <i>extents</i> of the Volume in <i>meters</i> .
Flow Axis	int	Defines which direction the “air pressure” will flow through this Overlap between Volumes. This should be defined to best match the apparent way in which the rooms are connected. An incorrect Flow Axis will cause breach events to appear to push air in the wrong direction. 0 - Flows along the X-Axis 1 - Flows along the Y-Axis 2 - Flows along the Z-Axis

DYNAMIC ROOM CONTAINER ASSET

As Asset that defines which Room Types this Room can be. The Room Type is chosen based on the randomized Properties of the Ship. If not eligible Room Type is found, the Default Room Data is used.

Parameter	Type	Description
Weighed Room Datas	List<Weighted Room Data>	Each entry in this list defines: Value - A Room Data Asset that defines the type of Room and it's pressurization chance. Weight - When multiple entries in this list are eligible to be chosen, the weighting for each choice. Restrictions - The Ship Properties required for this entry to be eligible. If any Properties in this list are missing from the Ship's properties, the entry is ineligible to be chosen.
Default Room Data	Room Data Asset	The fallback Room Data Asset to use if there are no eligible choices in the Weighted Room Datas list.

ROOM DATA ASSET

An Asset that defines a Room Type, including its localized name, its pressurization chance and other FX-related parameters.

Parameter	Type	Description
Room Name	localized string	The Localization Database ID of the full Room Name (ie. "Crawlspace")
Short Room Name	localized string	The Localization Database ID of the Short Room Name, used on the room labels in Scan Mode (ie. "CRWL")
Room Type	Room Type Asset	Reference to the Room Type Asset that defines Room Type
Pressurization Probability	float	The percentage chance that this room will be pressurized when the ship spawns
Dust Level Min	float	Defines the minimum random amount of dust particles in the room
Dust Level Max	float	Defines the maximum random amount of dust particles in the room
Dust FX Name	string	The type of dust particles to spawn There is only one type of dust fx currently: "Dust". Thus, this is should always be "Dust".
Decompression Prefab	Object	The FX prefab to use when their is a breach (ie. a "Violent Decompression" in this room)
Fog Profile	Volume Setting	Define the fog colour and density used when the Player is in this Room
First Room Enter Event	wWise Audio Event	The audio event sent when the Player first enters this Room
Subsequent Room Enter Event	wWise Audio Event	The audio event sent when the Player subsequently enters this room

Room Exit Event	wWise Audio Event	The audio event sent when the Player exits this room
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ROOM TYPE ASSET

Parameter	Type	Description
Name	string	[Deprecated - Use Room Name on the Room Data Asset instead]
Short Name	string	[Deprecated - Use Short Room Name on the Room Data Asset instead]
Room Type State Value	wWise Audio Event	The audio event used to define the general ambience, music and audio behaviour of this Room